Learning Dexterous Manipulation

Hands, simulators, training, and evaluation

> *Compiled 2026-10-14. Every claim traces to a note in `papers/notes/`, every note to a parsed source in `papers/md/` or `code/md/`, and every source to a hash or commit in `corpus/manifest.json`. The method is in Appendix A.*

Abstract

A dexterous hand can change an object's pose without setting it down. This survey covers 113 learning-based methods that attempt it, the hands they run on, the simulators they train in, and how they report results. The field has largely converged on one recipe: reinforcement learning on privileged state in a GPU simulator, distilled into a vision-only student for transfer to hardware. Almost half the corpus trains that way, nearly all of it in one simulator, whose own successors are barely used. A second and faster-growing branch learns from teleoperated demonstration instead, through vision-language-action models, diffusion policies, and systems whose contribution is the collection rig itself. Published work has tripled since 2022. Grasping remains the largest task family and bimanual coordination the fastest-rising, and almost every experiment runs on one of four hands. What the survey adds comes from reading released code and vendor specifications beside the papers. Eight hands that can be bought or built from published designs appear in no method paper here. Where the depth a hand sinks into the object it holds is reported at all, it is not measured in the physics that produced the motion. Section VII proposes a protocol for the unreported quantities; the body gives the counts behind every claim above.

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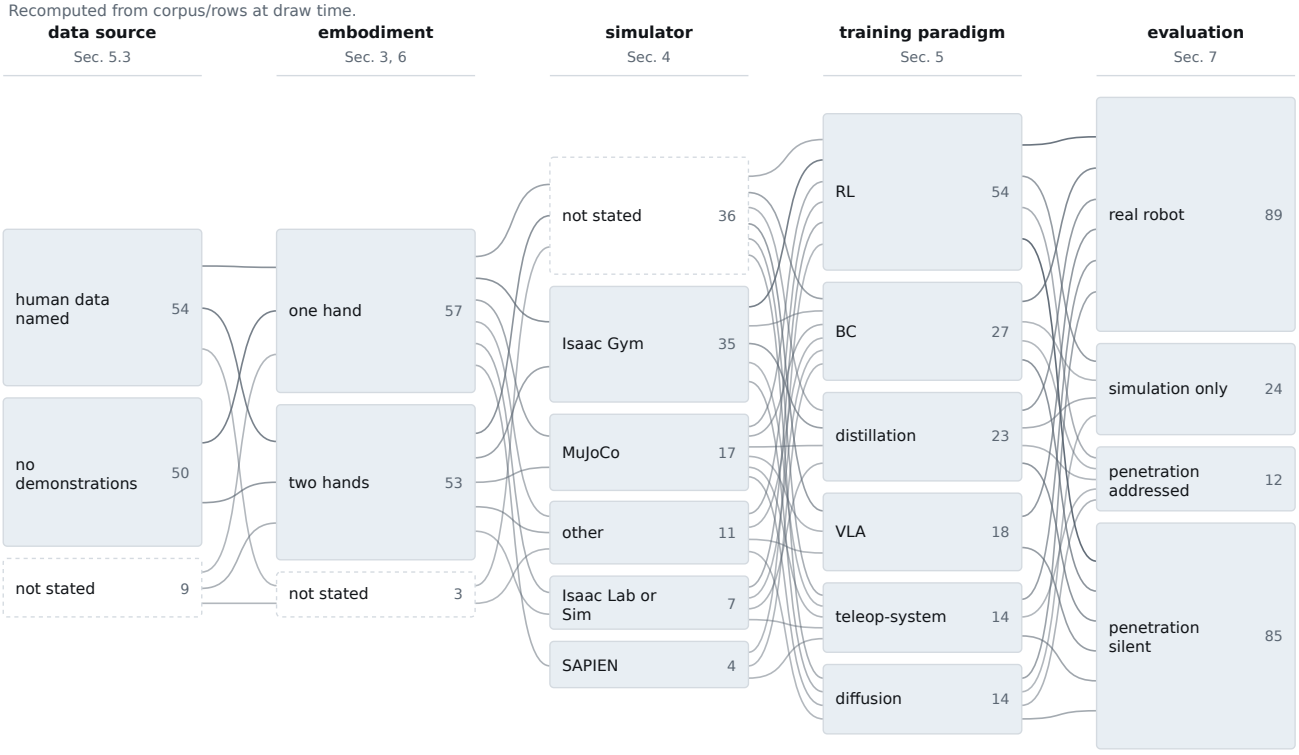
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A survey of 222 bibliography entries, 219 of which carry a structured row read from a note: 113 method papers, 33 hands, 15 simulators, 15 datasets, 14 benchmarks, 14 surveys, 8 tactile sensors and 7 evaluation protocols. "Method row" throughout means one of the 113, and every headline count here has one of those eight classes as its denominator.

1. Introduction

Figure 1. The field on one page, and the map of this survey

Every node is a count over the 113 method papers in the corpus, and every column head names the section that covers it. Dashed nodes are reporting gaps, not choices. A paradigm count may include a paper twice, because a method may use several. Recomputed from corpus/rows at draw time.



The routes converge. The heaviest single path is no demonstrations, one hand, Isaac Gym, reinforcement learning, real robot. Whatever enters on the left, almost everything leaves through a GPU simulator and arrives at a real robot without contact quality ever being measured.

A hand is dexterous when it can change an object's pose without putting the object down. That property, and not the finger count, is what separates a hand from a gripper. `bicchi_grasping_chapter_2001` draws the same line in Sec. 1.1 between restraining an object and manipulating it with the fingers: restraint is a static question about whether the contacts prevent motion, while in-hand manipulation is a dynamic one about whether contacts can be broken and remade while the object stays held. A parallel-jaw gripper can hold almost anything and reorient almost nothing.

The classical theory of grasping is about holding, and it is good at it. Given a set of contact points and an assumption about friction, it can decide whether an object is immobilized, and what force each finger has to apply to keep it that way. What it cannot do is say what the fingers should do next. Contact is what makes that hard: a finger is either touching or it is not, and it can push but not pull, so the equations of motion change form every time a contact is made or broken. The standard treatment says as much about its own results, that immobilizing an object "does not guarantee stability", and that "the nonsmooth nature of grasp dynamics, because of the unilateral constraints on displacements and forces, has made a thorough analysis very difficult" (`bicchi_grasping_chapter_2001`, Sec. 1.5 and 1.8). The tradition produced tests. You can check a candidate grasp against them; you cannot run them on a robot.

Learning changed what a practitioner has to produce. The old deliverable was a proof that a configuration holds. The new one is a policy that scores well over many simulated attempts, which means the simulator now stands in for the analysis and the score stands in for the proof. That is why the field moved, and it is also why this survey is organized as it is. It shifts two burdens onto the author. The simulator has to be faithful where fidelity matters. The measurement has to mean something. Much of what follows asks how well this literature carries those two.

One recipe now carries most of the field. Train a policy with reinforcement learning on privileged state, the object pose, velocities and physical parameters the robot cannot sense, in a GPU simulator running thousands of environments in parallel;

distill that teacher into a student that sees only what a camera and joint encoders provide; transfer the student. Fifty-four of the 113 method papers here train this way, most of them in one simulator, and about half of those add the distillation stage. In-hand cube reorientation is the recipe's benchmark, inherited from `openai_dexterity_2018` and still the task a new method is expected to show.

A second branch is growing underneath it and inverts the dependency. Instead of a simulator and a reward it takes human demonstrations through a teleoperation rig and fits a policy directly, as a vision-language-action model, a diffusion policy, or a data-collection rig whose design is itself the contribution. Section 5 counts them; Figure 1 shows how fast they have arrived. It is the newer literature, most of it from the last two years, and it trades the reward-design problem for a data-collection problem. Which of the two branches ends up feeding the other is not settled here, but the growth is the clearest trend in the corpus.

Both branches run on remarkably little hardware. Four hands carry almost every experiment in this corpus, led by the Allegro at 35 of the 113 papers; Figure 2 gives the rest of the distribution. Eight further hands can be bought today or built from published designs and appear in the experiments of none of them. The hardware exists and is purchasable, so this is not a manufacturing lag, and section 3 takes it up: a field whose results rest on four platforms is more fragile than its publication volume suggests.

What this literature cannot currently do is compare its own results. Nearly every paper states a success criterion, but they are not the same criterion, and only 70 of the 113 say how many trials a reported rate rests on. The criteria run from never dropping the object to reaching a pose within a tolerance each paper picks for itself. Two success rates from two papers on nominally the same task are usually not measuring the same event, and none of the surveys preceding this one supplies a protocol that would make them comparable. Section 7 derives one, with the trial counts a stated confidence interval actually requires.

Every number in this survey is traceable to a source, and every count in it is recomputed from those sources rather than typed in. Appendix A says how the corpus was built, what the method cannot do, and why each coverage figure here should be read as a floor rather than a rate.

One part of that method is unusual enough to flag now. Where a paper published a repository, its reward was read from the repository as well as from the paper, so the two can be compared. Reading an implementation against its own description is established practice outside robotics, in repeatability and reproducibility studies `collberg_repeatability_2016` `raff_reproducibility_2019`, in audits of how implementation detail moves reinforcement-learning results `engstrom_implementation_matters_2020`, in reward-design review `knox_reward_misdesign_2023`, in benchmark re-releases `metaworld_plus_2025`, and most directly in recent work on paper-code alignment `scicoqa_2026` `biocon_2026`. We are aware of none that does it for a robotics field's released reward functions. Section 5.6 reports what this one found.

The scope is single-hand and bimanual multi-fingered manipulation under learned control. Parallel-jaw work enters as a comparison and not as a subject. Two limits apply to every number here. The corpus is of the learned era, which is a selection and not a verdict on the analytic tradition; and where this survey failed to extract a value, the counts look the same as if the source had never stated one.

1.1 How to read this survey

Figure 1 puts the field on one page, and the sections work outward from the task. Section 2 sets out the task families, what makes each one hard, and the handful of terms this survey uses precisely. Section 3 covers the hands and their sensing, section 4 the simulators and what they expose about contact, section 5 how policies are actually trained, section 6 what changes when there are two hands, and section 7 how any of it is evaluated. Section 8 says what the corpus supports and what it does not. The appendices hold the method and the full tabulation: A the method and its limits, B the hardware and engine tables, C every paper-against-code comparison, D the surveys that precede this one, E the derivation behind the protocol.

If you are new to the field, read section 2 for the vocabulary, then the first half of section 5 for the two ways a dexterous policy gets trained, then section 3 to see what hardware any of it runs on. Those three give the shape of the field in about half

an hour. Section 7 is the one to read before designing an experiment, because it is about what the field's numbers do and do not mean.

If you already work here, the sections that are likely to tell you something you did not know are 4.2 on what simulators expose about contact, 5.6 on papers against their own released code, 6 on bimanual work as its own problem rather than two single hands, and 7.3 on what a success rate at twenty trials can support.

If you want the data rather than the prose, every table is generated from the corpus and every number in the text is recomputed from it at build time. Table 7 is the paper-by-paper tabulation, Tables 2 and 3 the hands, Table 4 the simulators, and the appendices hold the rest.

Three things to carry while reading. The corpus is of the learned era, which is a selection and not a verdict on the analytic tradition that preceded it. Every coverage figure is a floor: when a paper is silent and when this survey failed to extract a value, the counts look the same, and Appendix A says which direction that bias runs. And a success rate in this literature is rarely comparable across two papers, for reasons section 7 sets out, so prefer the sections that explain mechanisms to any ranking you could assemble from the numbers.

2. A taxonomy of the problem

2.1 Task families

Dexterous manipulation is not one task, and what separates its tasks is what makes some of them hard: what is held, what moves, which way gravity points, and what counts as failure. Six families cover most of the corpus, and the last paragraph here says what they miss. Grasping is the largest, with 58 of the 113 papers carrying the label; the labels are not exclusive, so one paper can sit in several families. Success is a lift that survives a hold, and the thresholds differ by more than an order of magnitude: one paper requires the object "held 10 cm above the table for 20 s", another "held at least 0.5 s in simulation". What makes grasping hard at scale is the continuum of starting configurations. `unidexgrasp_pp_2023` states it in Sec. 4.3, that "we are dealing with an infinite number of tasks considering the initial object pose can change continuously".

Functional and tool use is second with 45 papers. It is the family where a stable grasp can still be the wrong answer. `dexterous_functional_grasping_2023` gives the case in Sec. 2.1, that "grabbing a hammer from the head or handle are both equally valid ways of using it", and only one of the two lets the tool be used. Success is defined against the tool's function, and the field has no shared way to state it.

In-hand reorientation has 31 papers and the most settled criteria, because the community inherited one number. `openai_dexterity_2018` declared the goal achieved below 0.4 rad of orientation error, `dextreme_2022` keeps 0.4 rad at test time against a 0.1 rad training tolerance, and `eureka_2023` counts consecutive successes at 0.1 rad. A shared tolerance is not a shared protocol, because the stopping rule differs. `visual_dexterity_2022` measures error "when the controller predicts it has reached the goal and stops", which makes the policy a judge of its own trial.

Tracking a human reference has 12 papers and inverts the problem. The target is a trajectory rather than an endpoint, so success is a per-frame error band. `maniptrans_2025` requires all four of 30 degrees of rotation, 3 cm of translation, 8 cm of joint error and 6 cm of fingertip error. The difficulty is that the reference came from a human hand and was never dynamically feasible for the robot. `dexmachina_2025` warns in App. B.4 that scoring by timesteps inside the thresholds makes "the results highly sensitive to the threshold values".

Bimanual coordination has 43 papers and handover has 10. Handover is the smallest family and the one whose failure has a single moment, because the giver must release only after the receiver has the object. `dynamic_handover_2023` reports a hit rate above its success rate and blames the gap in Sec. 5.4 on "occasional challenges encountered during the grasping phase of the catcher". Not every paper here learns both sides. In `dexterous_handover_2025` only the receiver is learned and the giver is a scripted arm.

Those six families do not cover the corpus. Twenty-four of the 113 papers carry a label from outside them and eight carry no label from the six at all. Twenty fall in a catch-all class, mostly generalist policies evaluated on a task suite rather than on a dexterous task family, among them `pi0_2024`, `pi05_2025`, `pistar06_2025`, `openvla_2024` and `gemini_robotics_15_2025`. Three adjacent families are named here instead of being folded into the six. Locomanipulation is `humanplus_2024`, `omnih2o_2024` and `groot_n16_2025`, where the base is not fixed and the gravity argument below changes character. Piano playing is `robopianist_2023`, `rp1m_2024` and `pianomime_2024`, discussed in Section 6, where success is a per-timestep F1 against a MIDI score rather than an object pose. Deformable manipulation is `dexdeform_2023`, whose object carries its own state and its own physics. `ferrari_canny_1992` carries no task label because it is a grasp-quality measure rather than a task. Table 1's six families are the ones with enough papers to compare, and not a partition of the corpus.

2.2 Sources of difficulty

Contact is non-smooth, and that is a property of the problem rather than of any solver. `bicchi_grasping_chapter_2001` states in Sec. 1.2 that contact constraints are unilateral, and that losing a contact "involves an abrupt change of the structure of the model under consideration". Its Sec. 1.4 gives the sharper version, that a rod sliding on rough ground has configurations with no consistent solution and configurations with more than one. Simulators inherit the difficulty. The re-implementation

study contact_models_comparison_2023 concludes in Sec. V that "there is no fully satisfactory approach at the moment, as all existing solutions compromise either accuracy, robustness, or efficiency". pang_global_planning_2022 treats non-smoothness as the thing to remove, curating every contact pair in Sec. III-D so that contact points and normals change smoothly.

The hand has more actuators than the object has degrees of freedom and is still short of authority over it. The object moves only through contacts, and the contacts are what cannot be commanded. bicchi_grasping_chapter_2001 Sec. 1.4 states that in multifingered grasps "the number of independent contact forces is much larger than the number of actuators. Thus, from a controllability standpoint, not all the contact forces are controllable".

Vision is occluded by the hand doing the work. bai_unified_manip_survey_2025 names it in Sec. 4.3, that "occlusion hampers object tracking". dexpoint_2022 keeps vision and adds imagined hand points to the point cloud. rotating_without_seeing_2023 removes vision entirely and rotates objects from 16 binary touch sensors over the palm, links and fingertips.

Gravity direction changes the task rather than scaling it. visual_dexterity_2022 states that reorientation with the hand below the object "is much easier", because "with a downward-facing hand, the hand must manipulate the object while simultaneously counteracting gravity". anyrotate_2024 makes the direction a randomised variable by "randomly initializing hand orientations between episodes", and its Sec. 5.3 measures the cost, with performance dropping progressively from palm up and palm down, through base up and base down, to thumb up and thumb down.

Three different counts describe two hands, and they measure different things. The widest is the 53 papers whose robot has two hands at all. Narrower is the set whose task is coordinating them. Narrowest, and the denominator for every architecture count in this survey, are the 28 papers that put a learned closed-loop controller on two multi-fingered hands; section 6 opens by naming every exclusion that takes the wider count down to it. Every bimanual claim here says which of the three it uses.

Four things genuinely change when the second hand arrives. Contact stays non-smooth, occlusion stays, and gravity stays the same problem.

Role asymmetry is one. asymdex_2024 assigns a dominant hand with full finger and wrist control and a facilitating hand with 6-DoF base pose only, so that "the facilitating hand repositions and reorients one object, while the dominant hand performs complex manipulations".

Two hands on one object also close a kinematic chain through the object. In bicchi_grasping_chapter_2001 Sec. 1.4 the hand and object dynamics are separate and are "linked through the n rigid-body contact constraints". A second hand adds a second constraint set on the same object rather than a second independent problem. That is where physical plausibility fails first. bimangrasp_2024 is the one paper here that measures it, rejecting a synthesised grasp when "total penetrations exceeded 1.5 mm" and reporting in Sec. IV-C that "penetration remains the primary cause of grasp failure".

Two-arm collision does not exist for one hand at all. dydexhandover_2025 resets an episode when "any unintended arm contact with the environment or self-collisions" occurs. bunny_visionpro_2024 pays for it in the controller, where adding a sphere-approximated self-collision cost raises motion-control time from 0.74 ms to 7.85 ms. bidex_teleop_2024 solves it in hardware, mounting the hands so the human arm and the teacher arm "are perpendicular to each other and do not collide". dexmimicgen_2024 states plainly that it does not handle inter-arm collision at all.

The action space doubles as well. bidexhands_2022 runs 20 tasks on two Shadow Hands and finds single-agent PPO beating multi-agent RL on most of them, offering in Sec. 5.2 that "PPO algorithm is able to use all observations for training the policy, while MARL can only use partial observations". asymdex_2024 attacks the dimensionality from the other side, halving the observation and action dimension through its role split and a frame relative to the facilitating hand's object.

2.3 Terms used throughout

Six terms recur in every section that follows, and each is used here in one sense only.

Method paper. A paper whose own experiments train or evaluate a dexterous policy, as opposed to a hand, a simulator, a dataset, a benchmark, a tactile sensor, an evaluation protocol or an earlier survey. The 113 method papers are the unit every count in this survey is taken over.

Privileged state. Information a simulator can supply and a robot cannot: object pose and velocity, contact forces, mass, friction. A policy trained on it cannot be deployed as it stands, which is what makes the next term necessary.

Teacher-student distillation. Training one policy on privileged state, then fitting a second that sees only deployable observations, usually camera images and joint encoders, to reproduce the first. Section 5 finds 23 of the 113 methods do this, and it is the second half of the recipe section 1 describes.

Reference and rollout. A reference is a pose or a trajectory scored before anything executes: a retargeted human motion, a synthesized grasp, a planned path. A rollout is what a trained policy actually did. The distinction matters because a quantity can be measured carefully at the reference and never at the rollout, which section 7 finds is the case for interpenetration.

Interpenetration. The distance by which two bodies that should be touching instead overlap, a consequence of solving contact constraints approximately rather than exactly. Section 4 covers what each engine exposes; the depth is a property of the compliance someone chose, not of the engine.

Success criterion and per-cell trial count. The criterion is the predicate a paper uses to call an episode a success: a pose within a tolerance, an object never dropped, a task completed. The per-cell count is how many trials one reported rate is computed from. Section 7 finds 98 of them state a criterion and no two state the same one, and that 70 state a count.

Table 1. Task families against the properties that define success

task family	what is held	what moves	what gravity does	what counts as failure	shortest honest success criterion
In-hand reorientation	one rigid object, held by the fingers for the whole episode	the object's orientation in the palm frame, while contacts break and remake	a disturbance whose direction is set by hand orientation, and a load to counteract with the palm down, `visual_dexterity_2022`	the object leaves the hand, e.g. the fall reset at goal distance 0.24 m in `dextreme_2022`	orientation error below a stated tolerance under a stated stopping rule, 0.4 rad in `openai_dexterity_2018` and `dextreme_2022`, 0.1 rad in `eureka_2023`
Grasping	nothing at the start, the object rests on a support	the hand closes, then the object is lifted clear	the test itself, since the grasp exists to resist it	no lift, or a drop inside the hold window	a stated lift height held for a stated duration, 10 cm for 20 s in `dexgraspvla_2025`, 10 cm to episode end in `graspxl_2024`, 0.5 s in `omnigrasp_2024`
Functional and tool use	a tool, held in the pose its function requires rather than the most stable one	the tool's own joint, or the tool acting on a second object	secondary, the function dominates	a stable grasp in the wrong place, `dexterous_functional_grasping_2023`	no agreed criterion. Criteria range from never dropping the object, `dexterous_functional_grasping_2023`, to a four-point rubric, `dexvla_2025`. The one mechanical threshold is holding the tool joint past a commanded angle for the episode, `articulated_tools_inhand_2025`
Tracking a human reference	whatever the human held in the captured clip	hand and object follow a recorded trajectory under physics	full, and the reference never obeyed it, so the reference is not dynamically feasible	per-frame error leaves the band, or the object drops	all stated per-frame errors below threshold for the whole clip, 30 deg and 3 cm and 8 cm joint and 6 cm fingertip in `maniptrans_2025`. Thresholds are not comparable across papers, and `dexmachina_2025` calls the results "highly sensitive to the threshold values"
Bimanual coordination	one object by two hands, or one object per hand	both hands, the object, and the relative pose of the hands	can be carried by one hand while the other acts, `asymdex_2024`	the object drops, the arms collide, or the hands work against each other	no agreed criterion. Task-specific thresholds dominate, e.g. 0.035 m block-to-cup distance in `asymdex_2024`. The one criterion with a physical-plausibility gate is 2.0 s of hold with total penetration under 1.5 mm, `bimangrasp_2024`
Handover and in-hand transfer	by the giver at the start, by the receiver at the end	the object crosses between hands, released only after it is regained	unopposed during the exchange, and total in a thrown transfer, `dynamic_handover_2023`	a drop at the moment of release, and the catcher's grasp phase, `dynamic_handover_2023` Sec. 5.4	the receiver holds the object and moves it clear, more than 10 cm from the first hand in `hato_visuotactile_2024`, or holds it to episode end in `dydexhandover_2025`

Two cells say the field has no agreed criterion, and both are in families where a second body is involved. A criterion naming a distance, a duration and a trial count can be re-run by someone else. A rubric cannot, and Section 7 takes up what follows from that.

2.4 What has changed since 2022

The corpus roughly tripled between 2022 and 2024, and its composition changed while it grew. Three shifts are visible in the record, and none of them is small.

The demonstration branch overtook the simulator branch in 2024. Sorting each method by whether it trains in simulation, through reinforcement learning or distillation, or from human demonstration, through behaviour cloning, a vision-language-action model, a diffusion or flow policy, or a collection rig, the simulator branch led by better than three to one in 2023 and was behind a year later. It has trailed every year since; Figure 1 plots both. The recipe section 1 describes is this field's mature pipeline rather than its growth area, and a reader who knows only that recipe is reading the field as it was in 2023.

In-hand reorientation is being abandoned as the canonical task. It was 8 of the 11 method papers in 2022 and 3 of the 34 in 2025: the absolute count falls while the corpus triples, which is a change in what the field treats as worth publishing rather than a shift in emphasis. Bimanual coordination and tool use moved the other way over the same span, each from a handful of papers to more than a dozen. The cube-rotation benchmark that section 5 shows the reward literature organised itself around is the task the newer literature has left.

No paper earlier than 2025 uses Isaac Lab or Isaac Sim at all. Three do in 2025 and four in the partial 2026, against nine and two respectively on Isaac Gym, so on small numbers the successors already carry more of the most recent year than the original does. A survey assembled twelve months earlier would have recorded a field that had not moved, which is worth stating plainly: the concentration this survey reports on one simulator is a fact about a corpus with a cutoff, not a permanent property of the field.

3. Hands and who makes them

3.1 Design axes

A policy is only as dexterous as the hand it runs on, and the hands available divide sharply: a few that a laboratory can buy, a few more it can build from published designs, and a growing number that exist only in an announcement. What can be said about them is less than it looks. No degree-of-freedom, force, weight or price figure in Table 2 or Table 3 was measured by anyone outside the maker. Six entries are the exception, from peer-reviewed papers with stated protocols: ILDA, Pisa/IIT, ORCA, RUKA, LEAP and BiDexHand. Table 2 dates every other claim.

The degree-of-freedom count is the first number a vendor states and the least comparable one, because three different quantities get printed under that heading: joints, actuated joints and actuators. Shadow's own specification separates them, reading "20 actuated DOF and a further 4 under-actuated movements for a total of 24 joints" `shadow_dexterous_hand_2005`; Inspire's page gives a DoF figure and a joint count that differ by a factor of two `inspire_rh56dfx_2023`; a tendon hand can list more actuators than it has kinematic degrees of freedom `daxo_muscle_v0_2025`. Table 2 therefore prints joints and actuated DoF in separate columns, with actuators in a third, and the gap between them is the informative number.

The actuation ratio is the real decision. The Pisa/IIT SoftHand drives nineteen joints from one motor, and its authors state the cost plainly: "No in-hand dexterous manipulation is required for this prototype". Tendon drive moves the motors off the fingers and the mass follows them, so a hand with every motor in its palm can weigh a quarter of a tendon hand with a forearm; Table 2 gives both weights. What tendons cost is state estimation. The Faive Hand estimated joint angles from tendon length through a Kalman filter and its cube reorientation failed on hardware, which its authors blamed on "poor joint angle measurement from the EKFs, especially when there is contact" `faive_hand_2023`. Ruka-v2 adds detachable AS5600 encoders to measure that problem rather than close the loop on it, and those encoders put its open-loop linear joint-to-motor map at 8.26 degrees of error over seven joints `ruka_v2_2026`.

Direct drive trades torque density for transparency: LEAP's Dynamixel joints resist 19.5 N in a pull-out test against the Allegro's 8.5 N `leap_hand_2023`. Linkage drive is the argument against the forearm, putting motors, drivers and ball screws inside the palm and still reaching tens of newtons at the fingertip `ilda_hand_2021`. Linkages also buy joint coupling for free, as in BiDexHand's four-bar that drives each distal joint off the one behind it, though that hand's paper and its own README disagree about how many joints and servos it has `bidexhand_2025`.

Fingertip force is the number a hand buyer reads first, and across these hands it is six different measurements wearing one name. Pull-out resistance, a pinch between two fingers, fingertip normal force under a standard indenter, a calibrated multi-trial measurement, a peak vendor claim and two unlabelled columns on a spec sheet all appear as "fingertip force". One vendor's figure turns out to be the newton equivalent of a payload at a fixed motor current, which is a different quantity again. Table 2 therefore prints the protocol beside every force it reports, and leaves the cell empty where the protocol cannot be recovered. No two hands in it report payload under the same test either.

Compliance is repairability seen twice over, and both are mechanical fuses: the Pisa/IIT SoftHand's rolling-contact joints return to assembly after over-extension and ORCA's "poppable" pin joints dislocate rather than break `pisa_iit_softhand_2014` `orca_hand_2025`. ORCA also reports tendon drive as maintenance: "Prolonged use requires manual re-tensioning to maintain performance". What fails is rarely the motors. Its durability run found silicone skin degrading on two fingertips after about 2,000 to 4,000 grasp cycles and sensor wires snapping on three after about 4,500 to 7,000. The sensing wore out an order of magnitude sooner than the hand.

Two columns matter before any of the above, and no vendor comparison carries them. The first is the rate at which a control loop can actually be closed, which spans a factor of four across these hands and is not stated at all by some; one hand's quoted rate turns out to be a serial query ceiling while its own transferred policy runs at 20 Hz, which answer different questions. A reader deploying torque control is bitten by that first. The second column says whether a URDF or MJCF exists and who ships it, which is the bridge to section 4 and to this survey's recommendation: prefer a hand a simulator already carries. LEAP

is the warning there, because its paper says the URDF is released and the released API repository contains none `leap_hand_2023`.

Table 2. Hands that can be obtained

hand	maker	joints	act. DoF	actuators	actuation	weight g	force N	what the force is	payload	control rate	URDF or MJCF	tactile	price USD	open HW	status	source	claim date	corpus methods using it
`shadow_dexterous_hand_2005` [1]	Shadow Robot Company	24	20	20	tendon-driven, motors in the forearm (20 Smart Motor nodes, Maxon motors, PWM)	4300			4 kg in a power grasp, vendor claim, protocol not stated	EtherCAT at 1 kHz to the host, with a 5 kHz tendon-force loop inside each motor module	MuJoCo models for Hand E and variants, plus Gazebo, in sr_common	Shadow Tactile Fingertips (STF): 17x3 DoF Hall-effect taxels, 1000 Hz, uncalibrated; up to 5 fingertips; software also supports BioTac/MST/UBI0/PST fingertip options		no	sold	datasheet	specification dated 2024-12-04	21: `asymdex_2024`, `bidexgrasp_2026`, `bimangrasp_2024`, …
`sharpa_wave_2026` [2]	Sharpa Robotics (Sharpa Pte Ltd)	22	22	22		1300	20 / 12	vendor spec table with two unlabelled columns, probably two configurations; grip force is listed separately as 150 N / 90 N	40 kg / 24 kg (the same two unlabelled columns)	500 Hz over 1000BASE-T Ethernet	MuJoCo and Isaac Sim models	'Dynamic Tactile Array' (DTA), camera-type array at the fingertip: resolution 240x240 / 60x60, force resolution 0.02 N / 0.05 N, spatial resolution 1 mm / 2 mm, frame rate 180 fps / 30 fps, force range 0-30 N, latency 20 ms, 6-D F/T; number of sensors per hand not stated		no	sold	vendor page	undated page, copyright 2026, fetched 2026-09-17	6: `dexteleop0_2026`, `egoscale_2026`, `metis_2025`, …
`proception_prohand_2026` [3]	Proception Inc (YC W25)	22			tendon-driven, motors pull cables to move the fingers							integrated skin-like sensors detecting contact, supporting grip control; same 'sensor skin' used on the		no	sold	press	launch post and press, 2026-06-29	0

hand	maker	joints	act. DoF	actuators	actuation	weight g	force N	what the force is	payload	control rate	URDF or MJCF	tactile	price USD	open HW	status	source	claim date	corpus methods using it
												companion ProGlove; type/count not stated						
bidexhand_2025 [4]	Zhengyang Kris Weng, Center for Robotics and Biosystems, Northwestern University	21	16	15	cable-driven, N-configuration tendon routing (FeeTech servos, endless-loop antagonistic pull-pull); thumb CMC flexion driven via a 4-bar linkage instead of a direct tendon		2.14	fingertip normal force, calibrated sensor, five trials per finger, average peak	4.54 kg (10 lb) lifted	serial to an onboard ESP32, ROS 2 position control; rate not stated	URDF, consumed by the ROS 2 and MoveIt stack; no physics simulator	none mentioned		yes	open-source	paper	paper, 2025 (ICRA Dexterity workshop abstract)	0
wuji_hand_2025 [5]	Wuji Technology (founded 2019)	20	20	20	direct-drive rotary, backdrivable					1000 Hz across 20 axes over 100BASE-TX Ethernet, MIT force-position hybrid mode	MuJoCo and Isaac Sim models	'Multi-Axis Force/Torque Fingertip Sensing' confirmed present; no type, count or range given		no	sold	vendor page	undated page, copyright 2026, fetched 2026-09-17	2: toporetarget_2026, unidex_2026
ruka_v2_2026 [6]	Xinqi (Lucas) Liu, Ruoxi Hu, Alejandro Ojeda Olarte, Zhuoran Chen, Kenny Ma, Charles Cheng Ji, Lerrel Pinto, Raunaq Bhirangi, Irmak Guzey, New York University and NYU Shanghai	20	16	16	tendon-driven, forearm-mounted actuators; decoupled parallel 2-DoF wrist via a passive spherical ball joint; single-tendon-plus-spring abduction/adduction					Dynamixel over USB serial; no rate stated	MuJoCo XML in the shared v1 repo; not confirmed to carry the v2 wrist	optional e-flesh fingertips (magnetic touch-sensing form factor); not part of the base design	1500	yes	open-source	paper	paper, 2026	0
tesollo_dg5f_2024	Tesollo Inc. (Incheon HQ, Gwangmye	20	20	20	one integrated actuator per joint ('high-	1763			pinching 2.5 kg rated and 5 kg maximum;	250 Hz over Modbus RTU or TCP and		none standard; optional fingertip		no	sold	vendor page	undated page, copyright 2026,	0

hand	maker	joints	act. DoF	actuators	actuation	weight g	force N	what the force is	payload	control rate	URDF or MJCF	tactile	price USD	open HW	status	source	claim date	corpus methods using it
	ong R&D/factory)				torque actuation', absolute encoder); page does not say direct-drive				enveloping 10 kg rated and 20 kg maximum	Ethernet TCP/IP		sensors (6-axis F/T, 3-axis force, or tactile) available; count/type not given					download links dated 2026-08	
`unitree_dex5_2025`	Unitree Robotics (Yushu Technology Co., Ltd.)	20	16	16	in-joint geared motor ('hollow-cup motor' + high-precision encoder + low-damping small-clearance reducer), backdrivable; page does not use the words tendon or linkage	1100	10	fingertip normal force under a 1 cm diameter cylinder pressed vertically down (vendor footnote)	3.5 kg palm down and 4.5 kg palm left, on a 5 cm round hard object	1000 Hz over USB 2.0, with per-joint stiffness and damping commands		Dex5-1: none. Dex5-1P: 94 pressure sensors per hand (2x5 palm + 2x3x5 finger pad + 2x3x5 fingertip + 2x3x4 finger root), range 10 g-2500 g.		no	sold	datasheet	undated page, copyright 2016-2025, fetched 2026-09-17	0
`orca_hand_2025` [7]	Clemens C. Christoph, Maximilian Eberlein, Filippos Katsimalis, Arturo Roberti, Aristotelis Sympetheros, Michel R. Vogt, Davide Liconti, Chenyu Yang, Barnabas Gavin Cangan, Ronan J. Hinchet, Robert K. Katzschmann, Soft Robotics Lab, ETH Zurich	17	17	17	tendon-driven (antagonistic fishing-line tendon pairs per joint); wrist uses a GT2 timing belt drive	1200			10.5 kg on all four fingers and 2 kg on the index finger alone, both at a fixed 600 mA motor current	serial to Dynamixel or Feetech motors; rate not stated	URDF-derived kinematic constants only; no URDF or MJCF file in the released tree	yes: FSR-based binary tactile sensing on all 5 fingertips (RP-C7.6-ST), absolute detection threshold as low as 0.05 N		yes	open-source	paper	paper, 2025	0
`leap_hand_2023`	Kenneth Shaw, Ananye	16	16	16	direct-drive (Dynamixel servos, e.g.	595	19.5	pull-out resistance: the outward		up to 500 Hz querying over USB	URDF claimed in the paper;	none (future work only: 'we plan to	2000	yes	open-source	paper	paper, 2023	12: `bidex_teleop_2024`,

hand	maker	joints	act. DoF	actuators	actuation	weight g	force N	what the force is	payload	control rate	URDF or MJCF	tactile	price USD	open HW	status	source	claim date	corpus methods using it
	Agarwal, Deepak Pathak, Carnegie Mellon University				XC330-M288), joint velocity ~8 rad/s			force a flexed finger resists before slipping or deviating more than 15 degrees (Table III)		serial; the paper's own sim-to-real policy runs at 20 Hz	the released LEAP_Hand_API repo contains no URDF, xacro or MJCF	develop and integrate LEAP Hand with low-cost touch sensors')						bidexhd_2024, cross_embodiment_world_models_2025, …
allegro_hand_v4_2016 [8]	Wonik Robotics Co. Ltd. (Seoul, South Korea); earlier versions 1.0/2.0 made by SimLab Co. Ltd.	16	16	16						333 Hz, CAN, the hand's own real-time clock (driver README)	URDF and xacro, left and right, in the ROS driver repo	none		no	sold	driver repo	driver repo, versions 1.0-4.0 undated, fetched 2026-09-17	35: anyrotate_2024, anyteleop_2023, asymdex_2024, …
ruka_2025	Anya Zorin, Irmak Guzey, Billy Yan, Aadhithya Iyer, Lisa Kondrich, Nikhil X. Bhattasali, Lerrel Pinto, New York University	15	11	11	tendon-driven (11 Dynamixel actuators in the forearm: XM430-W210T for the thumb, XL330-M288-T for the other fingers)		2.74	pinch force, best of three trials, averaged over the left and right hands (Table III)	6.0 kg: weight added to a curled cloth-bag grip until joint-angle error exceeds 15 degrees, best of three trials	Dynamixel over a USB-to-serial bridge; rate not stated, data collection ran at 15 Hz	MJCF	none (stated limitation: 'lacks tactile sensing')	1300	yes	open-source	paper	paper, 2025	0
robotera_xhand1_2024 [9]	ROBOTERA	12	12		gear-driven force-controlled joint modules per finger segment, back-drivable	1100			25 kg, tracker's 'Strength' figure, over 25 kg gripping palm-up	not stated	URDF exists but is licensed; maniptrans_2025 withholds it	tactile/force sensors on every fingertip confirmed present (senses contact, force, temperature); count and array size not stated	14000	no	sold	third-party tracker	undated tracker page, fetched 2026-09-17; the vendor page was not fetched	8: cross_embodiment_world_models_2025, deximit_2026, dexmachina_2025, …
shadow_dex_ee_2024	Shadow Robot Company, in collaboration with Google DeepMind	12				4100				not stated; the page claims 'high bandwidth torque and position control loops'		stereo camera-based fingertip tactile sensors (hundreds of taxels each); multi-taxel		no	sold	vendor page	undated page, copyright 2026, fetched 2026-09-17	1: demostart_2024

hand	maker	joints	act. DoF	actuators	actuation	weight g	force N	what the force is	payload	control rate	URDF or MJCF	tactile	price USD	open HW	status	source	claim date	corpus methods using it
												3-DoF arrays on middle and proximal phalanges; counts not stated						
inspire_rh56dfx_2023 [10]	Beijing Inspire Robots Technology Co., Ltd.	12	6			540	10	fingertip strength, vendor spec, four fingers; thumb 15 N; force resolution 0.50 N		RS485; rate not stated		none (this variant is the 'without tactile sensors' table; a tactile FTP variant exists per bib but is not on this page)		no	sold	vendor page	undated page, fetched 2026-09-17	19: ace_teleop_2024, articulated_tools_inhand_2025, being_h05_2026, …
brainco_revo2_2025	BrainCo Inc.	11	6			383	15	pinch force, vendor minimum (>=15 N); the same page states a whole-fist grip of >=50 N	>=20 kg	RS485 and CAN FD; EtherCAT on Pro and Touch; rate not stated		Touch variant only: multi-dimensional fingertip tactile module with 'Tactile Adaptive Control'; count/type not stated. Basic and Pro variants: none.		no	sold	datasheet	undated page, fetched 2026-09-17	1: bidexgrasp_2026
agibot_omnihand_2025																unavailable	no source: both store.agibot.com URLs 404, fetched 2026-09-17	1: clutterdexgrasp_2025
linkerbot_l20_2025	LinkerBot (SDK author 'CHIUS INC')									USB-to-CAN at 1 Mbit/s, optional Modbus or RS485; rate not stated	URDF, with PyBullet and Isaac Gym examples in the SDK	L10/L20: per finger 4 channels (normal force, tangential force, tangential direction, proximity), 5 sites x 4 channels, 0-255 range, sensor type		no	sold	vendor page	undated page, fetched 2026-09-17	1: being_h05_2026

hand	maker	joints	act. DoF	actuators	actuation	weight g	force N	what the force is	payload	control rate	URDF or MJCF	tactile	price USD	open HW	status	source	claim date	corpus methods using it
												not stated. O6 (reseller): 16 capacitive tactile regions.						
`psyonic_ability_hand_2021` [11]	PSYONIC (San Diego, CA, USA)		6	6	linkage (repo file/function names indicate a four-bar finger linkage; not stated on the vendor page itself)					BLE, I2C, UART or RS485; rate not stated	URDF (left, right, large, small, no-FSR) plus MuJoCo and Isaac Sim paths in the vendor repo	30 touch sensor values streamed (FSR-based per API and URDF 'no_fsr' variant); per-finger split not stated		no	sold	vendor page	undated page, fetched 2026-09-17	7: `ace_teleop_2024`, `asymdex_2024`, `bunny_visionpro_2024`, …
`dexhand_open_source_2023` [12]	Rob Knight, The Robot Studio (design); electronics/firmware/ROS 2 by Trent Shumay, IoT Design Shop			18	tendon (fishing line, Sufix 832 80lb / 0.8mm kiteline); Emax ES3301/ES3302/ES3351/ES3352 micro-servos for fingers, Feetech SCS2332 or PWM servos for the wrist					Arduino SCServo or SBUS firmware, BLE; rate not stated	URDF, in the IoT Design Shop ROS 2 packages	none mentioned	300	yes	open-source	project page and GitHub README	page posts 2023-08-08 to 2023-10-01; no hardware release date	0

20 rows. 95 of 240 specification cells (39%) over the 12 specification columns are values no source stated; the key, maker, provenance and usage columns are excluded because they are never blank. Every figure here is the maker's or the authors' own claim. Nobody outside the maker has measured any DoF, force, weight or price cell in this table, except the rows sourced to a peer-reviewed paper with a stated protocol. The force column is not a ranking: read 'what the force is' first, because pull-out resistance, pinch force, a fingertip normal force under an indenter and an unlabelled vendor spec are different measurements. A blank cell means no source stated the value. No cell is truncated: a value wider than its column is wrapped at a word boundary, so a cell that runs to several rendered lines is one value and not several.

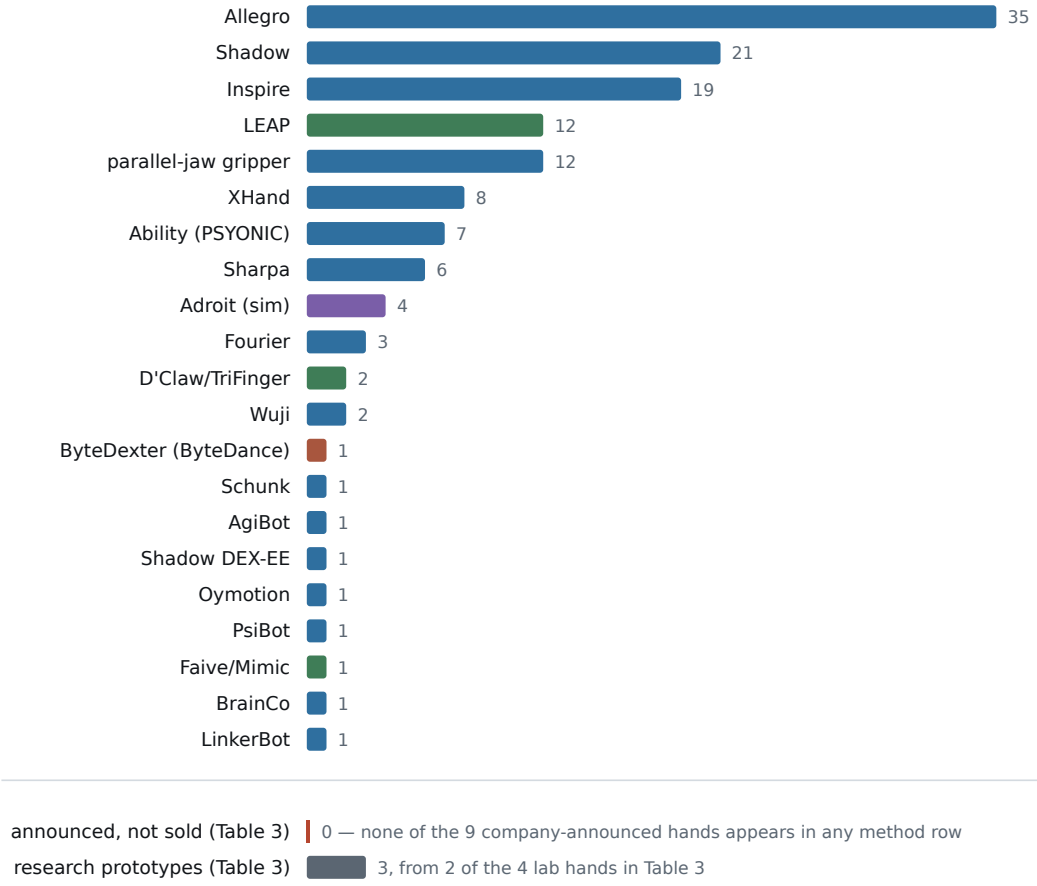
Figures with no reachable source, and other caveats on individual rows:

1. `shadow_dexterous_hand_2005`: the four distal joints are coupled and not independently controllable, so the '24 DoF' quoted in learning papers is the joint count
2. `sharpa_wave_2026`: the page footnotes 'Specifications may vary between products', and no column of the spec table is labelled
3. `proception_prohand_2026`: the evidence is a shipping announcement, not a datasheet: no weight, force, actuated-DoF count or control rate is stated anywhere
4. `bidexhand_2025`: the paper states 16 independently actuated DoF and 21 joints; the README states 15 servos driving 15 joints
5. `wuji_hand_2025`: the page footnotes that these figures 'reflect the current Beta1 product and the SPEC will continue to iterate'
6. `ruka_v2_2026`: the AS5600 encoders are attachable and detachable and are read for calibration and measurement only, not in the control loop
7. `orca_hand_2025`: the 19.6 N previously tabulated as a fingertip force is the newton equivalent of the 2 kg index-finger payload at a control-imposed current limit, and no source states it as a force
8. `allegro_hand_v4_2016`: allegrohand.com/v4 returned HTTP 404 and the Wonik wiki timed out; every cell here comes from the ROS driver repository, and weight, joint torque, payload and price have no reachable source
9. `robotera_xhand1_2024`: the 15 N in circulation comes from a survey table, not from the tracker or the vendor; the tracker states no fingertip force
10. `inspire_rh56dfx_2023`: the vendor's own page reads 'Degrees of freedom 6, Numbers of joints 12'; the joints column here prints the joint count, so the vendor's DoF figure is the actuated-DoF column
11. `psyonic_ability_hand_2021`: the four-bar finger linkage rests on repository file names and the five fingers on URDF meshes; the vendor page states neither
12. `dexhand_open_source_2023`: no joint count is given anywhere, so no DoF figure can be quoted; the servo count is 16 finger and thumb micro-servos plus 2 wrist servos, with a third optional. The $300 is 'additional total cost of components', excluding printing and the wrist servos

3.2 Hands in use, and what they cost

Figure 2. Which hands the surveyed methods actually run on

One count per method paper whose own experiments use the hand, from 113 method rows.
Colour: blue sold commercially, green open hardware, purple simulation-only model, orange built in-house and not sold.



The two oldest designs in Table 2 carry 49 of the 104 papers that name a hand at all, and seven papers use both. Figure 2 counts, per hand, the papers whose own experiments use it, and the shape of that figure is the finding: four hands take three quarters of the corpus and everything else shares the remainder.

The Allegro's position is the uncomfortable part. Its product page at allegrohand.com/v4 returned HTTP 404 while the Wonik wiki timed out `allegro_hand_v4_2016`. Everything Table 2 confirms about the most-used hand in the corpus comes from its ROS driver: 16 joints, four fingers, a torque interface, a 333 Hz CAN clock, 12 V, no tactile sensing. Its weight, joint torque, payload and price have no reachable source. The field's most common platform cannot be specified from a page a reader can open.

Concentration would matter less if the hand did not move the result, and it does. On the same simulated cube rotation LEAP turns the cube nearly three times faster than the Allegro `leap_hand_2023`, and on the same three-trial pinch test RUKA reports almost twice the Allegro's force `ruka_2025`. A method compared only on Allegro hardware is compared at one point in a space where a single axis moves the headline number two or three times over. The three newest Chinese hands have arrived quickly and take a growing share, none of them before 2024, and eight hands that can be bought or built take no papers at all.

The cost of entry has collapsed at the open-hardware end. Seven hands in Tables 2 and 3 publish their designs, and the five that also publish a price run from a few hundred dollars to a few thousand, against six figures for the hands this literature

actually runs on. Table 2 carries each price with what it includes, because the bases differ: the cheapest figure excludes printing and wrist servos `dexhand_open_source_2023`, and one vendor's own number sits against a third party's estimate several times higher `orca_hand_2025` `ruka_v2_2026`.

The expensive end of that comparison is secondhand throughout: the six-figure prices for a Shadow Hand come from other people's comparison tables, because the vendor's own page asks buyers to discuss pricing. So the collapse is real at the cheap end and inferred at the expensive one, and it has barely moved the literature. Twelve of the 104 hand-naming papers use an open-hardware hand, and all twelve of those are LEAP.

What the cheap hands give up is sensing. LEAP has none and names touch sensors as future work `leap_hand_2023`, RUKA states its design "lacks tactile sensing" `ruka_2025`, and BiDexHand's sources never mention touch `bidexhand_2025`. ORCA is the exception, with binary FSR fingertips whose threshold was measured as low as 0.05 N on a fresh fingertip, against the sensor's rated 0.29 N, and 6.38 N on a degraded one `orca_hand_2025`.

3.3 Tactile sensing

DIGIT set the cost floor, stating a manufacturing cost of about 15 USD per sensor in batches of a thousand `digit_2020`, which is where the price of touch sensing stopped being the obstacle. Durability was as much of the contribution: its gel barely degraded over an abrasion test that destroyed the two gels compared against it.

Digit 360 is the same lineage at a different operating point: millions of taxels, force resolution around a millinewton, and on-device processing that cuts event-to-action latency by most of an order of magnitude `digit360_2024`. It is also not for sale, and no paper in this corpus names it.

XELA's uSkin is the non-camera alternative, and the one that bolts onto hands the field already uses: a curved fingertip kit for the Allegro and LEAP, with three-axis sensing points at gram-force resolution running at a few hundred hertz `xela_uskin_2020`. Table 3 gives the specification, including which figures belong to the product family rather than to the fingertip kit. One paper in this corpus uses it, and its self-supervised encoders beat matched end-to-end models when both see a fraction of the labels, though its own bead-maze policies never complete the maze on the real robot `sparsh_2024`.

The usage number is the one to keep, and it needs its inclusion rule stated: a sensor physically on the hand, read by the deployed policy. Eight papers meet it: `anyrotate_2024`, `articulated_tools_inhand_2025`, `dexteleop0_2026`, `dexumi_2025`, `hato_visuotactile_2024`, `robot_synesthesia_2023`, `rotateit_2023` and `rotating_without_seeing_2023`. The rule excludes `penspin_2024`, whose 20 binary contacts are simulated on an Allegro that has no tactile hardware and whose released config sets `enable_tactile: False`, and it excludes `dexndm_2025` and `dexplore_2025` for the same reason. Sixty-three of the 113 method papers use the word tactile somewhere in their parsed text and 35 carry it in their structured note. Against the eight that meet the rule above, either number is the gap worth quoting.

Almost none of the eight uses a high-resolution sensor. `rotating_without_seeing_2023` removes vision entirely and rotates objects from 16 binary touch sensors over the palm, links and fingertips, deployed zero-shot to a real Allegro, and Robot Synesthesia uses 16 force-sensing resistors read as binary `robot_synesthesia_2023`. Two papers exclude touch deliberately, HORA reporting rotation "even without the usage of vision and tactile sensing" `hora_2022`, and ORCA's authors dropping it from their reinforcement learning "due to the additional complexity involved in accurately modeling them" `orca_hand_2025`. Taxel counts have risen by three orders of magnitude while the policies consuming them have stayed at binary contact.

3.4 The catalog against the literature

Behind Table 3's entries sit three kinds of evidence, and conflating them is how a DoF figure with no source ends up in a survey. Vendor prose carrying numbers is the strongest, as on 1X's page of 9 July 2026 `onex_neo_hand_2026`. Video is second and carries none. Press or bibliography assertion is third, as with Tesla's V3 hand, known here only through a paraphrase of patents because the USPTO PDF parsed empty `tesla_optimus_hand_2025`. Table 3 blanks the cells resting on the third class and footnotes who did the arithmetic.

A survey can go one step past recording that a claim is unverified, which is to say which claims are implausible on their face. One vendor's actuator count and mass work out to about six grams per actuator including structure, tendons, routing and skin `daxo_muscle_v0_2025`. Another's mass figure appears to exclude a half-kilowatt pump `clone_robotics_hand_2024`. A third's degree-of-freedom figure sits on the same page as a finger count that is exactly half of it, so it is almost certainly the pair of hands `figure_03_hand_2025`. A fourth number, widely repeated for the Optimus hand, is one article's arithmetic rather than a vendor statement, and the previous generation's own figure was half of it `tesla_optimus_hand_2025`.

Skepticism belongs to the evidence class, not to which table an entry lands in. The fully-actuated claims from the newest vendors are claims about unmeasured hardware, and they sit in Table 2 beside the footnotes their own pages carry: "Specifications may vary between products" on one, a spec that "will continue to iterate" on a beta product on another `sharpa_wave_2026` `wuji_hand_2025`. Even the best-specified vendor page in the corpus leaves its fingertip-force columns unlabelled.

Table 3. Hands announced but not purchasable

hand	maker	joints	act. DoF	actuators	actuation	weight g	force N	what the force is	payload	control rate	URDF or MJCF	tactile	price USD	open HW	status	source	claim date	corpus methods using it
clone_robotics_hand_2024 [1]	Clone Robotics	27			hydraulic artificial muscle ('Myofiber'; small water-filled tubes that contract when pressurized), carbon-fiber bones and ligament-style tethers, driven by a 500 W water pump and 36 electro-hydraulic valves; each Myofiber generates up to 1 kg grip force, survived 650,000 test cycles	<910						pressure pads in the palm detect grip firmness (count not given); 70 inertial sensors track angle/speed (proprioception, not tactile)		no	internal-only	press	press, 2025-11-24, on a video of 2025-11-15	0
onex_neo_hand_2026	1X Technologies	25	25		quasi-direct-drive tendons via the '1X Tendon Drive' at low gear ratios (~5:1-15:1); motors in the forearm, tendons pulled through the wrist		45	distal flexion force, peak vendor claim		not stated		high-resolution tactile sensing across fingertips and finger surfaces (normal force, contact location, shear); sensor count/technology not stated		no	announced	vendor page	vendor page, 2026-07-09	0
sanctuary_phoenix_hand_2024	Sanctuary AI (Sanctuary Cognitive Systems Corporation)	21			hydraulic, 'unique miniaturized hydraulic valves'							not in the press release text; sidebar-only headlines mention new		no	internal-only	press	undated press release; trade reprint 2024-12-17	0

hand	maker	joints	act. DoF	actuators	actuation	weight g	force N	what the force is	payload	control rate	URDF or MJCF	tactile	price USD	open HW	status	source	claim date	corpus methods using it
												touch/tactile sensors as a separate later announcement, not confirmed as part of this hand						
`leap_hand_v2_adv_2025` [2]	Kenneth Shaw, Deepak Pathak, Carnegie Mellon University	21	17	17	tendon (PIP/DIP coupled by a single tendon per finger); motor type not stated					not stated	URDF, loaded into PyBullet by the released teleoperation node	none mentioned	3000	yes	announced	vendor page	undated project page; API repo at commit a0936196, fetched 2026-09-17	2: `bidex_teleop_2024`, `dexwild_2025`
`ilda_hand_2021`	Ajou University, Korea Institute of Machinery & Materials, Korea University	20	15	15	linkage-driven, direct linear drive (3 Maxon DCX8M motors per finger with GPX8 16:1 gearboxes driving ball screws via parallel+serial four-bar linkages)	1100	34	fingertip normal force in the bent pose; 28 N stretched, and 25 N per finger measured while crushing an aluminium can	18 kg	CAN to a desktop; loop rate not stated		6-axis F/T sensor per fingertip (5 total), force resolution 62 mN, range +/-35 N; not a distributed taxel skin		no	prototype	paper	paper, accepted 2021-11-05	0
`pisa_iit_softhand_2014`	Centro E. Piaggio, University of Pisa and IIT (Catalano, Grioli, Farnioli, Serio, Piazza, Bicchi)	19	1	1	tendon-driven adaptive synergy, single motor via differential gears (6 W Maxon RE-max21, 84:1 reduction)				holding force about 20 N along z and holding torque 2 N m, 28 N and 3.5 N m with a stronger motor	not stated	URDF and Gazebo models in the ROS repo (BSD 3-Clause)	none (motor encoder only; a padded work glove supplied contact compliance during experiments)		no	prototype	paper	paper, IJRR 2014	0
`faive_hand_2023` [3]	Yasunori Toshimitsu, Benedek Forrai, Barnabas Gavin Cangan, Ulrich Steger, Manuel Knecht, Stefan Weirich,	16	11	16	tendon-driven, rolling-contact joints (16 Dynamixel XC330-T288-T servos, 6 antagonistic pairs)	1100			10 kg, whole-hand downward power grasp on a dumbbell	Dynamixel over serial; joint angles estimated from tendon length by EKF; rate not stated	MJCF, used directly by the released Isaac Gym training repo	none on the physical hand; a simulated 'fingertip force' (15-dim, critic-only/privileged) exists only in the RL training observation, not real			prototype	paper	paper, 2023	1: `graspxl_2024`

hand	maker	joints	act. DoF	actuators	actuation	weight g	force N	what the force is	payload	control rate	URDF or MJCF	tactile	price USD	open HW	status	source	claim date	corpus methods using it
	Robert K. Katzschmann, Soft Robotics Lab, ETH Zurich											tactile hardware						
paxini_dexh13_2024	PaXini Tech											1,140 ITPU multidimensional tactile processing units (press-release claim; unclear if PX-6AX-GEN3 sensor family)			announced	press	press, CES 2026-01-07	0
tesla_optimus_hand_2025 [4]	Tesla				V3: tendon-driven from forearm actuators, three tendons per finger through a crosstalk-managing wrist. Gen 2: in-hand actuators/sensors (unlocated in the article).							Gen 2: tactile sensing on all fingers, demonstrated matching contact location and force in an egg-pickup video; type/count not stated. V3: not stated.		no	prototype	press	Gen 2 video 2023-12-13; V3 patents filed 2024-10, relayed by press 2026-04	0
boston_dynamics_atlas_hand_2026	Boston Dynamics				electric (robot is fully electric); specific hand mechanism not stated							tactile sensing in the fingers and palms, confirmed present; type and count not stated		no	prototype	press	trade press, 2026-01-05 (CES); the Boston Dynamics blog of the same date has no hand content	0
figure_03_hand_2025 [5]	Figure AI											not stated on page for the hand itself; palm cameras in each hand are vision, not tactile		no	prototype	third-party tracker	undated tracker page, fetched 2026-09-17; figure.ai launch page returned HTTP 404	0

hand	maker	joints	act. DoF	actuators	actuation	weight g	force N	what the force is	payload	control rate	URDF or MJCF	tactile	price USD	open HW	status	source	claim date	corpus methods using it
daxo_muscle_v0_2025 [6]	Daxo Robotics			120	ultra-redundant tendon-driven; compliant structure with no rigid joints, flexible materials and tendon routing	750									prototype	third-party tracker	third-party catalogue post, 2026-09-16; daxo-robotics.com returned HTTP 404	0
xiaomi_cyberone_hand_2026	Xiaomi				motors located in the hand (compact motors generate heat requiring liquid cooling); no tendon/linkage statement, though contrast with tendon-driven hands' cycle life implies a non-tendon design							full-palm tactile sensing, area ~8,200 sq mm, detects pressure and contact across the whole palm not just fingertips; taxel count and type not stated		no	prototype	press	press, 2026-03-30	0

13 rows. 96 of 156 specification cells (61%) over the 12 specification columns are values no source stated; the key, maker, provenance and usage columns are excluded because they are never blank. Every figure here is the maker's or the authors' own claim. Nobody outside the maker has measured any DoF, force, weight or price cell in this table, except the rows sourced to a peer-reviewed paper with a stated protocol. The force column is not a ranking: read 'what the force is' first, because pull-out resistance, pinch force, a fingertip normal force under an indenter and an unlabelled vendor spec are different measurements. A blank cell means no source stated the value. No cell is truncated: a value wider than its column is wrapped at a word boundary, so a cell that runs to several rendered lines is one value and not several.

Figures with no reachable source, and other caveats on individual rows:

1. `clone_robotics_hand_2024`: 'under 2 pounds' is an upper bound, and the source does not say whether the 500 W pump and 36 valves are inside it
2. `leap_hand_v2_adv_2025`: the page claims 21 DoF with 17 powered motors and the README calls the same hand 17-DOF
3. `faive_hand_2023`: the row's open-hardware status is unconfirmed: the repo has LICENSE and LICENSE-NVIDIA files whose contents were not captured, and no open-hardware claim appears in the sources
4. `tesla_optimus_hand_2025`: the 22 DoF in circulation is a Teslarati article's arithmetic of four DoF on each of five fingers plus two at the wrist, not a Tesla statement; Gen 2's own figure is 11 DoF. This row's actuation cell describes V3 and its tactile cell describes Gen 2
5. `figure_03_hand_2025`: the tracker's 'Degrees of freedom, hands | 20' sits beside 'Number of fingers | 10', so the 20 is most likely the pair and about 10 per hand; the 16-per-hand figure in circulation has no reachable source
6. `daxo_muscle_v0_2025`: 120 actuators in 750 g is about 6 g per actuator including structure, tendons, routing and skin; no DoF number is stated, and a redundant tendon architecture with no rigid joints has fewer kinematic DoF than actuators

The bottom of Figure 2 carry the finding. None of the nine company-announced hands in Table 3 appears in a single paper whose own experiments use it. Tesla, Figure, 1X, Sanctuary, Boston Dynamics, Xiaomi, Clone, Daxo and PaXini account for zero of the 113 method papers' experiments. The nearest thing to a counterexample is `helix_2025`, a Figure blog post claiming a 35-DoF whole-upper-body action space at 200 Hz that includes individual finger control. It never names the hand, gives no per-hand DoF count, and reports no success rate or trial count for any task. It is the maker describing its own unreleased hand, which is the evidence class the finding is about. The four research prototypes in Table 3 are in a different position, since the Faive Hand and LEAP Hand v2 Advanced account for three papers between them `graspxl_2024` `bidex_teleop_2024`.

Table 3's emptiness is measurable and part of the same finding. Over the twelve specification columns, 61 percent of its cells are values no source stated, against 39 percent for the hands that can be bought. The hands with the highest advertised DoF counts have the least specification behind them.

The one hand that crosses the gap crosses it because its maker published rather than announced. ByteDexter reaches a paper through a ByteDance technical report stating 21 DoF per hand, 16 piezoresistive fingertip channels in the action vector and real-robot success rates `gr_dexter_2025`, not through a product page. That mechanism is available to every vendor in Table 3 and none has used it.

The gap runs the other way too, and the last column of Table 2 shows it. Eight hands that can be bought today or built from published designs take no papers at all: the Unitree Dex5 `unitree_dex5_2025`, the Tesollo DG-5F `tesollo_dg5f_2024`, the Proception ProHand `proception_prohand_2026`, ORCA, RUKA, Ruka-v2, BiDexHand and DexHand. Cheapness is established for only three of those eight, because the price cell is empty for the rest. A reader choosing a hand on this corpus's evidence has four options with published precedent behind them, and only one of the four, LEAP, has both a published design and a stated price. Below those four the record thins fast, and 19 of the hands in Tables 2 and 3 sit there on their specifications alone, with no paper in this corpus running on any of them.

What would close the gap is tooling rather than manufacturing. Nineteen of the 33 hands in Tables 2 and 3 appear in no paper here, and most of those are neither sold nor open, so the eight obtainable ones named above are the real measure of the lag.

The lag runs through the simulators too: fewer than half of them ship a first-party hand model at all, and every model they do ship is an Allegro or a Shadow. Two recent benchmarks compare a dozen hands between them without stating a degree-of-freedom count for any. What would close it is a conformance suite for hand models: one URDF or MJCF per hand, fixed joint-limit, mass and collision checks, and a published pass or fail per engine.

4. Simulators and the physics underneath

4.1 Requirements of a dexterous simulation

The clearest demonstration that hands are the hard case came from a benchmark that was not about hands. Erez et al. built a 35-DOF arm modeled on the Shadow Hand, closed it around a capsule with fixed spring-dampers, and asked four of the five engines for the largest timestep at which the object was still in the hand. The answers spanned a factor of 500, from milliseconds down to tens of microseconds, with MuJoCo at the slow-step end (`physics_engine_comparison_2015`, Sec. IV-D). Three caveats travel with that spread. The engines ran a deliberately restricted common model, hinge joints with sphere and capsule geometry and no meshes; the authors call it "an open question whether this model system can be tuned to work better in the gaming engines"; and they wrote MuJoCo and say so.

A grasp is hard for nameable reasons. It is many contacts at once, all persistent, all near stiction, on an object much lighter than the mechanism holding it. Persistence and multiplicity make the problem hyperstatic, and Le Lidec et al. show that per-contact solvers of the projected Gauss-Seidel family, and RaiSim's, then inject spurious jamming forces at stiction that vanish only once the object slides. The mass ratio makes it ill-conditioned, and in their stacked-cube test at a 10^3 to 10^-3 kg ratio those same methods fail to converge. Global methods with proximal regularisation stay robust in both cases (`contact_models_comparison_2023`, Sec. IV-A). What degrades a truncated solve is therefore conditioning and redundancy, not stiffness, and a grasp supplies both by construction. Locomotion is forgiving by comparison. On flat ground the contact model and solver choice "hardly affects" their quadruped's tracked base velocity, and only on rough, slippery terrain do RaiSim and CCP deviate. A walking robot makes and breaks a few contacts against ground far heavier than itself. A hand does neither.

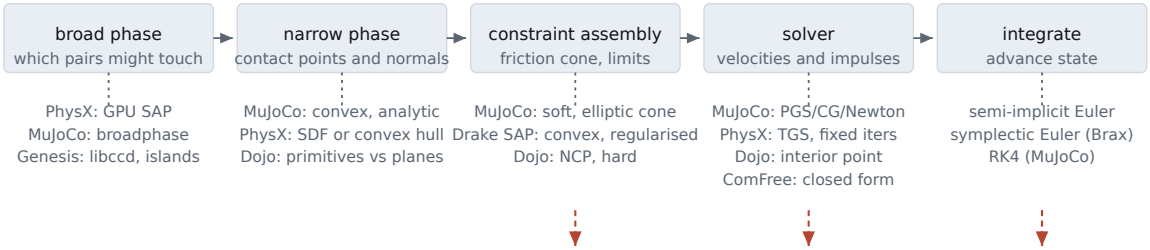
Speed enters by the same door. MuJoCo Playground reports that contact time scales with the number of possible contacts rather than the active ones, because JAX requires static shapes, which is why its tasks override the bound on contact points and the bound on geometry pairs by hand (`mujoco_playground_2025`, Sec. VI). Every override printed in that paper is a locomotion port. A hand's bound would have to be set the same way, for the worst case, but no hand configuration is printed.

Figure 3 sets out the stages of one simulation step. Three of them make overlap, and they do not answer to the same knob. In an engine that enforces non-penetration at the velocity level, integration turns any residual approach velocity into overlap of order v times Δt; an engine that enforces the gap at the next configuration carries no such term, which is why Dojo's hard-contact NCP keeps its feet above the floor at every timestep it was tested at. Constraint assembly fixes the compliance a loaded contact then rests at. A truncated solver leaves a residual that grows with conditioning. Which of the three dominates in a grasp is not measured anywhere in this corpus, and the three are not ordered here. A fourth thing is not on the figure, because it is not a stage of the step and not a source of overlap: a mismatch between the geometry the solver uses and the geometry the renderer draws, which runs in both directions.

4.2 Contact models and solvers

Figure 3. One simulation step, and where engines differ

The stages every rigid-body engine runs. Under each, the choices that separate the engines used for hands. Dashed red marks the three stages that make overlap. They do not answer to the same knob.



Where penetration comes from: three sources on different knobs. Which one dominates in a hand has not been measured.

- Velocity-level enforcement. Where non-penetration is enforced on velocities, any residual approach velocity is integrated into overlap of order v·Δt. Illustration under an assumption, not a measurement: a fingertip closing at 0.5 m/s — a speed no source in this survey reports — at Isaac Gym's 1/120 s Shadow Hand step gives about 4 mm before the next step's constraint acts. Engines that enforce the gap at the next configuration carry no such term: Dojo's hard-contact NCP holds its feet above the floor at Δt = 0.1, 0.01 and 0.001 s alike.
- The prescribed compliance. A regularised contact rests at a violation equal to the normal load times the compliance that was chosen for it. It is zero at zero load, and MuJoCo's closed form is independent of the object's mass. The depth is a setting, and no engine paper in Table 4 other than ComFree-Sim states the setting it ships.
- Solver truncation. A truncated solve degrades with the conditioning of the problem and with redundancy in the contact set, not with stiffness, which is prescribed. A grasp guarantees both, being many persistent contacts on a light object.

Not a source of overlap, but a reason a number is ambiguous: an outer convex hull or decomposition contains the visual mesh, so it blocks contacts and hides overlap measured against that mesh, while an inscribed primitive admits overlap the solver never sees. PhysX 5 uses SDF collision on non-convex bodies and skips the decomposition. A penetration number must name the geometry it used.

The depth is computable from the poses and the meshes. IsaacGymEnvs' IndustReal task already gates its policy update on it at 1 mm.

Table 4 is the engine-by-engine comparison, one line per simulator, built only from what a note confirmed. Le Lidec et al. supply the taxonomy that organises it, checking each formulation against the Signorini condition, Coulomb's law, and the maximum dissipation principle. Linear complementarity, the family of Bullet, ODE, PhysX and DART, satisfies Signorini alone, because linearising the friction cone to a pyramid biases friction toward its corners. The cone complementarity problem satisfies the other two but relaxes Signorini, so contact acts at a distance of size $\Delta t \cdot \mu \cdot \|c_T\|$. The full nonlinear problem satisfies all three and is non-convex (`contact_models_comparison_2023`, Table II). A contact model is not the algorithm that solves it. PhysX linearises the cone, which places its model in the LCP family, and it does not solve an LCP. Its Temporal Gauss-Seidel scheme folds substepping into the Gauss-Seidel sweep and exposes position and velocity iteration counts separately (`isaacgym_2021`, Sec. 3).

MuJoCo is inside that taxonomy. Le Lidec et al. classify it as CCP-MuJoCo, a convex relaxation solved by a Newton method on the primal QCQP, in the same family as CCP-Drake, the SAP-style scheme (`contact_models_comparison_2023`, Table III). The relaxation makes two errors of opposite sign in one engine. A loaded contact carries a violation, and a sliding contact carries force at a positive gap. Drake's SAP inherits the same pair, and its gliding effect at distance $\varphi \approx \delta t \cdot \mu \cdot \|v_t\|$ "unfortunately does not go away as $\delta t \to 0$" (`castro_sap_contact_2021`, Sec. V-A). Le Lidec et al. call that compliance a "numerical trick designed to circumvent the issues due to hyper-staticity or ill-conditioning at the cost of impairing the simulation".

The depth a loaded contact carries is a chosen number, not a property of the engine. In a regularised formulation the steady-state violation at a contact is the normal load times the compliance: zero at zero load, growing with what the contact carries. MuJoCo drives that violation with a critically damped stabiliser whose steady-state depth, for an object resting under gravity, has a closed form independent of the object's mass (`mujoco_convex_contact_2014`, Sec. V). Whoever sets the stiffness sets the depth.

Published penetration figures do not compare with each other, and the reason is worth knowing before anyone quotes one. Dojo solves a hard-contact problem and stays above the floor on an Atlas drop test where MuJoCo overlaps by tens of millimetres (`dojo_2022`, Sec. V-A); Drake's figures for the same quantity are orders of magnitude smaller, and they are

analytical bounds for a point mass at rest rather than measured depths (`castro_sap_contact_2021`, Sec. V-B); ComFree-Sim reports a depth directly, on small convex primitives dropped under their own weight (`comfree_sim_2026`, Sec. IV-A). A point-mass bound, a humanoid drop transient and a dropped primitive differ in load, inertia and regime, so the distances between those figures measure nothing about each other. What the set does establish is the part that matters for a hand: in a compliant formulation the depth follows from a stiffness somebody chose, and the GPU era moved that knob rather than removing it. The oldest engine comparison here puts the consequence plainly, reporting that engines pushed past their stable step "effectively simulate a different physics model which can no longer hold the object". No paper in this corpus reports the depth for a fingertip loaded by a grasp, which is the one configuration a dexterous result depends on.

Table 4's most interesting column is the one that records whether an engine's own paper reports a penetration depth, and it is nearly empty. Two engines report one, and both are engines whose paper is about contact accuracy. The word "penetration" appears nowhere in the Isaac Gym paper, and Isaac Lab's contact sensor reports force, duration and an average contact point with no measure of contact quality at all. That is a documentation gap rather than a capability gap: NVIDIA's own benchmark repository already computes an interpenetration depth per environment for its assembly tasks and scales the reward down as the depth approaches a millimetre. The quantity is a few lines of code away in the engine most of this literature trains in.

The rest of Table 4 is largely empty, and the emptiness is a result: two cells in five hold a value no source stated. No engine paper states a default physics timestep, and the three that report one report it for a named experiment, so the table's timestep column carries experiment settings rather than defaults. Fewer than half the engines state a solver iteration count or ship any dexterous hand. The license column answers a question a reader choosing an engine actually has, and most engines do not answer it.

Table 4. Simulators and physics engines

engine	contact model	solver	iters	diff.	GPU	dt s	penetration exposed	throughput	hands shipped	licence
brax_2021	rigid bodies in maximal coordinates; naive/quadratic-scaling pairwise collision detection (no broad-phase acceleration structure); no convex-decomposition or mesh-vs-primitive distinction described	not LCP-based; velocity-level collision updates with Baumgarte stabilization (inspired by the Tiny Differentiable Simulator); joints modeled as spring constraints rather than Featherstone-style articulated-body methods		yes	yes		no	millions of simulation steps/sec on a single accelerator (e.g. MuJoCo Ant equivalent on a single chip); scales to hundreds of millions of steps/sec distributed across a 4x2 TPUv3 and other TPU topologies; Ant on a TPUv3 8x8 estimated at ~hundreds of millions of steps/sec vs ~thousands of steps/sec for single-threaded CPU MuJoCo Gym (all figures qualitative/order-of-magnitude, no exact numeric table)	synthetic 4-fingered claw hand (Grasp environment, non-commercial, from-scratch morphology; not confirmed present in current repo snapshot)	
comfree_sim_2026	complementarity-free, analytical (closed-form) contact resolution in the dual cone of the Coulomb friction cone via a prediction-correction/impedance-style update, extended to a unified 6D model (normal, tangential, torsional, rolling friction)	no complementarity solve; a 4-kernel GPU pipeline (Algorithm 1: smooth-velocity prediction, per-contact/per-face dual-cone solve, generalized-impulse accumulation, velocity correction) computes the impulse in closed form per contact facet	none (closed-form, no per-step iterative solve)		yes	0.002	yes	AMD 32-core CPU + NVIDIA RTX 4090 GPU: ~3x faster (near-linear vs. MJWarp's superlinear) step time vs. contact count at 512 parallel envs; ~2x parallel-environment throughput vs. MJWarp at 256-4096 envs (Allegro-hand cube-grasping benchmark); on real LEAP-hand MPC (AMD 32-core CPU + RTX 4090), MPPI compute 13.9-28.2 ms vs. MJWarp's 34.0-68.8 ms (~2.4x average speedup)		
dojo_2022	hard-contact nonlinear complementarity problem (NCP) with an exact nonlinear (second-order-cone) friction cone, avoiding pyramidal/linearized approximation, to avoid interpenetration and creep artifacts	custom primal-dual interior-point solver (Algorithm 2), based on Mehrotra's predictor-corrector algorithm, extended for non-Euclidean quaternion variables, with cone-handling borrowed from CVXOPT; gradients via implicit differentiation (implicit function theorem) of the KKT/complementarity system, with central-	converges within 15 iterations for all three robots tested in the convergence study (Sec. V-A); default tolerances r_tol = kappa_tol = 1e-5	yes	no		yes	Intel Core i9-10885H, 32GB RAM, CPU only (no GPU support; described as future work). Table V, 1000-step forward+gradient wall-clock time at dt=0.01s: Humanoid - Dojo 1.750+/-0.135s vs MuJoCo 1.512+/-0.045s vs Drake 5.463+/-0.078s vs Brax 6.975+/-0.485s; Unitree A1 - Dojo 5.235+/-0.071s vs MuJoCo		

engine	contact model	solver	iters	diff.	GPU	dt s	penetration exposed	throughput	hands shipped	licence
		path parameter kappa trading gradient smoothness for physical accuracy						1.114+/-0.007s vs Drake 3.870+/-0.024s vs Brax 11.064+/-0.521s; Franka Panda - Dojo 1.159+/-0.077s vs MuJoCo 0.335+/-0.001s vs Drake 2.352+/-0.055s vs Brax 10.954+/-0.395s; Skydio X2 - Dojo 0.807+/-0.003s vs MuJoCo 0.047+/-0.002s vs Drake 0.571+/-0.011s vs Brax 7.953+/-0.484s. MuJoCo is fastest on every system.		
genesis_2024	Constraint-based rigid solver (equality/inequality constraints, plus an explicit noslip() post-pass, architecturally similar to MuJoCo's solver+noslip design) combined with multi-material coupling: FEM, MPM, particle-based PBD and SPH, a libuipc Incremental Potential Contact solver, an explicit inter-solver coupler, and a SAP (semi-analytic-primal-style) compliant-contact path used for grasp-coupling scenes.	ConstraintSolver with Newton-style line-search iterative resolve (linesearch.py) and constraint-island decomposition for parallelism (island.py); multiple pluggable physics solvers (Rigid, FEM, MPM, PBD, SPH, libuipc, SAP) compiled via the Quadrants (Taichi-derived) backend.		yes	yes				Shadow Hand	
isaacgym_2021	Rigid-body contacts via PhysX; contact geometry can be primitive shapes or meshes loaded from URDF/MJCF (Sec. 2.2); net contact forces exposed per-rigid-body via the Tensor API (Table 1, Get-only). No explicit statement of convex-vs-mesh contact generation or per-pair contact-point count. Shadow Hand tendons simulated via PhysX Fixed Tendon mechanics:	Temporal Gauss-Seidel (TGS) solver (Sec. 3, ref. 18), not classic PGS/LCP: folds sub-stepping into a per-body accumulated velocity-delta buffer projected onto the constraint Jacobians. Table 3 exposes two user-tunable iteration knobs, Position iterations (biased, velocity+positional-error-correcting) and Velocity iterations (unbiased, velocity-error-only-correcting);		no	yes	0.0083333333333333 33	no	Shadow Hand: 150,000 parallel environment steps/sec at 16,384 environments, single NVIDIA A100 GPU (Sec. 5.3); Shadow Hand OpenAI reproduction (feed-forward): >20 consecutive successes in <1 hour on 1x A100 vs. 30 hours on OpenAI's cluster of 384x16-core CPUs (6144 cores total) + 8x NVIDIA V100 GPUs (Sec. 6.4.1).	Shadow Dexterous Hand, Allegro Hand, TriFinger (3-finger, 9-DoF manipulator; the paper itself says this is not a hand)	

engine	contact model	solver	iters	diff.	GPU	dt s	penetration exposed	throughput	hands shipped	licence
	spring+damping force proportional to deviation from rest length, propagated through a tendon-joint tree (Sec. 3, Appendix A.1).	no default numeric values for these are given in the main text.								
`isaaclab_2025`	Rigid contacts by default via PhysX 5 (SDF collisions, Featherstone articulation solver, Sec. 2.2); `Factory` assembly envs specifically use 'SDF-based contact generation, a contact reduction technique, and a Gauss-Seidel solver' (Sec. 6.4). ContactSensor exposes net normal force per body, optional filtered pairs, contact duration, average contact point, and a short history (Sec. 3.3.1); no contact-quality/penetration metric is defined (survey note).	NVIDIA PhysX 5's internal rigid-body/Featherstone-articulation solver, used as-is; Factory assembly envs specifically named as using a Gauss-Seidel solver with SDF-based contact generation and contact reduction (Sec. 6.4). No timestep, decimation, or solver-iteration values are stated anywhere in the paper for the general (dexterous) suite; the only note is qualitative, that the Digit humanoid 'requires a higher solver iteration count for stable simulation' (Fig. 14).		no	yes		no	DextrAH teacher task (state-based grasp-and-lift): over 900,000 FPS training throughput with 8 GPUs (RTX Pro 6000, the paper's only multi-GPU platform) and 16,384 environments (Sec. 4.1.1); Franka cabinet drawer task: over 1.6 million FPS at the same 8-GPU/16,384-environment setting. Single-GPU dexterous-suite numbers are not stated.	KUKA Allegro hand (first-party dexterous suite, Sec. 7.2.3), ShadowHand (third-party GraspQP grasp evaluation only, Sec. 6.4), AbilityHand (third-party GraspQP grasp evaluation only, Sec. 6.4), Fourier GR1-T2 dex hands, Unitree Inspire hand, Unitree trihand (XR teleop retargeters in code only)	
`maniskill3_2024`	PhysX (via SAPIEN) rigid-body contacts on GPU; no statement of convex-vs-mesh contact generation or contact-point count in the note. AllegroHandRightTouch reads FSR contact impulse (`get_fsr_impulse`) for touch sensing (App. VII-H, code).	PhysX (via SAPIEN), used as-is; no LCP/PGS/TGS/Newton solver-algorithm name given. Benchmark (cartpole) settings state 'Solver Position Iterations: 4 / Solver Velocity Iterations: 0' (App. XI-A), but per-task `_default_sim_config` values are not exported in the parsed source.	4 position / 0 velocity iterations (benchmarked cartpole configuration only, App. XI-A); per-task defaults not exported in the parsed source		yes			Up to 30,000+ FPS (RGBD+segmentation) on a single RTX 4090 GPU, environment count for that headline figure not stated; at 128 parallel environments with cameras, ManiSkill3 uses 3.5GB GPU memory vs. Isaac Lab's 14.1GB (Sec. III-B); PPO uses up to 4096 parallel environments for state-based training and 256-1024 for RGB training (App. IX-A), all on RTX 4090 (App. XI-A benchmark hardware).	Allegro Hand (incl. touch-sensor variant, AllegroHandRightTouch), Ability Hand, Inspire Hand, Delto 3-finger hand, TriFinger	Apache-2.0
`mujoco_2012`	soft, convex velocity-stepping contact; three interchangeable solvers replace the standard LCP-with-friction-pyramid	implicit-complementarity solver (customized non-smooth Newton method, most accurate); convex		no	no			up to ~400,000 dynamics evaluations/sec on a 12-physical-core machine (2x 6-core Intel X5860 3.33GHz,		

engine	contact model	solver	iters	diff.	GPU	dt s	penetration exposed	throughput	hands shipped	licence
	approach: implicit-complementarity, convex (kinetic-energy minimization with a soft non-penetration cost), and diagonal (mass-aware spring-damper)	solver (interior-point method used for reported timings; projected Newton/CG/Gauss-Seidel also mentioned as unverified-faster alternatives); diagonal solver (per-contact critically-damped spring-damper using the diagonal of the A matrix, fastest/least accurate)						24 threads via hyper-threading), 3D humanoid with 18 DOF and 6 active contacts (Abstract, Table 3)		
`mujoco_convex_contact_2014`	soft, convex, complementarity-free; a unified impulse vector covers joint dry friction, joint/tendon/distance limits, and frictional contacts together, with an elliptical friction cone whose dimensionality per contact is 2 (sliding), 3, or 5 (sliding+torsional+rolling)	GPGS (generalized projected Gauss-Seidel, described as the authors' own unpublished-at-the-time method handling cone and pyramid constraints) for the contact/limit impulse; a separate 'soft Gauss principle' derivation for equality/holonomic constraints; an analytically-invertible closed-form solver for a restricted circular-cone special case	5 and 50 (both tested, Fig. 3, Sec. VI-B)	no	no	0.01		single-core Intel i7-3930K (Windows 7): forward dynamics of a 27-dof humanoid with 10 contacts evaluated in 0.1 ms (100x real-time at a 10 ms timestep); inverse dynamics of the same humanoid under 100 impulses in 30 microseconds (10 microseconds with no impulses); no parallel/multi-environment throughput reported		
`mujoco_warp_2025`	MuJoCo's native solref/solimp-parameterized soft-constraint contact model ported to Warp; per-contact Jacobian construction (dense and sparse variants, with flex variants) and equality/limit/friction constraint rows; not a re-derived contact model	MuJoCo's default (implicit CCP-style) solver and Newton constraint solver (via a 'newton' flag threaded through equality/limit/friction kernels) are supported; legacy PGS solver and noslip post-processing pass are 'not yet supported' per the README		no	yes					
`newton_2025`	Solver-dependent: MuJoCo-style soft-constraint (solref/solimp-style) contact via SolverMuJoCo/MJWarp; Proximal-ADMM / Dual-Variational-Inequality contact resolution via SolverKamino; position-based-dynamics-style constraint projection	Multi-solver architecture: SolverMuJoCo (wraps MJWarp), SolverKamino (Proximal-ADMM + DVI), XPBD (rigid), VBD (cloth/deformables), Style3D (cloth), plus a separate IKSolver (LM and L-BFGS backends).			yes				Allegro Hand	

engine	contact model	solver	iters	diff.	GPU	dt s	penetration exposed	throughput	hands shipped	licence
	for the XPBD and VBD solvers.									
`orbit_2023`	PhysX SDK 5 signed-distance-field (SDF) collision checking for rigid bodies (handles non-convex geometry such as screw threads); FEM-based solver using a constraint-based stable neo-hookean material formulation for deformable bodies; PBD (position-based dynamics) solver for cloth	PhysX SDK 5's internal rigid-body and FEM solvers, used as-is; no LCP/PGS/TGS, Newton, or CG iteration counts given		no	yes		no	125,000 FPS physics-only ceiling (no env count stated); ~10x rigid-body throughput vs. CPU-vectorized frameworks (robosuite, ManiSkill2, IsaacGymEnvs) swept 256-4096 envs; ~3x deformable/cloth throughput vs. DEDO; 270 FPS aggregate for 10 cameras at 640x480; hardware: 16-core AMD Ryzen 5950X, 64GB RAM, NVIDIA RTX 3090 (Sec. V-E-b, VII)	Allegro hand	
`pybullet_2016`					yes					
`raisim_2018`										requires a valid license and activation key from the RaiSim Tech website; the repo README also states RaiSim itself is no longer supported in favor of a RaiSim2 repo
`sapien_2020`	Rigid-body contact with convex-decomposed collision meshes (PhysX 4.1); three joint systems offered (kinematic, dynamic, and PhysX articulation) trading control accuracy against speed.	PhysX 4.1's internal rigid-body solver, used as-is; no LCP/PGS/TGS name or iteration count given in the paper.			no			~5000 Hz engine, ~700 Hz OpenGL render; single-instance CPU-physics figure with no parallel environments, on a laptop with a 2.2 GHz Intel i7-8750 CPU and an Nvidia GeForce RTX 2070 GPU.		

15 rows; 69 of 165 cells (41%) are values no source stated. No cell is truncated. A value wider than its column is wrapped at a word boundary, so a cell that runs to several rendered lines is one value and not several.

4.3 GPU-parallel engines and their throughput

Isaac Gym set the pattern for the GPU-parallel turn. Physics, observations, rewards and actions stay on the GPU, and PhysX resolves contacts with the Temporal Gauss-Seidel sweep described above. Its per-task timesteps are published, which is rare. The result that reorganised the field was that reproducing OpenAI's Shadow Hand cube reorientation took under an hour on a single GPU, against 30 hours on a CPU cluster (`isaacgym_2021`, Sec. 6.4.1). Thirty-five of the papers examined run on it.

Orbit and Isaac Lab moved the stack to PhysX 5, and the dexterous offering is thinner than the predecessor's: the first-party suite is lifting, grasping and reorienting with the KUKA Allegro hand, while Shadow and Ability hands appear only through third-party grasp evaluation (`isaaclab_2025`, Sec. 7.2.3). MJX and MuJoCo Playground took the other route, putting MuJoCo's own solver on the GPU through JAX at the cost described above. MuJoCo Warp ports the same solref and solimp soft-constraint entries to NVIDIA Warp, and its README states that the legacy PGS solver and the noslip pass are unsupported and that differentiability "is not yet available" (`mujoco_warp_2025`). GPU MuJoCo is not a differentiable MuJoCo. Genesis is the broadest, pairing a rigid constraint solver with FEM, MPM, PBD and SPH solvers and a differentiable backward path, and it ships a Shadow Hand. Its README quotes no throughput number at all, so the large FPS figures that circulate for Genesis trace to nothing in `genesis_2024`.

Newton changes how these names should be read. It is explicitly multi-solver: SolverMuJoCo wraps MuJoCo Warp, SolverKamino is a proximal-ADMM and dual-variational-inequality contact solver, and XPBD and VBD do position-based constraint projection (`newton_2025`). Its contact model is solver-dependent, not a property of the engine, which is how Table 4 records it. Isaac Lab's code already exposes a `--physics newton_mjwarp` backend switch in its hands demo, and its roadmap announces Newton integration. An experiment reported as Isaac Lab may be running PhysX 5 with TGS, or MuJoCo's soft-constraint entries under Warp, and those two make different contact errors. The name also fails to fix the physics inside one engine. MuJoCo's convex solver is a family, interior point or projected Newton, conjugate gradient or Gauss-Seidel (`mujoco_2012`, Sec. II-D), and MuJoCo Warp supports neither PGS nor the noslip pass. PhysX 4 and PhysX 5 differ in whether non-convex rigid bodies get SDF collision, which is a contact-geometry difference between Isaac Gym and Isaac Lab under one vendor name (`orbit_2023`, `isaaclab_2025`). Naming a simulator no longer names its physics. Papers should report the backend and the solver beside the framework.

Reported throughput does not compare across engines, because four headline figures measure four different quantities. One engine reports parallel environment steps per second with physics, observations and rewards on device. Its successor reports frames per second in training, which includes the learning update, and on eight GPUs rather than one. A third reports a rendering-heavy figure with reward and termination removed from the timing and no environment count (`maniskill3_2024`). A fourth reports a physics-only ceiling, also without an environment count (`orbit_2023`). Table 4 records each figure with the quantity it measures, which is the only form in which any of them is usable.

The one number reported cleanly enough to reuse is MuJoCo Playground's, in training steps per second on one GPU over five seeds. A hand reorientation task runs about six times slower there than a parallel-jaw pushing task on the same hardware in the same codebase. That factor is not a property of the task alone: Playground tunes solver iterations, line-search iterations, timestep and contact bounds per environment, and does not print the two configurations side by side, which is what makes even those two figures incomparable. A training-loop rate is also mostly not a measurement of physics. On Playground's own breakdown the policy update dominates the frame, with physics a few percent of it.

Cross-framework comparisons add further free parameters. One widely cited table takes the best result over a range of environment counts for each system (`maniskill2_2023`, Table 1a), which tunes a free parameter per competitor; and the two systems being compared are not the same kind of thing, one running rigid-body physics on CPU workers behind a shared render server and the other on the GPU. Because the measured loop includes rendering at a fixed control rate, what is compared is a visual sample-collection loop rather than two physics engines. Its authors say so themselves, and Playground is

equally explicit that its own cross-simulator plot borrows two of its numbers from another paper. Several sources give no figure at all, pointing instead at an external dashboard or omitting throughput from their documentation entirely.

The failure reaches past the engine papers into the methods. Of the 47 corpus method papers that name a GPU-batched simulator, 19 state no environment count anywhere, and Isaac Lab's own paper states no physics timestep, no decimation and no solver iteration count for any task. A throughput figure is interpretable only with the environment count, the GPU, the timesteps, the solver iteration budget, and a statement of whether rendering and the learning update sit inside the measurement. Almost nobody reports all five.

4.4 Tactile simulation and the sim-to-real gap

The three tactile simulators in this corpus calibrate against three different things, and none of them is a manipulation outcome. TACTO is a rendering layer over a host engine, by default PyBullet's rigid contact model. It reads post-solve link poses and the engine's reported normal force and maps that force to gel-mesh deformation at the rendering level, so it contributes no contact physics of its own. Its only sim-to-real number is a tactile pose-estimation task, where the simulated-data model is about twice the error of one trained on a small set of real datapoints (`tacto_2020`, Table II).

Taxim is example-based rather than simulated, calibrated from real indentations, and it beats TACTO on every optical-similarity metric against real images. Its marker-motion model is calibrated against a finite-element reference, and it tracks that reference about three times more tightly than the reference tracks reality, so its residual is bounded below by its own calibration target. The authors also state that they simulate quasi-static contact only, with slip left as future work (`taxim_2021`, Sec. V). Slip is what a hand needs.

Tactile Genesis inverts the thread. It implements two penetration-depth backends on the collision geometry, an analytic SDF query and a BVH raycast, and uses the depth as the sensor signal rather than as an error, so the quantity every other engine treats as a defect sets this one's operating point.

Most of what is written about this gap is attribution without measurement. PenSpin asserts that the pure physics gap "cannot be bridged by extensive domain randomization alone" while reporting no experiment that isolates it (`penspin_2024`). MuJoCo Playground attributes its LEAP hand failures to physical flex in low-cost hardware and names more accurate collision geometries as the fix, without measuring either. DeXtreme lists four candidate causes for its shortfall, including a malfunctioning Allegro thumb used in most trials, and disambiguates none by experiment (`dextreme_2022`).

What has been measured is narrower, and section 7 collects the transfer numbers. The one result that links a quantified model error to transfer ranks system-identification runs by dynamics-model error and pairs each with real success: the best-identified model grasps 8 of 10 attempts, the worst 0 of 10 (`humanoid_sim2real_recipe_2025`, Table 1). Where a cause has been pinned down elsewhere it is usually perception or actuation rather than contact, with pose estimators and camera trackers named as the unmodelled source in the real numbers (`pddm_2019`, App. C).

The one paper that looks at the contact side is usually read backwards. DeXtreme's real-to-sim replay interpenetrated because the replayed poses carried the pose estimator's error, not because the contact model failed, and the paper says so: it could not calibrate physics against the real system because the state it replayed was already wrong. The commonly cited reading, that the contact model was at fault, is not what the sentence says.

5. Training a dexterous policy

5.1 Sources of supervision

A dexterous policy is defined by what supervises it, and four sources are in use across the corpus's 113 papers: a reward function supervises reinforcement learning, a human demonstration supervises imitation, a human reference trajectory supervises a physics-based tracker, which is imitation with a simulator in the loop, and nothing supervises a model-based planner, which is handed a cost and a model instead.

The labels do not partition the corpus. Reinforcement learning is the largest at 54 of the 113 papers, behaviour cloning next, distillation after that, and ten further tags cover the rest. They sum to far more than the corpus, because most methods published since 2024 sit on two branches at once: a reinforcement-learning teacher and a behaviour-cloned student are one paper, not two. Figure 4 draws the tree and the cross-links, and Table 7 is the paper-by-paper version of the same thing.

What the deployed policy looks like once training is done matters more than the name a paper gives itself, and on that axis the field has converged hard. Section 5.5 shows why.

Table 7's emptiest columns are the ones a reader most needs. Only 70 of the 113 papers state how many real trials are behind their headline number, and fewer still state an environment count or how many unseen objects were tested; section 7.1 takes up what that means for comparison. A mostly empty line is not a weak method, but it is one that cannot be compared with any other here.

Table 7. Methods

method	yr	task	paradigm	algorithm	hand	DoF	bi	sim	real	trials	unseen obj	penetration	code
ferrari_canny_1992	1992												no
dapg_2017	2017	grasp, reorient, functional/tool, other	RL+demo	NPG (Natural Policy Gradient) + BC pretraining (DAPG)	ADROIT	24	no	MuJoCo	no			not addressed	yes
openai_dexterity_2018	2018	reorient	RL	PPO	ShadowRobot Dexterous Hand (EDC electric-motor version)	24	no	MuJoCo	yes	10		not addressed	no
openai_rubiks_cube_2019	2019	reorient	RL, distillation	PPO with Automatic Domain Randomization (ADR); Policy Cloning (DAGGER/distillation-style) for architecture changes	Shadow Dexterous E Series Hand (E3M5R)	20	no	MuJoCo (+ ORRB/Unity3D for rendered vision training images)	yes	10		not addressed	no
pddm_2019	2019	reorient, other	MPC	PDDM: ensemble of feed-forward dynamics models + MPPI-style filtering/reward-weighted-refinement trajectory optimizer (no policy network, no policy-gradient training)	Shadow Hand (24-DoF, real and sim); D'Claw (9-DoF, sim only, valve turning)		no	MuJoCo	yes			not addressed	yes
dexpilot_2020	2020	reorient, grasp, functional/tool, other	teleop-system	DART model-based hand tracker bootstrapped by two learned neural priors (GloveNet ResNet-50 keypoint regressor; PointNet++-based keypoint network + JointNet FC mapping), online fingertip-vector kinematic retargeting solved via SLSQP (NLopt) warm-started each frame, Riemannian	Wonik Robotics Allegro hand (16-DoF, 4x4-joint fingers), retrofitted with 4 SynTouch BioTac tactile sensors at the fingertips	16	no		yes	150		not addressed	no

method	yr	task	paradigm	algorithm	hand	DoF	bi	sim	real	trials	unseen obj	penetration	code
				Motion Policies for arm motion generation									
castro_sap_contact_2021	2021	bimanual-coord		SAP (Semi-Analytic Primal) solver: a custom Newton-type solver on an unconstrained convex primal reformulation of compliant contact, with analytic gradients, line search, and sparsity exploitation	Allegro hand (16 DoF each), two	16	yes	Drake	no			measured	yes
dexmv_2021	2021	grasp, functional/tool	RL+demo, data-collection	DAPG / SOIL / GAIL+ on top of TRPO, using demonstrations translated from human video via TSV retargeting + inverse-dynamics action estimation	Adroit Hand	30	no	MuJoCo	no		500	not addressed	yes
dexpoint_2022	2022	grasp, other	RL	PPO	Allegro Hand	16	no	SAPIEN	yes	260	40	not addressed	yes
dextreme_2022	2022	reorient	RL	PPO (rl-games implementation, LSTM actor/critic, asymmetric/state-privileged critic)	Allegro Hand	16	no	Isaac Gym (PhysX)	yes	10		not addressed	yes
dexvip_2022	2022	grasp	RL	PPO with a video-mined consensus grasp-pose auxiliary reward (R_pose) combined with an affordance reward (R_aff) and a lift-success reward (R_succ)	Adroit Hand	30	no	MuJoCo	no			not addressed	no
dgrasp_2022	2022	grasp	RL	PPO (own implementation)	MANO hand model	51	no	RaiSim	no		3	measured	yes
dime_2022	2022	reorient, grasp	teleop-system, BC, RL+demo	INN/VINN (nearest-neighbor imitation) vs. BC; DAPG/PPO/BCRL (simulation RL-finetuning)	Allegro Hand		no	MuJoCo	yes	10		not addressed	yes

method	yr	task	paradigm	algorithm	hand	DoF	bi	sim	real	trials	unseen obj	penetration	code
holo_dex_2022	2022	reorient, grasp, functional/tool	teleop-system, BC	VINN (BYOL nearest-neighbor) vs. Behavior Cloning / BC-Rep	Allegro Hand	16	no		yes	10	10	not addressed	yes
hora_2022	2022	reorient	RL, distillation	PPO	Allegro Hand (Wonik Robotics)	16	no	IsaacGym	yes	240	30	not addressed	yes
mjpc_2022	2022	reorient	MPC	Predictive Sampling: zero-order, derivative-free sampling-based shooting MPC over a spline-parameterised control sequence; the framework (MJPC) also implements iLQG and Gradient Descent as alternative planners, but the Hand result is attributed to Predictive Sampling	Shadow Hand (MuJoCo Menagerie model)		no	MuJoCo	no			not addressed	yes
pang_global_planning_2022	2022	reorient, grasp, bimanual-coord, other	trajopt	iMPC (iLQR-style trajectory optimizer over a locally-smoothed convex quasi-dynamic contact model, CQDC) for local case studies; an RRT variant (RRT-through-contact) using a smoothed Mahalanobis distance metric, single-step dynamically-consistent extension, and contact sampling for global planning on harder tasks	Allegro Hand (full 3D model, in-hand rotation / pen placement / plate pickup / door opening); a 2-DoF-per-finger planar hand (2 fingers) for 2D reorientation tasks		yes	Custom Convex Quasi-Dynamic Differentiable Contact (CQDC) model implemented in Drake (MultibodyPlant/SceneGraph), validated against Drake's own high-fidelity second-order contact solver	yes			constrained	yes
videodex_2022	2022	grasp, reorient, functional/tool	BC, data-collection	two-stream Neural Dynamic Policy (NDP) trajectory regression (L1 loss): pretrained on human-video-	LEAP Hand	16	no		yes			not addressed	no

method	yr	task	paradigm	algorithm	hand	DoF	bi	sim	real	trials	unseen obj	penetration	code
				retargeted trajectories, fine-tuned on real teleoperated demonstrations; R3M visual encoder									
visual_dexterity_2022	2022	reorient	RL, distillation	PPO (teacher, privileged state) + DAgger imitation distillation to a depth-point-cloud student (two-stage: synthetic-PC pretrain then rendered-occluded-PC fine-tune)	D'Claw (open-source, low-cost; 3-finger and modified 4-finger versions)		no	Isaac Gym (PhysX)	yes	20	12	not addressed	yes
aloha_act_2023	2023	bimanual-coord	BC	ACT (Action Chunking with Transformers, CVAE)	parallel-jaw gripper (not dexterous), two		yes	MuJoCo	yes	25		not addressed	yes
anyteleop_2023	2023	grasp, functional/tool, handover	teleop-system, RL+demo	DAPG (downstream IL); keypoint-vector retargeting optimization (dex_retargeting)	Allegro Hand		no	SAPIEN	yes	100		not addressed	yes
artigrasp_2023	2023	grasp, functional/tool, bimanual-coord	RL	PPO (D-Grasp-based implementation)	MANO hand model, two	51	yes	RaiSim	no		1	not addressed	yes
dexdeform_2023	2023	bimanual-coord, other	BC, trajopt	latent skill model (VAE over demonstrations) for skill-planned trajectory initialization, refined by gradient-based trajectory optimization through a differentiable MPM simulator, with refined rollouts fed back as new demonstrations	Shadow Dexterous Hand (simulated)	28	yes	PlasticineLab (MLS-MPM, CUDA), built on Hu et al. 2018's Moving Least Squares Material Point Method	no			not addressed	yes
dexpbt_2023	2023	grasp, reorient, bimanual-coord	RL	PPO (rl_games) with decentralized Population-Based Training (PBT) over RL hyperparameters and reward-	Allegro Hand	16	yes	Isaac Gym (PhysX)	no			not addressed	yes

method	yr	task	paradigm	algorithm	hand	DoF	bi	sim	real	trials	unseen obj	penetration	code
				shaping coefficients									
dexterous_functional_grasping_2023	2023	functional/tool	RL	PPO	LEAP hand	16	no	IsaacGym	yes	70	5		no
diffusion_policy_2023	2023	grasp, functional/tool, bimanual-coord, other	BC, diffusion	conditional DDPM over action sequences (CNN or transformer noise-prediction network), DDIM at inference			yes	task-specific: Robomimic (MuJoCo-backed), Push-T (custom 2D physics per IBC), BlockPush (per BET), Franka Kitchen (per Relay Policy Learning); no single engine/timestep stated for the combined benchmark	yes	20		not addressed	yes
dynamic_handover_2023	2023	handover, bimanual-coord	RL	MAPPO	Allegro Hand	16	yes	IsaacGym	yes	15	14	not addressed	no
eureka_2023	2023	reorient, bimanual-coord, other	RL	PPO with LLM-authored (GPT-4) evolutionary reward search	Shadow Hand (also Allegro Hand; bimanual pairs of Shadow Hands for Dexterity tasks)	20	yes	IsaacGym	yes			not addressed	yes
pgdm_2023	2023	grasp, functional/tool	RL	PPO (Stable-Baselines3) with an object-trajectory-tracking reward plus pre-grasp-based exploration initialization (PGDM)	ShadowHand	24	no	MuJoCo	yes			not addressed	no
physhoi_2023	2023	track-human-ref	RL	PPO	SMPL-X humanoid (simulated, not a robot hand)	90	no	Isaac Gym	no			not addressed	yes
robot_synesthesia_2023	2023	reorient	RL, distillation	PPO	Allegro Hand	16	no	Isaac Gym	yes	5		not addressed	no
rotateit_2023	2023	reorient	RL, distillation	PPO	AllegroHand (Wonik Robotics)	16	no	IsaacGym	yes		15	not addressed	
rotating_without_seeing_2023	2023	reorient	RL	PPO	Allegro Hand	16	no	IsaacGym	yes	30	5	not addressed	
unidexgrasp_2023	2023	grasp	grasp-synthesis, RL, distillation	PPO (teacher) + DAgger (student distillation)	ShadowHand	26	no	Isaac Gym	no		241	penalised	yes

method	yr	task	paradigm	algorithm	hand	DoF	bi	sim	real	trials	unseen obj	penetration	code
unidexgrasp_pp_2023	2023	grasp	RL, distillation	PPO + DAgger (with critic distillation)	Shadow Hand	24	no	Isaac Gym	no			not addressed	yes
ace_teleop_2024	2024	grasp, functional/tool	teleop-system, BC	3D Diffusion Policy (DP3) for xArm platforms; ACT for H1 humanoid (downstream IL); IK-based scale/recenter end-effector mapping + AnyTeleop fingertip-vector hand retargeting	Ability Hand / Inspire Hand / parallel-jaw gripper (embodiment-dependent)		yes		yes	10	18	not addressed	no
anyrotate_2024	2024	reorient	RL, distillation	PPO	Allegro Hand	16	no	IsaacGym	yes		10	not addressed	no
asymdex_2024	2024	bimanual-coord, grasp	RL	PPO	Shadow Hand (30-DoF: 24-DoF hand + 6-DoF floating wrist) in simulation; 16-DoF Allegro hand + 6-DoF Ability Hand (mismatched, hardware-availability artifact) in real-world experiments	30	yes	NVIDIA Isaac Gym	yes	20		not addressed	yes
bidex_teleop_2024	2024	bimanual-coord, handover, functional/tool	teleop-system, BC	ACT (Action Chunking Transformer)	LEAP Hand (16 DoF); LEAP Hand V2 (21 DoF) used for 'extreme dexterity' experiments	16	yes	none	yes	20		not addressed	no
bidexhd_2024	2024	functional/tool, bimanual-coord	RL, distillation	IPPO (teacher) + DAgger (student distillation)	LEAP Hand, left and right	16	yes	Isaac Gym (PhysX)	no			not addressed	yes
bimangrasp_2024	2024	grasp, bimanual-coord	grasp-synthesis, diffusion	MALA (Metropolis-adjusted Langevin) energy optimisation + DDPM (BimanGrasp-DDPM)	Shadow Hand, pair	22	yes	Isaac Gym	no		225	penalised	no
bunny_visionpro_2024	2024	grasp, functional/tool, bimanual-coord	teleop-system, BC	ACT, Diffusion Policy, DP3 (downstream IL); SQP fingertip-vector hand retargeting +	Ability Hand	24	yes		yes	10		not addressed	yes

method	yr	task	paradigm	algorithm	hand	DoF	bi	sim	real	trials	unseen obj	penetration	code
				unified IK/singularity/collision arm-motion optimization									
cyberdemo_2024	2024	grasp, reorient, functional/tool	BC, teleop-system, data-collection	ACT (Action Chunking with Transformers) BC policy, trained on sim teleoperation demonstrations augmented via Sensitivity-Aware Kinematics Augmentation, automatic curriculum learning, and action aggregation, fine-tuned on real demonstrations	Allegro Hand	16	no	SAPIEN	yes	20	2	not addressed	no
demostart_2024	2024	grasp, functional/tool, reorient, other	RL, distillation	MPO (teacher, auto-curriculum) + BC distillation (student, Perceiver-Actor-Critic)	DEX-EE Hand	12	no	MuJoCo	yes	100		not addressed	no
dexcap_2024	2024	grasp, bimanual-coord, functional/tool	BC, diffusion, data-collection	Diffusion Policy with Perceiver point-cloud encoder (DP-perc)	LEAP Hand	16	yes	none	yes	60	9		yes
dexmimicgen_2024	2024	bimanual-coord, grasp, functional/tool	data-collection, BC	MimicGen-style per-arm subtask transform/replay generation; downstream BC-RNN / BC-RNN-GMM / Diffusion Policy	Inspire dexterous hand (6-DoF, real-world GR1); unnamed 'dexterous hands' in the two simulated dexterous-hand embodiments	6	yes	RoboSuite (MuJoCo)	yes	20			yes
dp3_2024	2024	reorient, grasp, functional/tool, bimanual-coord, other	BC, diffusion	Diffusion Policy backbone (DDPM training / DDIM inference) conditioned on a compact 3D point-cloud feature (small MLP + max-pool encoder) instead of image features	Shadow Hand (Adroit, Bi-DexHands, DexDeform, DexMV domains); Allegro Hand (DexArt, HORA domains; also real Roll-Up/Dumpling/Drill); parallel gripper (MetaWorld; real Pour)		yes	MuJoCo (Adroit, MetaWorld), IsaacGym (Bi-DexHands, HORA), Sapien (DexArt, DexMV), PlasticineLab (DexDeform)	yes	40	5	not addressed	yes

method	yr	task	paradigm	algorithm	hand	DoF	bi	sim	real	trials	unseen obj	penetration	code
dreureka_2024	2024	reorient	RL	PPO with LLM-authored (GPT-4) reward search + reward-aware physics prior domain randomization	LEAP hand	16	no	Isaac Gym	yes			not addressed	yes
egomimic_2024	2024	grasp, functional/tool, bimanual-coord	BC	shared-backbone ACT variant co-trained on paired egocentric human video and teleoperated bimanual robot demonstrations, with dual pose-space and joint-space action heads	parallel-jaw gripper (not a dexterous hand; see note scope flag)	1	yes		yes	135		not addressed	yes
graspxl_2024	2024	grasp	RL	PPO	MANO (also Shadow Hand, Allegro Hand, Faive Hand)	45	no	RaiSim	no		503409	not addressed	yes
hato_visuotactile_2024	2024	handover, bimanual-coord	diffusion	Diffusion Policy (DDPM, CNN-based)	Psyonic Ability Hand, two	6	yes		yes	10		not addressed	yes
hudor_2024	2024	grasp, functional/tool	RL	open-loop IK replay of a single retargeted human fingertip trajectory plus online residual RL (DrQv2) trained with an object-centric point-tracking trajectory-matching reward	Allegro Hand	16	no		yes	10		not addressed	no
humanplus_2024	2024	track-human-ref, locomanipulation	RL, BC	PPO (HST shadowing policy) + decoder-only Transformer BC (HIT)	Inspire-Robots RH56DFX, two	6	yes	Isaac Gym-based legged_gym/rsl_rl	yes	10		not addressed	yes
objdex_2024	2024	track-human-ref, bimanual-coord	BC, RL, distillation	BC (Transformer high-level planner) + PPO (low-level controller)	Shadow Hand		yes		yes	80	1	not addressed	no
okami_2024	2024	grasp, functional/tool, bimanual-coord	trajopt, BC	single-video imitation: SLAHMR-extended SMPL-H body+hand reconstruction, GPT-	Inspire Hand (x2)	6	yes	RoboSuite (2 of 6 tasks only)	yes	12		not addressed	yes

method	yr	task	paradigm	algorithm	hand	DoF	bi	sim	real	trials	unseen obj	penetration	code
				4V/Grounded-SAM/Cutie object-centric reference-plan construction, factorized arm IK (Pink) + dex-retargeting hand-joint mapping, SE(3) trajectory warping to new object poses at test time; optional downstream ACT BC policy trained on OKAMI-generated rollouts									
omnigrasp_2024	2024	grasp, track-human-ref	RL, distillation	PPO (hierarchical: PHC-X finger imitator → PULSE-X distilled latent prior → PPO over latent action space)	SMPL-X whole-body humanoid (simulated, not a robot hand)	90	yes	Isaac Gym	no		5	not addressed	yes
omnih2o_2024	2024	track-human-ref, locomanipulation	RL, distillation	PPO (teacher) + DAgger (student) + diffusion policy (autonomy layer)	Inspire hands (open-loop, DoF not stated)		yes		yes	20			yes
open_television_2024	2024	bimanual-coord, functional/tool	teleop-system, BC	ACT (Action Chunking Transformer) with DinoV2 ViT backbone	Inspire Robots hand (6 actuated DoF, 12 total); Fourier GR-1 embodiment instead uses a 1-DoF parallel-jaw gripper	6	yes	none	yes	20		not addressed	yes
openvla_2024	2024	other	VLA	autoregressive next-token prediction over discretized (256-bin per-dimension, quantile-based) action tokens, single-step (no action chunking)			no	LIBERO benchmark (simulated Franka) for fine-tuning experiments; engine/version not stated	yes	230		not addressed	yes
penspin_2024	2024	reorient	RL, distillation	PPO + BC	Allegro Hand	16	no	Isaac Gym	yes	50	7	not addressed	yes
pi0_2024	2024	other	VLA, flow	conditional flow matching (linear-Gaussian/optimal-transport probability path), forward-Euler integration at			yes		yes	10		not addressed	yes

method	yr	task	paradigm	algorithm	hand	DoF	bi	sim	real	trials	unseen obj	penetration	code
				inference (10 steps), shifted-Beta noisy-timestep sampling									
pianomime_2024	2024	music	RL+demo, distillation	PPO (per-song specialists); DDPM diffusion (generalist)	Shadow Hand E3M5	23	yes	ROBOPIANIST (MuJoCo)	no			not addressed	yes
rdt1b_2024	2024	bimanual-coord	diffusion	RDT (DiT + cross-attention, QKNorm/RMSNorm, MLP decoder, Alternating Condition Injection)	parallel-jaw gripper (ALOHA-style, Cobot Mobile ALOHA), two		yes		yes	139	2	not addressed	yes
resdex_2024	2024	grasp	RL, distillation	PPO + DAgger	ShadowHand	18	no	IsaacGym	no		241	not addressed	yes
twisting_lids_2024	2024	functional/tool, bimanual-coord	RL	PPO (asymmetric actor-critic)	Allegro Hand, two	16	yes	Isaac Gym	yes	20	15	not addressed	no
umi_2024	2024	functional/tool, bimanual-coord, other	BC, diffusion, data-collection, teleop-system	Diffusion Policy (unmodified loss/decoder); contribution is the hand-held data-collection hardware and the observation/action interface (relative-trajectory representation, inference-time latency matching)			yes		yes	260			yes
articulated_tools_inhand_2025	2025	functional/tool	RL, BC, distillation	PPO (oracle) + BC (student + CATFA online adaptation)	Inspire Hand	6	no	IsaacLab	yes	50		not addressed	no
being_h0_2025	2025	grasp, functional/tool	VLA, BC	autoregressive next-token prediction over discretized MANO motion tokens (Grouped Residual Quantization) for pretraining; L1-loss MLP regression head over learnable query tokens for post-training imitation learning	6-DoF Inspire hand mounted on a 7-DoF Franka Research 3 arm (real robot); MANO parametric hand model used as the pretraining representation (no physical hardware)	6	no		yes	20		not addressed	yes

method	yr	task	paradigm	algorithm	hand	DoF	bi	sim	real	trials	unseen obj	penetration	code
clutterdexgrasp_2025	2025	grasp	RL, diffusion, distillation	PPO (teacher) + DP3 diffusion policy (student)	AgiBot dexterous hand	6	no	Isaac Gym	yes	167	2029	constrained	no
cross_embodiment_world_models_2025	2025	other	world-model, MPC	DPI-Net (graph neural network particle-dynamics world model) + sampling-based receding-horizon MPC	PSYONIC Ability Hand (6-DoF) and Robot Era XHand (12-DoF) for real-world deployment; trained across 6 simulated hands (Ability, Allegro, XHand, Leap, Shadow, forearm-less Shadow variant) for the embodiment-scaling study		no	SAPIEN (rigid-body Object Pushing data) and Rewarped (differentiable multiphysics, deformable Plasticine Reshaping data)	yes	20		not addressed	no
dexgraspvla_2025	2025	grasp, other	VLA, diffusion	hierarchical VLA: frozen VLM high-level planner (bounding-box affordance) + DiT diffusion-policy controller conditioned on frozen DINOv2 features and a SAM+Cutie object mask, trained end-to-end from scratch by imitation learning (MSE noise-prediction loss); Immiscible Diffusion noise assignment; DDIM sampling (16 of 50 training steps)	PsiBot G0-R, 6-DoF, single (right) hand	6	no		yes	1287	360	not addressed	yes
dexmachina_2025	2025	track-human-ref, bimanual-coord	RL	PPO (rl-games)	six URDFs: Inspire, Allegro, XHand, Schunk (main), Ability, DexRobot Dex Hand		yes	Genesis	no			constrained	yes
dexman_2025	2025	track-human-ref, bimanual-coord	RL, trajopt	PPO (RL_GAMES) residual policy on top of IK-retargeted human motion	Shadow Dexterous Hand		yes	NVIDIA Isaac Gym	no				no
dexndm_2025	2025	reorient	RL, distillation	PPO (oracle) + BC (generalist) + supervised residual policy	LEAP hand		no	Isaac Gym	yes			not addressed	no

method	yr	task	paradigm	algorithm	hand	DoF	bi	sim	real	trials	unseen obj	penetration	code
dexplore_2025	2025	track-human-ref	RL, distillation	PPO (teacher) + DAgger-style VAE distillation (student)	Inspire hand (also Allegro hand)	6	no	Isaac Gym	yes		2	not addressed	yes
dexremoe_2025	2025	reorient	RL	PPO	GX11 three-fingered dexterous hand (custom)	11	no	IsaacGym	no		50	not addressed	no
dexteritygen_2025	2025	reorient, functional/tool, grasp	RL, diffusion	RL-generated Anygrasp-to-Anygrasp transitions used to train a UNet DDPM diffusion foundation controller (DexGen) with gradient-guided diffusion sampling that projects an external (teleoperated) command onto a safe, high-likelihood learned action	Allegro Hand	16	no		yes	240			no
dexterous_handover_2025	2025	handover, grasp	RL	PPO	Allegro Hand		no	IsaacLab	no		3	not addressed	
dextrack_2025	2025	track-human-ref	RL+demo	PPO + IL action-supervision loss	Allegro hand (sim); LEAP hand (real)	16	no	Isaac Gym	yes			measured	yes
dexumi_2025	2025	grasp, functional/tool	BC, data-collection	diffusion policy (DDPM-style action-chunking) trained on exoskeleton-collected demonstrations converted to robot-hand-looking video via a SAM2 segment/inpaint/composite pipeline	Inspire Hand (12 DoF, 6 active) and XHand (12 active DoF)		no		yes	20			yes
dexvla_2025	2025	grasp, functional/tool, bimanual-coord	VLA, diffusion	ScaleDP (Scale Diffusion Policy, transformer-based diffusion action head, up to 1B parameters, multi-head per embodiment) plugged into a Qwen2-VL-2B	Robotiq parallel-jaw gripper (Franka rig a, Bimanual UR5e rig c); Inspire multi-fingered dexterous hand, 6-DoF hand joint space (Franka rig b); AgileX bimanual arm			LIBERO (secondary, App. A.2 only; main results are real-robot)	yes	10	30	not addressed	yes

method	yr	task	paradigm	algorithm	hand	DoF	bi	sim	real	trials	unseen obj	penetration	code
				VLA backbone, trained with a 3-stage 'Embodied Curriculum Learning' recipe; loss L = Ldiff + αLntp (α=1)	end-effector unnamed (rig d)								
dexwild_2025	2025	grasp, functional/tool, bimanual-coord	BC, data-collection	diffusion U-Net policy (also compared against ACT) co-trained on human wearable-rig demonstrations and teleoperated robot demonstrations via fixed-ratio batch sampling	LEAP Hand / LEAP Hand V2 Advanced	17	yes		yes		11		yes
dydexhandover_2025	2025	handover, bimanual-coord	RL	MAPPO (human-regularized, CTDE, with hybrid advantage estimation)	11-DoF (6 actuated) dexterous hand, per side (vendor not stated)	11	yes	NVIDIA Isaac Sim / Isaac Lab	no		9	not addressed	no
egozero_2025	2025	grasp, functional/tool	BC	closed-loop Transformer policy (BC, Gaussian NLL loss) over a morphology-agnostic 3D-point state-action space (triangulated object keypoints + thumb/index 3D coordinates + a thresholded grasp scalar)	Franka Panda parallel-jaw gripper (not a dexterous hand; see note scope flag)	1	no		yes	15		not addressed	yes
gemini_robotics_15_2025	2025	other	VLA		ALOHA parallel gripper (by platform convention, not re-specified); Bi-arm Franka parallel gripper; Apollo humanoid multi-finger dexterous hand (grasp types shown: single-finger push, cylindrical side grasp, medium wrap, five-finger power grasp, bimanual grasp/rotate; no		yes	MuJoCo (used to generate evaluation scenes at scale, not for training; engine timestep/contact model not stated)	yes			not addressed	no

method	yr	task	paradigm	algorithm	hand	DoF	bi	sim	real	trials	unseen obj	penetration	code
					DoF or vendor model given for any embodiment)								
gemini_robotics_2025	2025	other, handover	VLA, distillation		ALOHA 2 parallel gripper, two fingers (primary embodiment); bi-arm Franka parallel gripper; Apollo humanoid five-fingered dexterous hand (no model/DoF given, evaluated only qualitatively)		yes		yes	20		not addressed	no
geometric_retargeting_2025	2025	grasp	teleop-system	per-finger MLP retargeting network (GeoRT) trained offline against five geometric losses: motion-direction preservation, C-space coverage (Chamfer), flatness, pinch correspondence, and a learned self-collision classifier	Allegro Hand		no		yes			not addressed	yes
gr_dexter_2025	2025	bimanual-coord, grasp	VLA	GR-Dexter (Mixture-of-Transformer VLA, flow matching + next-token prediction, following GR-3)	ByteDexter V2, two	21	yes		yes	125	23	not addressed	no
groot_n16_2025	2025	grasp, functional/tool, bimanual-coord, locomanipulation	VLA	flow-matching DiT action head (32 layers, per page and cross-confirmed by code changelog) fed by a Cosmos-2B VLM backbone (per page); predicts state-relative action chunks for most embodiments; no algorithm name beyond "DiT" is given			yes	Galaxea R1 Pro simulated on the BEHAVIOR suite; physics engine, timestep, and env count not stated	yes			not addressed	yes
groot_n1_2025	2025	grasp, functional/tool,	VLA, flow	dual-system architecture:	Fourier dexterous hands,		yes	RoboCasa (Kitchen and	yes	10	5	not addressed	yes

method	yr	task	paradigm	algorithm	hand	DoF	bi	sim	real	trials	unseen obj	penetration	code
		bimanual-coord, handover		Eagle-2 VLM (System 2, reasoning) feeding a DiT flow-matching action head (System 1, acting) via alternating cross-/self-attention; K=4-step forward-Euler denoising at inference; latent-action VQ-VAE for labeling action-less video	on the Fourier GR-1 humanoid; DoF/finger count not stated			GR-1 Tabletop benchmarks); DexMimicGen (cross-embodiment suite and sim pretraining-data generation); underlying physics engine and timestep not stated					
h_rdt_2025	2025	grasp, functional/tool, bimanual-coord	flow	2B-parameter flow-matching diffusion transformer (H-RDT), pretrained on EgoDex human hand-pose trajectories then fine-tuned per robot embodiment by reinitializing the state/action adaptors and action decoder	parallel-jaw / 2-jaw grippers at deployment (Aloha-Agilex, ARX5, Franka-Panda, UR5+UMI); no dexterous hand deployed, though pretraining action space includes bimanual fingertip positions from EgoDex		yes	RoboTwin 2.0	yes	25		not addressed	no
helix_2025	2025	grasp, functional/tool, handover	VLA	dual-network "System 1, System 2" VLA: S2 (7B VLM, 7-9Hz) produces a continuous latent conditioning vector consumed by S1 (80M-param cross-attention encoder-decoder transformer, 200Hz); trained end-to-end with a "standard regression loss" (no diffusion/flow-matching, no formula given)			yes		yes			not addressed	no
human2sim2robot_2025	2025	track-human-ref	RL	PPO	Allegro Hand	16	no	IsaacGym	yes	70		not addressed	yes
humanoid_policy_human_policy_2025	2025	grasp, functional/tool	BC	Human Action Transformer (HAT), an ACT-style action-	Inspire Hand (x2, 5-fingered)	6	yes		yes	230		not addressed	yes

method	yr	task	paradigm	algorithm	hand	DoF	bi	sim	real	trials	unseen obj	penetration	code
				chunking transformer co-trained jointly on unified human (PH2D) and teleoperated-robot demonstrations in a shared 54-D wrist-pose + fingertip state-action space									
humanoid_sim2real_recipe_2025	2025	grasp, bimanual-coord, handover	RL, distillation	PPO (asymmetric actor-critic) for per-task specialist policies, distilled into a generalist via Diffusion Policy trained on filtered successful rollouts	Fourier GR1 hand (6 actuated + 5 underactuated DoF); cross-embodiment check also uses Inspire hand (6 actuated + 6 underactuated DoF)	6	yes	NVIDIA Isaac Gym	yes	10			no
maniptrans_2025	2025	track-human-ref, bimanual-coord	RL, BC, diffusion	PPO (two-stage: frozen generalist imitator + per-task residual policy)	Inspire Hand (12-DoF sim); also Shadow(22), MANO(22), Allegro(16)	12	yes	Isaac Gym	yes			not addressed	yes
metis_2025	2025	grasp, functional/tool, bimanual-coord	VLA	Autoregressive next-token cross-entropy over discretized 'motion-aware dynamics' tokens (VQ-VAE visual dynamics + RQ-VAE motion dynamics), plus a supervised continuous-action regression head (Action Decoder); combined loss L = Lar + lambda*Laction	Inspire dexterous hand (paired, main real-robot eval); cross-embodiment eval uses 22-DoF SharpaWave Dexterous Hand	6	yes		yes	20		not addressed	no
pi05_2025	2025	other	VLA, flow	hierarchical high-level discrete-subtask prediction (FAST tokenizer, autoregressive) + low-level flow-matching action expert; two-stage schedule pretrain (alpha=0, 280k			yes		yes	40		not addressed	yes

method	yr	task	paradigm	algorithm	hand	DoF	bi	sim	real	trials	unseen obj	penetration	code
				steps) then post-train (alpha=10.0, 80k steps)									
pistar06_2025	2025	other	VLA, RL, flow	RECAP: advantage-conditioned policy extraction on a flow-matching VLA, using a distributional value function (cross-entropy over 201 discretized return bins) to compute an N-step advantage and a binarized improvement indicator, with classifier-free-guidance-style advantage-conditioning dropout (30%)			yes		yes	750		not addressed	yes
wm_dex_human_videos_2025	2025	grasp	world-model, MPC	DexWM: a deterministic latent-space (DINOv2) world model with a CDiT-based predictor conditioned on MANO fingertip-keypoint-difference actions, trained with a state-prediction loss plus an auxiliary hand-consistency heatmap loss; controlled via CEM/MPC planning in latent space	Allegro Hand		no	RoboCasa	yes	12			no
being_h05_2026	2026	bimanual-coord, other	VLA, flow	Rectified Flow (continuous action velocity-field prediction) + Masked Motion Token Prediction (discrete cross-entropy channel) inside a Mixture-of-Transformers /	Inspire Hand (Franka FR3); LinkerBot O6 (Unitree G1); unnamed 6-DoF dexterous hands on PND Adam-U and BeingBeyond D1; parallel	6		LIBERO, RoboCasa (engine/timestep not stated)	yes	20		not addressed	yes

method	yr	task	paradigm	algorithm	hand	DoF	bi	sim	real	trials	unseen obj	penetration	code
				Mixture-of-Flow (MoF) action expert with shared 'Foundation Expert' layers and routed Top-K specialized experts	gripper on LeRobot SO-101 (not dexterous)								
bidexgrasp_2026	2026	grasp, bimanual-coord	grasp-synthesis, diffusion	bi-level QP-ADMM optimisation (region-pair init + decoupled per-hand force closure) + DDPM generator	Shadow Hand pair (sim); Inspire and BrainCo hands (real)		yes	MuJoCo	yes	260	30	measured	no
deximit_2026	2026	bimanual-coord, grasp, functional/tool	data-collection, BC, grasp-synthesis, trajopt	4D hand-object reconstruction (ST2+FPose) + LLM-based (Qwen3-VL) subtask decomposition with Action-Centric Scheduling + BODex-style force-closure grasp synthesis and keyframe motion planning, feeding a 3D Diffusion Policy (DP3)	XHands (real-world deployment); exact simulation hand not explicitly named		yes		yes			constrained	no
dexora_2026	2026	grasp, functional/tool, bimanual-coord	VLA, diffusion, teleop-system	Decoder-only diffusion transformer (28 layers, hidden size 1024, 16 attention heads) with T5 (language) and SigLIP (vision) encoders feeding conditional tokens; DDPM training, DPMSolver++ sampling for inference; training clips reweighted by a discriminator trained with a positive-unlabeled objective (DWBC-style	XHAND dexterous hand (paired), 12 fully actuated joints per hand, thumb and index additionally support lateral ab/adduction	12	yes	MuJoCo (digital twin of the real platform, used to generate the synthetic corpus and mirrored in real-time teleoperation)	yes	20		not addressed	

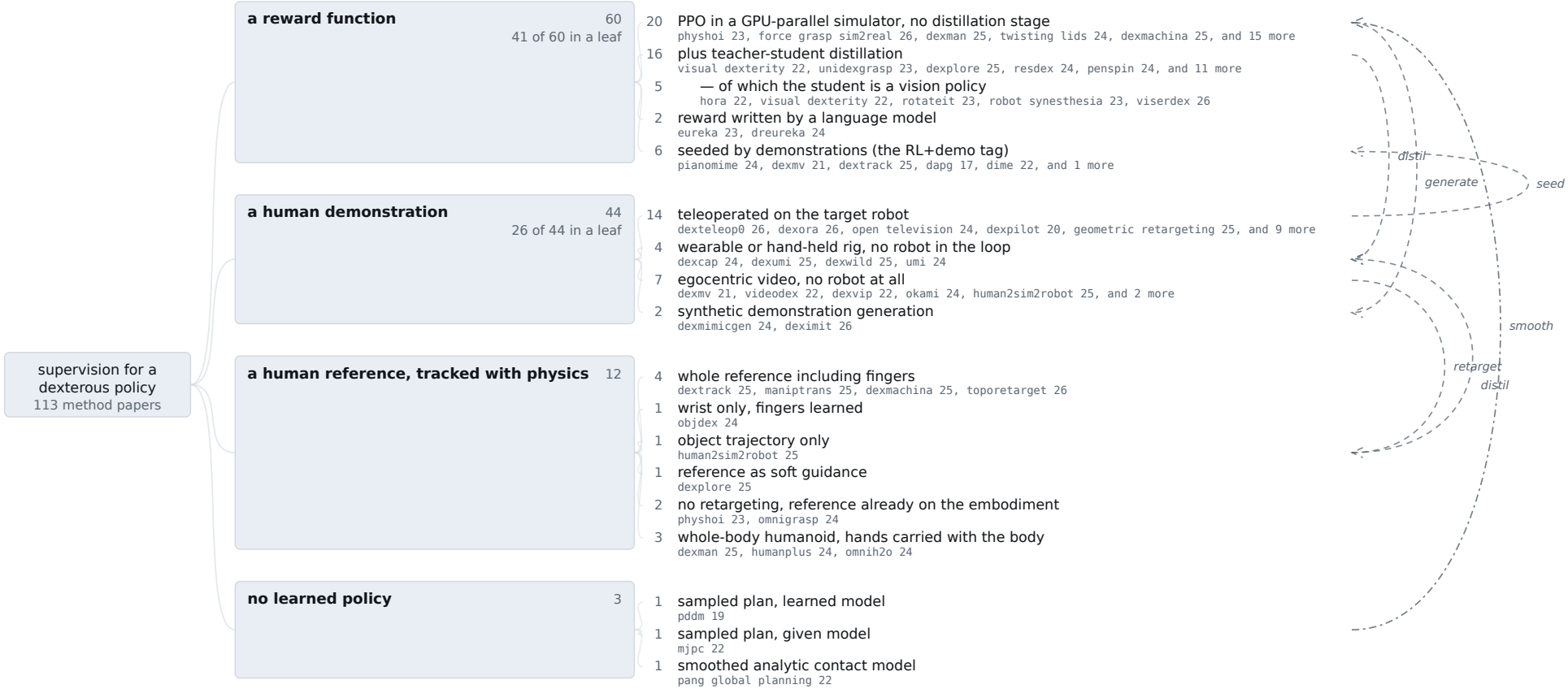
method	yr	task	paradigm	algorithm	hand	DoF	bi	sim	real	trials	unseen obj	penetration	code
				score-to-weight mapping) that scores demonstration quality									
dexteleop0_2026	2026	grasp, functional/tool, bimanual-coord	teleop-system, MPC	box-constrained QP shared-autonomy residual controller (force tracking + multi-contact force-torque balance + nominal stabilization terms) on top of DexPilot-style vector retargeting	Sharpa Wave (x2)	22	yes	NVIDIA IsaacSim 4.5	yes	35			no
egoscale_2026	2026	grasp, functional/tool, bimanual-coord	flow, VLA	flow-based VLA (GR00T N1-style): VLM backbone + DiT action expert trained with a flow-matching objective, in a three-stage recipe (large-scale human-video pretraining, aligned human-robot mid-training, task-specific post-training)	Sharpa Wave (22-DoF); cross-embodiment target Unitree G1 (7-DoF tri-finger hand)	22	yes		yes	10			no
force_grasp_sim2real_2026	2026	grasp, reorient	RL	PPO (asymmetric actor-critic)	xHand	12	no	IsaacLab	yes	70	2		no
poise_2026	2026	reorient	RL	asymmetric PPO	Sharpa Wave hand	22	no	Isaac Lab	yes	10	1	not addressed	no
simtoolreal_2026	2026	functional/tool	RL	SAPG (PPO variant)	Sharpa five-fingered hand	22	no	IsaacGym	yes	120	12	not addressed	no
teledexter_2026	2026	reorient, functional/tool	RL, teleop-system	single-stage RL (SAPG) with consecutive subgoal co-tracking reward	SharpaWave (22-DoF, headline); also LeapHand (16-DoF)	22	no	Isaac Gym	yes	15		penalised	no
toporetarget_2026	2026	track-human-ref	trajopt, RL	constrained Laplacian-optimization retargeting + PPO tracking	Wuji Hand		no		yes			constrained	no
unidex_2026	2026	grasp, functional/tool	flow, VLA, data-collection	UniDex-VLA: a pi0-style flow-matching VLA	Inspire, Leap, Shadow, Allegro, Ability, Oymotion,		no		yes	20	1		no

method	yr	task	paradigm	algorithm	hand	DoF	bi	sim	real	trials	unseen obj	penetration	code
				with a Uni3D pointcloud encoder and Gemma-based backbone, predicting actions in a unified Function-Actuator-Aligned Space (FAAS) across 8 dexterous hands	XHand, Wuji (8 hands in UniDex-Dataset); real-robot eval uses Inspire, Wuji, and Oymotion hands								
viserdex_2026	2026	reorient	RL, distillation	PPO (RSL-RL)	Allegro Hand	16	no	Isaac Lab	yes	50		not addressed	no

113 rows; 225 of 1582 cells (14%) are values no source stated. No cell is truncated. A value wider than its column is wrapped at a word boundary, so a cell that runs to several rendered lines is one value and not several.

Figure 4. What supervises a dexterous policy

Four kinds of supervision, subdivided to the level at which the 113 method papers actually differ.
Every leaf count is the rows its stated predicate selects, recomputed from corpus/rows; a paper may appear more than once.



The branches are not exclusive, and the dashed curves on the right are the overlaps. Most work after 2024 trains with a reward in simulation and ships something shaped like an imitation policy, so it belongs to the first branch and the second at once. Eight further rows carry a trajectory-optimisation or MPC tag inside a learned pipeline and are counted on the branch that learns. The double-dashed smoothing edge is the one overlap that is a theorem rather than a pipeline: `pang_global_planning_2022` proves that a policy gradient and an analytic log-barrier relaxation compute the same local model of contact.

5.2 Reinforcement learning

Sixty of the 113 papers learn from a reward, and the recipe they use is remarkably uniform. PPO is the algorithm in nearly all of them; the next most frequent choice appears four times. Most run in a GPU-parallel simulator, about a third distill a privileged teacher into a deployable student, and 16 papers combine all three. That intersection is the recipe as it is actually practised, and Table 7 gives the per-paper detail.

The hand is almost always an Allegro. It appears in 35 of them, and its hold on in-hand reorientation is tighter still: more than half of those papers use it, against roughly a fifth for the Shadow and an eighth for LEAP.

Privilege is handled two ways. `openai_dexterity_2018` gives the value network object and target orientation, joint angles, joint velocities and object velocities the policy never sees, with a footnote recording that current object orientation was left out of the policy inputs by accident, and `dextreme_2022` uses the same asymmetric critic with a 2048-unit LSTM against the actor's 1024. `hora_2022` instead compresses nine privileged object properties into an eight-dimensional vector regressed from 30 steps of proprioception, and `visual_dexterity_2022` distills twice, from a state teacher to a synthetic point cloud and then to a rendered one, for a fivefold speedup.

Domain randomisation is the part of the recipe with the least discipline. The best practice in the corpus makes it adaptive, widening each range when performance at its boundary is good enough and narrowing it when it is not, and publishing the ranges that search discovered (`openai_rubiks_cube_2019`, `dextreme_2022`). Below that standard, two shipped configurations disagree with their own papers about what was randomised at all: one ships a joint-noise range four times wider than the paper states, in a commit its own README says is not the one that reproduces the paper, and the other ships a disturbance force of zero where the paper describes one. Both are in Table 11 with the file and the two values.

Table 5 puts the 21 in-hand reorientation methods against nine recurring term families and marks each cell by where the term was found: in the paper, in the code, in both, or in the code with every shipped configuration setting its weight to zero.

Table 5. Reward terms across in-hand reorientation methods

method	yr	goal or rotation tracking	object velocity	finger-object distance	hand-pose deviation	action rate or magnitude	torque, work or joint velocity	drop or failure	contact or force	success bonus	terms	code	paper/code mismatch
openai_dexterity_2018	2018	paper						paper		paper	3	no	
openai_rubiks_cube_2019	2019	paper						paper		paper	3	no	
dextreme_2022	2022	both				both	both	code (0)		both	6	yes	yes
hora_2022	2022	both	both		both		both				5	yes	yes
visual_dexterity_2022	2022	both		both	paper		both	both	paper	both	7	yes	yes
dexpbt_2023	2023	both		both		code	paper	code (0)		both	4	yes	yes
eureka_2023	2023	paper	paper	paper								yes	yes
robot_synesthesia_2023	2023	paper	paper	paper		paper	paper	paper			6	no	
rotateit_2023	2023	paper	paper		paper		paper				6		
rotating_without_seeing_2023	2023	paper	paper	paper			paper	paper			6		
anyrotate_2024	2024	paper	paper		paper		paper	paper	paper	paper	10	no	
demostart_2024	2024									paper	1	no	
dreureka_2024	2024	paper	paper		paper			paper			4	yes	yes
penspin_2024	2024	both	both		both	code (0)	both				7	yes	yes
dexndm_2025	2025	paper	paper		paper		paper			paper	7	no	
dexremoe_2025	2025	paper	paper			paper				paper	5	no	
dexteritygen_2025	2025											no	
force_grasp_sim2real_2026	2026	paper			paper	paper	paper	paper	paper	paper	7	no	
poise_2026	2026	paper					paper	paper	paper	paper	4	no	
teledexter_2026	2026	paper								paper		no	
viserdex_2026	2026	paper	paper			paper	paper	paper		paper	10	no	

21 rows; 97 of 189 term cells (51%) are families the method does not use. `paper` *means the term is in the paper and either no code was released or it is absent from the released reward code;* `code` *means it is in the released code and not in the paper's stated reward;* `both` *means it is in both. Marks are read from papers/notes/, term by term; the per-method source section is in corpus/reward_matrix.json.*

The families are not equally popular. Almost every method has a goal or rotation tracking term, which is the task itself. After that the common families are an effort penalty, a sparse success bonus, an object-velocity penalty and a drop penalty, each in about half of them. Only four carry a contact or force term at all. Table 5 gives the full matrix.

Three cells an earlier draft marked *code* print *code (0)* instead, because in each the term is in the released code with every shipped configuration setting its weight to zero: `dextreme_2022`'s timeout reward, `dexpbt_2023`'s fall penalty, and `penspin_2024`'s action penalty. Marking them *code* would tell a reader the code optimises something the paper does not state, and leaving them blank would hide a term that is in the file. The repository carries the config key and the zero beside each of the three marks.

Only three of those four contact terms are what they look like. `anyrotate_2024` scores good and bad fingertip contacts, `poise_2026` rewards a friction-cone wrench margin, and `force_grasp_sim2real_2026` tracks a commanded grasp force. The fourth penalises the *object* touching the table, which is task shaping against using the table as a third finger rather than a statement about hand-object contact, and the flag that enables it defaults to off. So three of the 21 reorientation methods put a hand-object contact quantity in the reward, and none puts interpenetration in it: the one method that penalises interpenetration does so while building its references offline, not in the policy's reward (`teledexter_2026`). The physical quality of the contact is not something this literature optimises.

Term counts range from one to ten. `demostart_2024` gives a single binary terminal reward of 1 and lets a curriculum over demonstration states do the shaping. `anyrotate_2024` and `viserdex_2026` each publish ten weighted terms. `viserdex_2026` is the cleanest specification in the table, with every term, weight and equation in one appendix table, and a statement that the same weights are used for every object.

Weights are frequently unrecoverable. Three of the 21 papers give no numeric weight that survives PDF conversion, because the equation blocks are images, which is a limit of this survey and is recorded as such. `robot_synesthesia_2023` is a worse case. Its six coefficients are described only as "tuned hyper-parameters" and no number is printed anywhere in the paper, so the reward it reports cannot be reconstructed from the paper. `dexremoe_2025` names an angular-velocity penalty weight in prose with no entry in its hyperparameter table, and `dextrack_2025` names an affinity reward that is missing from its weight table.

Curricula in this literature either relax the physics or tighten the tolerance. `dexpbt_2023` ratchets a success tolerance down once the policy starts succeeding at it; `maniptrans_2025` starts in zero gravity with high friction and restores both while narrowing its fingertip threshold; `dexmachina_2025` attaches virtual spring-dampers to the object, lets them do the task at first, then decays their gains to zero. The two papers that ablate the curriculum find it load-bearing. `rotateit_2023` has to ramp its rotation penalty in from zero, because applying it from the start teaches the policy only to hold the object still, and `viserdex_2026` fails completely without the regularisation leg of its curriculum.

Population-based training appears once. `dexpbt_2023` trains populations of up to 32 agents, mutating the middle third of each population and replacing the bottom with mutated copies of the top, and reports 0.32 trillion environment steps for its largest run. The technique is routine outside robotics and nothing else in this corpus uses it.

A language model can write the reward outright, which removes the hand-tuning and replaces it with a search whose objective nobody has read. `eureka_2023` has GPT-4 write the reward function directly, constrained to return a total and a dictionary of named components, and feeds per-component statistics back between iterations. Its appendix gives an example whose reward correlates at −0.26 with the human-written one and still scores 1.45 on the human-normalized metric, the strongest argument in the corpus that hand-tuned weights are not a ceiling. `dreureka_2024` moves the safety requirement into the prompt instead of a penalty term, asking in words for a cube rotating at about 0.25 radians per second with fingers penalized for leaving their initial pose. Neither can be checked against a file. Eureka's generated rewards are written at run

time to a gitignored directory and never committed, so the reward behind any reported number cannot be recovered from the repository, though its README documents the paper's budget of five iterations and sixteen samples as the default and marks the shipped one-iteration block a fast-test preset. DrEureka's repository contains only the locomotion and globe-walking trees, so its LEAP-hand reward has no code at all.

Reinforcement learning stalls on objectives that are not learnable as stated. `anyrotate_2024` found the angular-velocity objective unlearnable in the multi-axis setting and replaced it with a moving keypoint target. `rotateit_2023` reports that its multi-axis policy "does not converge when training with only reinforcement learning" and adds an imitation loss against single-axis oracles. `eureka_2023` cannot spin a pen from scratch and needs a pretrain-then-finetune split, with the from-scratch ablation failing to complete one cycle. Section 5.3's contact-graph term exists because of the general shape of the problem: contact moves the object off the reference, so return goes down, so the policy learns not to touch the object.

It stalls on geometry too. `hora_2022` fails on objects under 4 cm across because the fingers collide with each other. What it does not stall on is publication: the real-trial counts behind these headlines are small enough that section 7.1 treats them as the reporting problem they are.

5.3 Learning from human data

Table 6 lists the 30 papers in this corpus that describe a system for getting human motion onto a robot hand, with the retargeting objective compressed to one clause each.

Table 6. Teleoperation and human-data systems

system	operator interface	robot hand	retargeting objective	latency	rig USD	data collected
dexpilot_2020	four RealSense D415 RGB-D cameras tracking a bare hand, markerless; a coloured glove is worn only offline to train the neural hand-pose priors	Wonik Robotics Allegro hand (16-DoF, 4x4-joint fingers), retrofitted with 4 SynTouch BioTac tactile sensors at the fingertips	minimise a weighted squared distance between human and Allegro fingertip task-space vectors, with switching weights that close primary-finger-to-thumb gaps and enforce a minimum finger separation, plus a regulariser toward the open hand, solved online with SLSQP	about one second, end to end (Sec. IV)		
dexmv_2021	human demonstrator recorded on video (cubic capture rig, two RealSense D435 cameras), no live teleoperation; hand pose (MANO) and object pose extracted offline from the recorded video	Adroit Hand	match human (MANO forward-kinematics) and robot task-space vectors (palm-to-fingertip and palm-to-mid-phalanx pairs) per frame via SLSQP with a temporal-consistency term, then fit a minimum-jerk joint trajectory and compute torques via inverse dynamics			700 traj, 7 h
dexvip_2022	no live operator; a consensus hand pose is mined offline from curated HowTo100M video frames via FrankMocap 3D hand-pose estimation and k-medoid clustering	Adroit Hand	deterministic 4-stage geometric mapping (not an optimization) from FrankMocap's 21-joint hand pose to Adroit's 30-DoF joint space, applied once offline to produce a single static consensus target pose per object category, used only as a reward target			
dime_2022	vision-based, single RGB camera, MediaPipe hand detector, no glove/headset/exoskeleton	Allegro Hand	map human fingertip 2D image locations directly to fixed-height robot fingertip 3D targets (no depth estimation), then solve per-finger inverse kinematics for joint angles	not stated; targets at 30 Hz, PD loop at 300 Hz		30 traj
holo_dex_2022	VR headset (Meta Quest 2), built-in 4-camera hand tracker, no glove/exoskeleton	Allegro Hand	direct joint-angle copy for index/middle/ring fingers; thumb fingertip position matched via inverse kinematics; pinky ignored (no cost function or optimization)	under 100 ms on a local network (Sec. IV-D)	399	
videodex_2022		LEAP Hand	hand poses mapped via a distilled MLP replicating a Robotic-Telekinesis-style fingertip/palm keypoint-vector energy function; wrist trajectory computed via a PnP + monocular SLAM + heuristic gravity-alignment + workspace-rescaling pipeline, both applied offline to build pretraining trajectories, not run online at deployment			965 traj
anyteleop_2023	vision-based, camera only (RGB or RGB-D, single or multi-camera), MediaPipe hand-keypoint detection, no glove/headset/exoskeleton	Allegro Hand	minimize the difference between human and robot fingertip keypoint vectors via forward kinematics, with joint-limit constraints and a temporal-smoothness penalty	26-35 ms hand pose, 9-10 ms retargeting (Table II); no end-to-end figure		
pgdm_2023		ShadowHand	for pre-grasps only: inverse kinematics fits the robot hand's fingertip positions to a single human hand pose frame near first contact, used solely to initialize RL exploration; full object trajectories			40 traj

system	operator interface	robot hand	retargeting objective	latency	rig USD	data collected
			are used directly as the RL goal with no hand-trajectory retargeting			
ace_teleop_2024	visual exoskeleton: 3D-printed bimanual 6-DoF-per-arm exoskeleton with wrist-mounted cameras for MediaPipe finger tracking; not a glove or VR headset	Ability Hand / Inspire Hand / parallel-jaw gripper (embodiment-dependent)	wrist/end-effector pose mapped via a scale-and-recenter IK transform (Normal/Mirror/Bimanual modes) from exoskeleton forward kinematics; finger joints retargeted via AnyTeleop's fingertip-keypoint-vector optimization with joint limits and temporal smoothing	27 Hz hand tracking; no end-to-end figure	600	
bidex_teleop_2024	Manus Meta motion-capture gloves for the fingers plus GELLO-style teacher arms for wrist and arm pose; no headset	LEAP Hand (16 DoF); LEAP Hand V2 (21 DoF) used for 'extreme dexterity' experiments	inverse kinematics from glove fingertip positions to LEAP fingertip targets, with the arm commanded directly in joint space from the teacher arm rather than in end-effector space	claimed low-latency, no number given	6000	50 traj
bunny_visionpro_2024	VR headset (Apple Vision Pro) hand/wrist tracking, plus a custom ERM haptic feedback device driven by fingertip FSR tactile sensors; no glove or exoskeleton	Ability Hand	minimize scaled human-robot fingertip keypoint-vector distance via forward kinematics with joint limits and temporal smoothness (hand), solved online via SQP with a reduced-dimension reformulation for loop joints; separately, a unified IK + singularity-avoidance + sphere-approximated self-collision objective for arm motion	3.43 ms retargeting, 15.93 ms motion control, >60 Hz overall (Table 1)		
cyberdemo_2024	vision-based teleoperation (single RealSense camera observing operator hand motion, real-time hand detection and retargeting via a cited third-party system), not a glove or headset	Allegro Hand	delegated entirely to the cited third-party teleoperation system; CyberDemo does not describe or modify the human-to-robot mapping itself			
dexcap_2024	wearable backpack rig: Rokoko EMF gloves, one SLAM camera per wrist and a chest-mounted RGB-D LiDAR; no robot in the loop during collection	LEAP Hand	fingertip-position inverse kinematics into the 16-dim LEAP joint space anchored by the mocap wrist pose, with the little finger discarded		4000	787 traj, 4.5 h
egomimic_2024	human data: head-worn Project Aria glasses (egocentric RGB + onboard SLAM/hand-tracking, no glove or exoskeleton); robot demonstrations: standard bimanual ViperX teleoperation	parallel-jaw gripper (not a dexterous hand; see note scope flag)	no per-finger kinematic retargeting (end effector is a 2-finger gripper); domain alignment via a shared per-timestep camera-centered reference frame, per-embodiment Gaussian normalization of proprioception/actions, and SAM2-based visual masking with a directional line overlay applied identically to human and robot streams		1000	2,150 traj, 4 h
hudor_2024	VR headset (Meta Quest 3) built-in hand tracking for fingertip 3D positions, ArUco-marker rigid-body calibration into the robot frame; demonstrator is in the same scene as the robot, not internet video	Allegro Hand	direct Cartesian-space correspondence: human fingertip 3D positions are treated as target Cartesian positions for the corresponding robot fingertips with no kinematic-chain optimization, converted to joint commands via a custom Jacobian-based gradient-descent IK			4 traj

system	operator interface	robot hand	retargeting objective	latency	rig USD	data collected
okami_2024	single human demonstrator recorded on a static RGB-D camera (Intel RealSense D435i), no live teleoperation and no glove/headset; single video per task	Inspire Hand (x2)	factorized retargeting run once per demonstration video: arm/shoulder-elbow-wrist IK (via the Pink library, wrist position dominating the objective) plus hand joint angles computed via dex-retargeting from SMPL-H hand keypoints; the resulting trajectory is affinely SE(3)-warped at test time to match newly localized object start/end poses, with fingers retargeted independently of arm warping			
open_television_2024	VR headset (Apple Vision Pro or Meta Quest 3) with active stereo video streamed back from a 2-3 DoF camera gimbal on the robot's neck	Inspire Robots hand (6 actuated DoF, 12 total); Fourier GR-1 embodiment instead uses a 1-DoF parallel-jaw gripper	dex-retargeting keypoint-vector optimisation over five wrist-to-fingertip and two thumb-to-finger vectors, solved with SLSQP, with end-effector position taken relative to the head and orientation absolute	60 Hz stereo round trip and 60 Hz control		
dexteritygen_2025	human teleoperator tracked via a Manus Glove (retargeted to the Allegro hand at 300Hz via an unreleased, confidential fast retargeting method) plus a Vive tracker for 6-DoF wrist pose driving the arm	Allegro Hand				
dexumi_2025	hand-specific 3D-printed wearable exoskeleton (per-robot-hand optimized joint-to-fingertip kinematics), joint encoders reading exoskeleton angles directly (no visual fingertip retargeting), a wrist-mounted camera (ARKit-tracked wrist pose), and matching tactile sensors installed on both exoskeleton and target robot hand	Inspire Hand (12 DoF, 6 active) and XHand (12 active DoF)	no visual/kinematic-chain retargeting at inference; a learned per-joint regression model maps exoskeleton encoder values directly to robot motor values (calibrated by visually overlaying exoskeleton and robot at matched motor values), with the exoskeleton's mechanical linkage itself optimized offline so its fingertip workspace matches the target robot hand's forward-kinematics workspace	per-sensor latency measured and compensated offline, no figure		1,355 traj
dexwild_2025	wearable, calibration-free rig: motion-capture glove + two palm-mounted stereo cameras (pinky-side and thumb-side) + wrist ArUco marker tracking, worn by untrained operators; no VR headset	LEAP Hand / LEAP Hand V2 Advanced	robot hand kinematics optimized to match observed human fingertip positions (fixed hyperparameters, no per-user tuning); a relative (delta) end-effector state-action representation removes the need for a global coordinate frame, and identical palm-camera mounting on human and robot hands makes RGB observations look the same across embodiments			9,290 traj, 46.2 h
doglove_2025	haptic force-feedback exoskeleton glove (21-DoF encoder-based motion capture, 5-DoF cable-driven force feedback, per-fingertip LRA haptic feedback), plus an HTC Vive Tracker for wrist localization; not vision-based	LEAP Hand	map glove-FK fingertip positions to robot fingertip targets via the third-party Mink differential-IK solver, with a single scalar hand-size scaling factor; no paper-original cost function	30 Hz minimum system rate (120 Hz mocap, 30 Hz haptics)	600	
egozero_2025	Project Aria smart glasses (egocentric fisheye RGB + SLAM	Franka Panda parallel-jaw gripper (not a dexterous hand; see note	no kinematic hand-to-robot retargeting; both human and robot			700 traj

system	operator interface	robot hand	retargeting objective	latency	rig USD	data collected
	cameras + onboard hand/camera pose tracking), no glove, no live teleoperation of a robot during data collection; at inference an iPhone provides the egocentric view instead of Aria	scope flag)	are represented in a shared morphology-agnostic 3D-point space so the policy transfers zero-shot without any cross-embodiment pose-mapping step			
geometric_retargeting_2025	Manus glove for fingertip keypoints + HTC Vive tracker for wrist pose (real-world evaluation); the retargeting method itself is tracking-source-agnostic	Allegro Hand	learned per-finger neural network trained offline (3-5 minutes) to jointly satisfy motion-direction preservation, C-space coverage, flatness, pinch correspondence, and self-collision avoidance, replacing per-frame online optimization with a single 1kHz forward pass at inference	1 kHz retargeting inference, against 60-100 Hz for optimisation baselines		
h_rdt_2025	for pretraining: none (consumes EgoDex's pre-existing released 3D hand-pose annotations, no live capture by this paper); for fine-tuning: standard teleoperation of each robot platform, not described in detail	parallel-jaw / 2-jaw grippers at deployment (Aloha-Agilex, ARX5, Franka-Panda, UR5+UMI); no dexterous hand deployed, though pretraining action space includes bimanual fingertip positions from EgoDex	no explicit kinematic retargeting; the 48-D bimanual wrist+fingertip human action space is treated as a superset of most end-effector action spaces, and the human-to-robot correspondence is instead learned implicitly by reinitializing and fine-tuning the state/action adaptors and action decoder on paired robot data while keeping the pretrained vision/language backbone			338,000 traj, 829 h
humanoid_policy_human_policy_2025	consumer VR headsets (Apple Vision Pro built-in camera with ARKit hand/head tracking, or Meta Quest 3/Vision Pro with a 3D-printed ZED Mini stereo mount via an OpenTelevision web app), <$700; robot-side demonstrations collected via standard humanoid teleoperation	Inspire Hand (x2, 5-fingered)	no offline kinematic retargeting network; human and robot share an identical 54-D state-action space via a bijective fingertip-to-fingertip correspondence (both are 5-fingered hands), bridged only by action-speed interpolation (human trajectories slowed ~4x) and instructing operators to minimize torso movement		700	
wm_dex_human_videos_2025		Allegro Hand	no explicit robot retargeting network; both human hands (MANO 21-keypoint) and robot end-effectors (Allegro fingertips via forward kinematics; parallel-jaw grippers approximated with dummy circular keypoints) are mapped into the same fingertip-keypoint-difference action representation, with the missing pinky filled by reusing the ring-finger-equivalent Allegro keypoints			829 h
dexteleop0_2026	VR headset (Meta Quest 3), egocentric hand/wrist tracking (26 tracked hand-joint transforms per hand), no external motion capture	Sharpa Wave (x2)	DexPilot-style vector matching of critical inter-joint/finger-to-palm target vectors for the hand, plus analytical IK on wrist pose for the arm, producing a raw joint command that a real-time QP then corrects to keep fingertip contact forces in a safe window and balance net object force/torque	30 Hz QP control cycle		
egoscale_2026	large-scale pretraining: in-the-wild egocentric video with off-the-shelf	Sharpa Wave (22-DoF); cross-embodiment target Unitree G1 (7-	human 21-keypoint (MANO-style) hand pose retargeted via an			20854 h

system	operator interface	robot hand	retargeting objective	latency	rig USD	data collected
	SLAM + hand-pose estimation, no dedicated capture rig; aligned mid-training: Vive trackers for 3D wrist pose + Manus gloves for 25-joint in-hand pose, same camera rig as the robot	DoF tri-finger hand)	optimization-based procedure enforcing joint limits and kinematic constraints into the 22-DoF Sharpa Wave hand joint space; wrist motion is represented as a shared relative SE(3) transform identical across human and robot data			
`teledexter_2026`	NOKOV motion-capture system tracking operator hand pose and object 6D pose in real time (not vision-based, not a glove)	SharpaWave (22-DoF, headline); also LeapHand (16-DoF)	two-stage geometry-aware retargeting: Stage 1 fingertip-vector alignment to human hand geometry; Stage 2 object-mesh-aware refinement combining a surface-contact term, a differentiable-SDF interpenetration penalty, a self-collision sphere term, and a temporal-smoothness term, producing the fingertip+object-pose subgoals the RL controller tracks			
`unidex_2026`	human-video side: no live capture rig, converts existing egocentric RGB-D datasets (H2O, HOI4D, HOT3D, TACO) via offline retargeting; fine-tuning demonstrations collected via OpenTeleVision + dex-retargeting on an Apple Vision Pro	Inspire, Leap, Shadow, Allegro, Ability, Oymotion, XHand, Wuji (8 hands in UniDex-Dataset); real-robot eval uses Inspire, Wuji, and Oymotion hands	two-stage human-in-the-loop kinematic retargeting: an automatic PyBullet multi-end-effector IK stage aligns a 6-DoF offset between a virtual human-hand-fixed base and the real robot base to match fingertip positions (respecting joint limits, with iterative mimic-joint correction), followed by an interactive GUI stage where a human manually corrects alignment; a separate visual-alignment step masks the human hand from the pointcloud and re-renders the retargeted robot-hand mesh into the scene			52,000 traj

30 rows; 60 of 210 cells (28%) are values no source stated. No cell is truncated. A value wider than its column is wrapped at a word boundary, so a cell that runs to several rendered lines is one value and not several.

Retargeting splits four ways. The dominant family matches task-space vectors between the human and the robot hand, weighting fingertip distances and adding a minimum separation between the primary fingers, solved online; `dexpilot_2020` set that pattern, and `anyteleop_2023`, `open_television_2024`, `bunny_visionpro_2024` and `ace_teleop_2024` use the same formulation. The second copies joint angles wherever the kinematics allow it and solves only the thumb by inverse kinematics (`holo_dex_2022`). The third learns the map offline, which buys about an order of magnitude in rate over online optimisation and, in `geometric_retargeting_2025`, raises configuration-space coverage from 38 to 90 percent. The fourth removes retargeting from the loop altogether, building an exoskeleton whose fingertip workspace is designed to match the target hand and regressing encoder values straight to motor values (`dexumi_2025`).

The table's empty cells are the point. Fewer than half of the teleoperation papers state a latency at all, and only a handful state a rig cost. Where costs are stated they are low and falling, from a few hundred dollars for a headset or a glove (`holo_dex_2022`, `doglove_2025`) to a few thousand for a full mocap backpack (`dexcap_2024`, `bidex_teleop_2024`). Latency is not one quantity either: some figures are end to end, others are a single retargeting stage, and several cells hold a control rate instead, so the column's spread is not a range. Fewer than half report how many trajectories they collected, and fewer still report hours of data.

Where throughput is measured, human collection beats teleoperation by a wide margin. `humanoid_policy_human_policy_2025` times a grasp demonstration at 3.79 seconds for a bare human and 19.72 seconds through a teleoperated humanoid, and a pour at 4.81 against 37.31 seconds. It attributes the gap to retargeting latency and to the workspace of a 7-DoF arm rather than to anything about the hand.

Four architectures cover the corpus. Action chunking came from `aloha_act_2023`, which predicts a chunk of joint targets with a CVAE and blends overlapping chunks, reaching high success on fine bimanual tasks from about 50 demonstrations each. Diffusion came from `diffusion_policy_2023`, which denoises an action sequence and reports a 46.9 percent average improvement over the previous generation of imitation baselines; `dp3_2024` swaps the image encoder for a small point-cloud encoder and improves on it again, most sharply on real tasks. Flow matching is the default for large models from 2024 onward and carries eight papers. Discrete action tokens are the fourth, and the only one of the four that needs no continuous action head at all.

Data generation is a separate lever and is underrated. `dexmimicgen_2024` turns 60 human demonstrations into 21,000 simulated ones, and a real two-handed humanoid trained on the generated set succeeds where the same policy trained on the source demonstrations never does. `dex1b_2025` iterates optimisation, a proposal model and a simulator filter into roughly a billion synthetic grasps, and beats the prior best by a wide margin on its own benchmark.

Seven papers train a dexterous-hand policy from video with no teleoperation at any stage. `dexmv_2021` retargets 700 self-recorded demonstrations by matching palm-to-fingertip task-space vectors and estimating actions by inverse dynamics, `videodex_2022` mines 965 trajectories from EpicKitchens to pretrain a neural dynamic policy, and `dexvip_2022` goes further into the reward, clustering 715 curated HowTo100M frames into one consensus grasp pose per object category and adding it as a reward term. `okami_2024`, `human2sim2robot_2025` and `hudor_2024` work from a single video each, and `wm_dex_human_videos_2025` pretrains a world model on 829 hours of EgoDex.

Three papers measure the gap directly rather than asserting it. `okami_2024` runs the same reference-plan and warping pipeline with and without human body and hand retargeting, and the version without it loses 75.0, 42.1, 82.0 and 74.0 points across four task and setting pairs. That is the largest clean embodiment-gap number in the corpus.

`humanoid_policy_human_policy_2025` ablates two mitigations separately on the same vertical-grasp task. Removing the interpolation that slows human trajectories down cuts success to a quarter of the full method's, and removing the unified state space collapses it to zero, because the policy is handed a shortcut for telling the two embodiments apart. `egozero_2025` reports the sharpest version of the same effect: removing its 3D augmentation drops every one of its seven tasks to no

successes at all, and substituting monocular depth for triangulated depth does the same, because the best metric depth models it tested carry more than 5 cm of error.

More human correspondence is not reliably better, and one paper here tested it directly. `objdex_2024` supplies the counterexample. It retargets only the wrist and lets reinforcement learning discover finger motion, and its ablation that adds a fingertip-matching reward "does not yield benefits and even leads to lower performance". Richer human correspondence is not uniformly better, and this is the one result in the corpus that says so with an ablation.

Twelve papers sit in the human-reference tracking family. Eight track a human hand on an object and are covered here; the other four sit at the edges, `human2sim2robot_2025` tracking only the object's trajectory, `dexman_2025` retargeting bimanual video onto a full humanoid, and `humanplus_2024` and `omnih2o_2024` tracking whole-body motion. A reference is retargeted onto the robot and a policy trained to make the simulated body follow it: the reference supplies the shaping reward engineering would otherwise have to invent, and the simulator the physical consistency pure imitation lacks.

The multiplicative reward that most human-reference methods now use has a single source. `physhoi_2023` is the origin of the form. It multiplies a body term, an object term, an interaction-graph term and a contact-graph term, and reaches 95.4 percent success on GRAB against 27.0 percent for a DeepMimic baseline. The contact-graph term exists to stop the policy learning not to touch the object. `omnigrasp_2024` replaces the action space with a pretrained latent motion prior and reports 94.6 percent grasp success and 84.8 percent trajectory success on GRAB, while stating outright that it omits penetration metrics.

The next step was to stop treating demonstrations as fixed and let the method generate its own. `dextrack_2025` adds an imitation loss to PPO and mines its own demonstrations through a homotopy search over easier neighbouring trajectories, beating the best baseline by about eight points at either threshold. It has a penetration-depth formula and applies it only to its input references, never to its own rollouts, and presents tolerance of "severe hand-object penetrations" as robustness. `maniptrans_2025` freezes a generalist hand imitator and trains a per-task residual on top, with single-hand success about half again its bimanual figure. Its stance on contact is to raise the friction coefficient above the real value rather than model skin deformation, and its real deployment is open-loop replay.

Two papers in this corpus ran the same curriculum and disagreed about whether it helps. `dexmachina_2025` moves to Genesis with 12,000 environments and six hand URDFs, and drives the object with virtual controllers that decay to zero. Its re-implementation of `maniptrans_2025`'s curriculum does not beat no curriculum on its own setup, which is a direct disagreement between two papers worth quoting in both directions. `dexplore_2025` drops the explicit retargeting stage and treats the reference as soft guidance, with termination thresholds derived from the failure rate, reaching 87.7 percent on GRAB with an Inspire hand.

`toporetarget_2026` is the only corpus method that quantifies penetration as an evaluation quantity, reporting maximum depth and the fraction of frames past 2 mm, and constraining it during retargeting with a 1 mm soft tolerance and a 30 mm hard bound. It reaches 87.5 percent on its own pen-spinning set against 46.9 for the best baseline. It does not re-measure penetration after the tracking policy runs, so the property it constrains is a property of the reference and not of the behavior.

The pattern holds across all eight, and it is the reference-versus-rollout split in its sharpest form. Penetration is handled at the reference, if at all, and never at the rollout. `objdex_2024` completes the set with the only real-robot numbers among them, from 100 percent on a microwave and a laptop down to 41.2 percent on a ketchup bottle over 20 trials each.

Human data does not port across hands, and the map is usually left unstated. Fifty-four papers use human data and name 45 distinct hand strings between them; twenty of those appear in Table 6 with a stated retargeting objective. The other 33 never say how the human motion reached the hand, and the objectives that are stated do not converge, running from a fingertip keypoint-vector energy in `anyteleop_2023` through a per-joint regression from exoskeleton encoders in `dexumi_2025` to no retargeting at all in `egozero_2025`. What would close it is a retargeting benchmark in TopoRetarget's form `toporetarget_2026`, reporting contact precision, contact alignment, maximum penetration and the share of frames past 2 mm, per hand and per dataset.

5.4 Generalist and vision-language-action policies

The finding is the size of the hand. Eighteen papers carry the generalist tag. Fourteen of them settle whether the reported evaluation ran on a multi-fingered hand. Eight of those fourteen did, six report no hand result at all, and the remaining four never say, `groot_n16_2025` naming no end effector anywhere on its page.

Five of the eight state a hand size, and that comparison has to be made in actuated degrees of freedom rather than joints, for the reason section 3 opens with. Four of the five use a six-actuator hand and one a twelve, so their median is six. The reinforcement-learning literature of section 5.2 works an order of magnitude above that: its median hand actuates 16 joints. No generalist policy here that settles the question evaluates on a hand in that band, and the two that would reach it on paper are the two that never say whether the hand was in the evaluation at all.

The six with no hand result are parallel-jaw throughout (`pi0_2024`, whose released code encodes each ALOHA gripper as one scalar, `pi05_2025`, `pistar06_2025`, `openvla_2024`, `helix_2025`, which claims individual-finger control of a 35-DoF upper body with no success rate or trial count for any task, and `gemini_robotics_2025`, whose Apollo hand appears in a figure caption with no number attached). Its successor `gemini_robotics_15_2025` closed that gap with Apollo progress scores, not success rates, of 0.74 down to 0.62 across five generalisation axes. `dexvla_2025` is the borderline case the other way: a hand in one real-robot task family, grippers in the other five and in all 91 pretraining tasks.

The released code also drifts. `groot_n1_2025`'s repository is a later N1.7 generation with a different backbone, a different action horizon and embodiments the paper does not describe, and `pi05_2025`'s supports only the flow-matching head, not the hybrid recipe the paper describes.

The gap is size rather than absence. A generalist policy that does touch a hand runs it at roughly a third of the actuation the reinforcement-learning literature assumes, and the papers that never settle the question are the smaller half of the same gap. What would close it is one dexterous-hand task inside the standard generalist suite, with the hand's actuated degrees of freedom and its vendor printed beside the number.

5.5 Model-based baselines and hybrids

Three corpus methods solve dexterous tasks without learning a policy, and they are the control group the rest of this section lacks.

Not every method here learns a policy. A smaller line plans through a model instead, and it reaches dexterous tasks on a fraction of the data. `pddm_2019` learns an ensemble of dynamics models and plans through it with an MPPI-style optimiser, replanning every step, on a few hours of data in simulation and a few more on a real Shadow Hand. `mjpc_2022` removes the learned model too, sampling rollouts through MuJoCo itself at millisecond latency, and reorients a cube in real time from scratch. It also reports no success criterion, no trial count and no real-robot result.

One of the three does more than plan: it explains why the learned and analytic routes arrive at the same place. `pang_global_planning_2022` is the most substantial. It proves that the randomised smoothing implicit in reinforcement learning and an analytic log-barrier relaxation compute the same local linear model of contact, then uses the analytic version inside a trajectory optimiser and an RRT. Allegro in-hand rotation takes 19.59 seconds to optimize and its plate-pickup task 117.16 seconds of planning on a 16-core CPU, against the GPU-days of section 5.2, and it is the only method here that imposes non-penetration as a hard constraint. Its own limitation is the honest part: its 3D systems transfer to hardware far worse than its 2D ones, because the quasi-dynamic assumption breaks and planned grasps miss contacts under a second-order solver.

The recurring shape in recent work is reinforcement learning in simulation distilled into something that looks like an imitation policy, and it appears in four variants. The first distills a privileged teacher into a vision student inside one paper, as `hora_2022` and four others do. The second uses reinforcement learning as a demonstration factory: `dextrack_2025` mines demonstrations with per-trajectory RL and trains a generalist on them, and `maniptrans_2025` does the same at dataset scale. `dexteritygen_2025` goes furthest, training a diffusion controller on billions of simulated grasp-to-grasp transitions and projecting a human teleoperator's coarse command onto it, which takes four reorientation tasks from no successes at all under

raw teleoperation to roughly half. The third generates data without reinforcement learning at all. The fourth wraps an RL policy in a residual, as `resdex_2024` does across thousands of objects in hours of GPU time.

In the variants that dominate 2025 and 2026 work, the reward is no longer the objective of the deployed policy. It is the objective of the process that produced the deployed policy's training data. Reward engineering has moved upstream into data curation, and every reward-shaping pathology in section 5.2 now reaches the shipped policy through a dataset rather than a gradient.

5.6 Paper against released code

A paper that states its reward and also ships the code that trained it can be read twice, and the two readings sometimes disagree. In eight of the papers here they state different things. The sharpest case is `physhoi_2023`: its Table 4 gives the object's rotation error a weight for the GRAB dataset, and in the released reward function that error is set to zero with the computation that would produce it commented out on the same line, so no weight on it could move the reward. Table 11 states all eight the same way, each as a repository, a commit, a file, the value the paper prints and the value the file contains, so a reader with a browser can confirm or refute any line of it without this survey and without anyone's agreement. What the comparison does not do is name a cause. A repository at some commit is not the code that produced a paper's numbers, and nothing here is a claim about intent or about work that released nothing to read.

The authors of every paper named in the table were written to on 19 September 2026, before this survey was posted, with the claim, its evidence and the sentences the survey would print, and were given until 10 October 2026 to answer. One reply changed a comparison and removed it from the count. A second confirmed one: the first author of `pianomime_2024` states that the released code is the implementation behind that paper's results and that its description of the reward "might be imprecise" (personal communication, C. Qian, Oct 2026), which settles the direction of the comparison in the code's favour. The rest are recorded as they came, and silence is recorded as silence rather than as agreement. The letters as they went out are in `outreach/`, so a reader can see exactly what every author was asked, and `outreach/RECIPIENTS.md` records who each one went to and how the address was found. A disputed line is corrected the same way it was made, in public: the route is the address in the author block, and Appendix C prints every comparison beside the review note that revised it.

The practical lesson for a reader of this literature is small and worth keeping: read the reward function before the reward table, and treat a printed weight as a hypothesis about the code.

Table 11. Paper and released repository, the nine rows that state different values

method	confidence	repository, fetched commit	file in it, and where	what the paper prints	what that file contains
pddm_2019	high	google-research/pddm 06b88cdbaf	pddm/envs/cube/cube_env.py _get_obs	observation dimension 46 for in-hand reorientation, Table 2	the six fields _get_obs concatenates sum to 39 on the code's own inline comments
dextreme_2022	high	isaac-sim/IsaacGymEnvs aeed298638	isaacgymenvs/cfg/task/AllegroHandDextremeADR.yaml actionDeltaPenaltyScale	action-delta penalty weight -0.25, Table 2	actionDeltaPenaltyScale: -0.2, and -0.01 in the ManualDR yaml beside it
visual_dexterity_2022	high	Improbable-AI/dexenv ad9634e9d2	dexenv/conf/dclaw.yaml alg.num_envs	32,000 teacher training environments, Table S1	num_envs: 8000, and 16384 in the parent hand_default.yaml
dexpbt_2023	high	NVIDIA-Omniverse/IsaacGymEnvs aeed298638	isaacgymenvs/tasks/allegro_kuka/allegro_kuka_base.py compute_kuka_reward	four staged reward terms, Sec. III-C	eight summed components, hand_delta_penalty multiplied by 0 with the comment currently disabled
physhoi_2023	high	wyhuai/PhysHOI 6095c605e2	physhoi/env/tasks/physhoi.py compute_humanoid_reward	object-rotation weights 0.1 and 0.01 for GRAB, Table 4	eor and eorv set to torch.zeros_like(ep), the computation commented out on the same lines
unidexgrasp_2023	high	PKU-EPIC/UniDexGrasp 36c9bfcf7c	dexgrasp_policy/dexgrasp/tasks/shadow_hand_grasp.py compute_hand_reward, goal_cond branch	seven named weights from 0.1 to 10, Table 7	a torch.where cascade on literals -0.5, -1.0, 0.9, 0.1, 0.2, none of the seven present
penspin_2024	medium	HaozhiQi/penspin 5035c52dc9	configs/task/AllegroHandHora.yaml forceScale	a disturbance force of 0.2 times object mass at probability 0.25, Table 8	forceScale: 0.0 and randomForceProbScalar: 0.0, the only shipped config naming either
pianomime_2024	high	sNiper-Qian/pianomime c4abefac8d	single_task/piano_with_shadow_hands_res.py _set_rewards	two weighted terms, 2/3 and 1/3, Table 3	five summed terms; _compute_energy_reward ends return 0 and _compute_fingering_reward return 0.0

8 rows. The repository and the commit are the ones `corpus/code_manifest.json` records, and the file is in the parsed copy at `code/md/<key>.md`. `what the paper prints` names the table or equation the value was read from. `confidence` is the row's own `mismatch_confidence` field, 7 high, 1 medium; `penspin_2024` is held below high pending a direct code read this survey has not made, and Appendix C prints the review note. No cell states a cause, and none is a claim about what the work's authors did: a reader with a browser settles every line of this table without asking anyone.

6. Bimanual dexterous manipulation

Two hands are not two instances of the same problem. They share one object, they can collide with each other, and the action space doubles, so neither the hardware result nor the policy carries over from one hand. Counting the work is the first difficulty, because the widest count, the 53 papers whose robot has two end effectors, is not a count of bimanual dexterous manipulation at all. Thirteen of those put no dexterous hand on the robot: five are parallel-jaw throughout (`aloha_act_2023`, `rdt1b_2024`, `egomimic_2024`, `h_rdt_2025`, `umi_2024`), four name no hand and run grippers on every embodiment (`pi0_2024`, `pi05_2025`, `pistar06_2025`, `diffusion_policy_2023`), two report every bimanual number on grippers and show five-fingered hands only qualitatively (`gemini_robotics_2025`, `gemini_robotics_15_2025`), and two name no end effector anywhere (`helix_2025`, `groot_n16_2025`). Three more run a gripper on one embodiment and a hand on another (`ace_teleop_2024`, `dp3_2024`, `open_television_2024`). The list of exclusions is in Appendix A, so the denominator can be audited rather than taken on trust.

: the method papers whose notes place a learned closed-loop controller on two multi-fingered hands. It is the 53 less those 16; less two static grasp-synthesis methods, `bimangrasp_2024` and `bidexgrasp_2026`; less three systems with no learned policy, `castro_sap_contact_2021`, `dexteleop0_2026` and `pang_global_planning_2022`; and less four papers where no learned policy holds both hands: `omnih2o_2024`, whose fingers are mapped open-loop from the Vision Pro and sit outside its 19-DoF policy; `okami_2024`, whose headline pipeline is open-loop retargeting with a learned policy only in a side experiment; `dexdeform_2023`, a skill model refined by trajectory optimization rather than a closed-loop controller; and `omnigrasp_2024`, a simulated human body with no bimanual task. `dexterous_handover_2025` never enters, because its entry records one hand. Benchmarks and datasets are outside the 28 by class, `bidexhands_2022`, `bench2dex_2026` and `robopianist_2023` being benchmarks and `rp1m_2024` and `humanoidgen_2025` datasets; they are quoted here as evidence and never counted. Every paper below runs two multi-fingered hands unless said otherwise.

6.1 Coupling, roles and collision

When both hands hold the same object, the object closes a kinematic loop between them. Neither hand can move without changing what the other must do. `bidexgrasp_2026` reports that the coupled objective "often yields imbalanced solutions, where one hand dominates stability while the other contributes marginally." `artigrasp_2023` reports simulation speed scaling "roughly quadratically with the number of contacts," and trains each hand alone before pairing them.

Role asymmetry is an assumption most of these systems make silently. `bidexhd_2024` states it outright: "we assume the robot to be right-handed by default, i.e., the left hand handles the target object and the right hand handles the tool," and `twisting_lids_2024` bakes the same split into its reward, putting reference contact keypoints on the bottle base for the left fingertips and on the lid for the right. The bias is in the data first: `taco_2024` recruited 14 right-handed subjects and measures the right hand moving consistently faster.

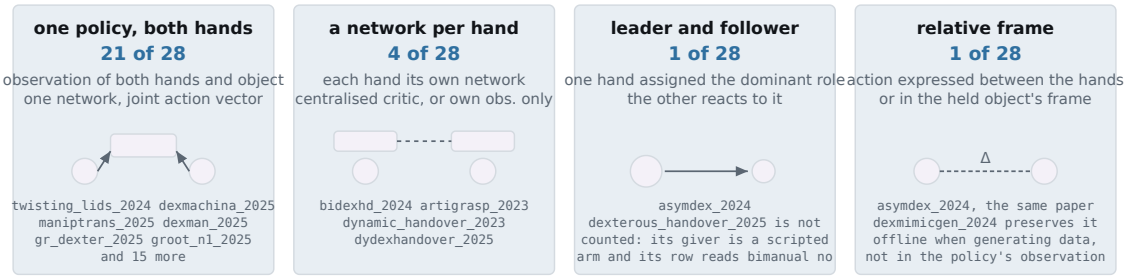
The arms collide, and the corpus handles this by construction rather than by control. `robopianist_2023` ships a forearm-forearm collision term that its own reward table never lists. Two papers name arm collision as unsolved. `dexmimicgen_2024` "does not explicitly handle collisions," and `dexman_2025` says its control parameterisation "overlooks full arm posture," which it needs to avoid them.

The action space doubles and the observation grows faster: a two-hand benchmark carries action vectors in the tens and observations in the hundreds of dimensions, and a recent generalist emits a chunk of arm joints, end-effector poses, hand joints and fingertips together. What the larger vector does not carry is any statement of who does what. The usual answer is one scalar reward summed over both sides, and the cost is a compounding failure rate: where success requires both hands to succeed, `maniptrans_2025`'s rate falls by about a third from single-hand sequences to bimanual ones.

6.2 Coordination architectures

Figure 5. Four ways to control two dexterous hands

Counts are the 28 corpus rows that put a learned closed-loop controller on two dexterous hands.
The panels are not a partition: asymdex_2024 is drawn in two of them, and the two rows whose notes never say which they use are in none.



The field is concentrated in the first panel. The two on the right are where the structure of the problem is actually used, and asymdex_2024 is the only corpus paper that puts either one inside a policy.

Figure 5 sets the four mechanisms side by side. Every one of the 28 carries exactly one architecture, read from its note: one policy over both hands, a network per hand, leader and follower, or a note that does not say. The fourth panel is not a fourth architecture but the relative frame, and it is the same single paper as the third.

Twenty-one of the 28 put one policy over both hands, which is three quarters of the set: the trackers and tracking-adjacent methods `dexmachina_2025`, `dexman_2025`, `maniptrans_2025`, `objdex_2024` and `humanplus_2024`; the reinforcement-learning tasks `dexpbt_2023`, `twisting_lids_2024`, `eureka_2023`, `pianomime_2024` and `humanoid_sim2real_recipe_2025`; the demonstration pipelines `bidex_teleop_2024`, `dexcap_2024`, `dexwild_2025`, `dexmimicgen_2024`, `hato_visuotactile_2024` and `humanoid_policy_human_policy_2025`; and the generalist policies `gr_dexter_2025`, `groot_n1_2025`, `dexora_2026`, `metis_2025` and `egoscale_2026`. In every one of them a single network takes a concatenated two-hand observation and emits a two-hand action. Two papers do not say which they are, `bunny_visionpro_2024` and `deximit_2026`, whose notes describe the rig and the data pipeline but never the policy's own decomposition. The concentration is not the outcome of a comparison that was won.

Four of the 28 give each hand its own network, and the two papers that compare the choice disagree. `bidexhands_2022`, a benchmark entry and so outside the 28, ships the MARL baselines and finds PPO over the full observation beats HAPPO and MAPPO "in most cases," with the gap narrowing on tasks that need both hands, because PPO "can use all observations" where MARL sees only part. `bidexhd_2024` concludes the opposite. Its independent PPO teachers score 74.59 percent stage-two tracking rate on trained tasks against 53.88 for a centralised policy over both observations. Both are simulation only, on different tasks and hands, so neither settles it. `artigrasp_2023` trains one PPO policy per hand, and `dynamic_handover_2023` and `dydexhandover_2025` use MAPPO with one agent per arm-hand system.

One paper assigns explicit leader and follower roles, and it is the same one that expresses its policy in a relative frame. `asymdex_2024` gives the dominant hand full finger and wrist control, restricts the facilitating hand to a base pose, and writes both the dominant hand and the object in a frame attached to the object the other hand holds, which halves the observation and the action. It is also the only paper here that ablates the two mechanisms separately, and neither works without the other: relative frames without asymmetry and asymmetry without relative frames both collapse to near zero where the full method succeeds most of the time. Its real rig pairs an Allegro with a cheaper hand because only one Allegro was available, so the roles are confounded with the hardware. Two other papers use a relative quantity without giving either hand a role, one appending the inter-hand pose to its observation and one applying a shared transform to both arms while generating data, which is the mechanism offline rather than in a policy.

One architecture carries three quarters of the field, and the choice behind it has been compared twice, with opposite outcomes, and ablated once. What would close it is a single handover task run on three architectures with the same hand and the same seeds, reporting giver and receiver returns separately rather than one shared number.

6.3 Benchmarks, datasets and the shared axes

Two purpose-built bimanual dexterous suites exist in the corpus, four years apart.

The two bimanual benchmarks in the corpus trade coverage against measurement, in opposite directions. `bidexhands_2022` is 20 tasks on two Shadow Hands in Isaac Gym, ordered by the infant age at which humans acquire the skill, at 2048 environments and a reported 30,000-plus FPS. Its measurement discipline is weaker than its coverage. It reports reward and normalized score, never a success rate. Its only success flag in code tests the object-to-goal distance against 3 cm, which ignores orientation and exists only in the four catching tasks. Any success rate later work attributes to Bi-DexHands comes from that flag or its own definition.

`bench2dex_2026` is the more instrumented of the two, running 26 long-horizon tasks across 12 arm-and-hand embodiments with thousands of teleoperated demonstrations in eight modalities, including a ray-cast tactile image that maps every hand geometry into one representation. Its headline result is the gap between conditions rather than the leaderboard: the matched leader takes about half the rollouts and falls to a fifth under combined perturbation, which compresses the field until the ranking no longer holds. Tasks are not crossed with embodiments, so no contrast isolates the hand, and the authors state that their hand and tactile models are not calibrated against matching physical hardware.

Ten papers in this corpus are bimanual datasets, five of them two-hand interaction capture, and their annotations are where the contact problem first appears. `arctic_2022` deliberately labels over-shooting vertices as in contact, so its ground truth records interpenetration as contact. `taco_2024` fits its sequences under an attraction loss and a penetration loss that the paper says trade against each other. `oakink2_2024` is the only one publishing a penetration figure for its own annotations, at a mean depth of a few millimetres. The two largest egocentric sets publish none at all (`hot3d_2024`, `gigahands_2024`). Those five feed most of the physics-based tracking work in this section, so an interpenetrating reference is where several of these pipelines start.

Table 7 carries the bimanual papers on the same columns as every other method here, and three of those columns are worth reading together.

Real trials. `asymdex_2024` reports 20 per task, `hato_visuotactile_2024` 10 per condition, `dexmimicgen_2024` 20 on a Fourier GR1 and `bidexgrasp_2026` 260 across 30 objects. `maniptrans_2025` reaches real Inspire hands by open-loop replay with no trial count and no success rate. Seven of the 28 have no real robot at all: `artigrasp_2023`, `bidexhd_2024`, `dexmachina_2025`, `dexman_2025`, `dexpbt_2023`, `dydexhandover_2025` and `pianomime_2024`. The two piano papers outside the denominator, `robopianist_2023` and `rp1m_2024`, are simulation-only as well.

Penetration. Most bimanual papers read "not addressed." `bimangrasp_2024` gates on it, failing any grasp whose total penetration across hand-object, self and inter-hand exceeds 1.5 mm, and reports that "penetration remains the primary cause of grasp failure." `bidexgrasp_2026` reports it as a number, 0.15 to 0.20 cm maximum depth for its own grasps. `deximit_2026` carries a named hand-hand penetration term in its grasp-synthesis objective. `dexmachina_2025` resolves it once, replaying retargeted joints against a fixed object "to eliminate object penetrations" and measuring nothing during rollout. `artigrasp_2023` declines the metric outright, because its baselines "include a physics simulation which exhibits no interpenetration."

Comparability. `bimangrasp_2024` reports 54.03 percent success in Isaac Gym at friction 3. `bidexgrasp_2026` re-runs the same grasps in MuJoCo at friction 0.6 and gets 26.80 percent, at 1.52 cm penetration depth. Neither number transfers, and section 5.3's re-implementation result is the same lesson on the training side.

6.4 Handover, and what transfers from one hand

Handover is the one bimanual task where the hands are unambiguously asymmetric, because one gives and one receives. All three corpus handover papers use a single shared reward across giver and receiver. None defines separate objectives for the two roles.

Whether the two hands optimize the same objective is usually left implicit. `dynamic_handover_2023` states one formula, r = r_dis + r_linvel + r_torque, and never says whether thrower and catcher receive different decompositions of it.

dydexhandover_2025 is explicit that they do not: "Both hands share aligned objectives, forming a fully cooperative relationship," with one weighted sum feeding both agents. Only the thrower gets an extra KL regulariser toward a policy pretrained on human throws, which constrains style and not role.

The third case is weaker still. In dexterous_handover_2025 only the receiver is learned. The giver is a UR5e with an Allegro hand that "holds the object without moving during the whole episode." There is one agent, one reward and no second policy, its entry in Table 7 accordingly records one hand, and it is one of the papers the 28 excludes. The 94 percent often attached to this paper needs its conditions. It is Total Success, which counts "Indetermination" cases where the simulator failed to resolve collisions and the object clipped through the giver's hand, on the short prism, in simulation, over 100 episodes, with no real robot in the paper.

Nothing in the handover literature measures contact quality. Contact appears only as a positive signal, a boolean per-phalange touch in dexterous_handover_2025 and a boolean contact reward in dydexhandover_2025. The two systems that do handover well sidestep the problem. hato_visuotactile_2024 reaches 10 of 10 on a real slippery handover by imitating teleoperation with fingertip touch sensing and no physics model. humanoid_sim2real_recipe_2025 stages its reward with a discrete variable switching which hand's fingertips are scored, and still calls handover its hardest task at 52.5 percent real success.

The training recipe transfers intact. PPO with thousands of parallel environments, an asymmetric critic on privileged state, domain randomisation and distillation into a vision policy is the same in twisting_lids_2024 and humanoid_sim2real_recipe_2025 as in section 5.2. Reward forms transfer literally, and the contact-goal reward r_contact = Σ 1/(1+α d) appears in both.

Multi-task learning does not transfer, and the bimanual benchmarks show it clearest. bidexhands_2022 finds multi-task PPO and ProMP fail on MT4, MT20, ML4 and ML20. robopianist_2023 finds F1 on the shared training song drops "from roughly 0.7 F1 for 1 song to almost 0 F1 for 16 songs," with no positive transfer "regardless of the size of the pre-training tasks."

Single-hand evaluation does not transfer either. Success criteria compound, because maniptrans_2025 fails a trajectory "if either hand fails to meet these conditions." And object-centric metrics cannot say which hand caused the failure. bidexhd_2024 scores the steps where both objects track their references, and dexmachina_2025 averages ADD across object parts before an AUC. Both say whether the task worked and nothing about which hand lost it.

One failure mode has no single-hand counterpart. Two hands can penetrate each other, and the corpus almost never looks. Three papers carry an inter-hand penetration term, and all three are static grasp synthesis: bimangrasp_2024, bidexgrasp_2026 and deximit_2026. Not one learned bimanual controller in the corpus measures or penalizes hand-hand penetration during a rollout. pianomime_2024 ships a flag that turns inter-hand collision off in the physics.

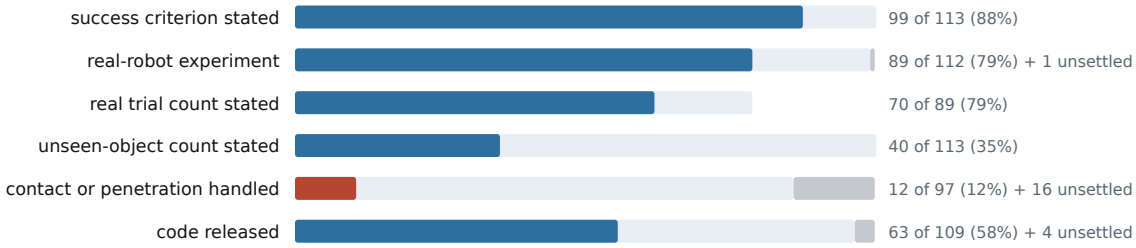
7. Evaluation, and a protocol for comparison

7.1 Reporting practice

Most of this literature does test on hardware, which is the check that matters and the one a simulator cannot substitute for: 89 of the 113 papers report a real-robot experiment. What they report about those experiments is much less even. Figure 6 draws six reporting axes, each against the denominator that belongs to it, and two of the six are the ones a reader most needs. Without a count of unseen test objects there is no denominator for a claim about generalisation; without some statement about contact there is no way to tell whether the hand passed through the object. Those two are the axes the field states least often. Every share in the figure is a lower bound rather than a rate, for reasons Appendix A gives.

Figure 6. What the method literature reports

Share of the 113 surveyed method papers whose row records each item, each against its own denominator: the pale track is the rows the item can apply to, the grey tail the rows whose note did not settle it. A bar is evidence the item was stated, not that the work did it well.



Red marks an item fewer than a third of the rows it applies to state. Every bar is a floor: a hand audit of the nulls recovered 15 trial counts, 19 criteria and 7 object counts (§7.1).

The plainest gap is the trial count. Nineteen of the papers that ran a real robot never say how many times they ran it, and a percentage with no denominator cannot be given an interval, so it cannot be compared with anything.

Where a count is stated it is small, and it is not always the same quantity. Some papers report a per-cell count, a fixed number of trials on each task or each condition; others report a grand total over every cell; a few cannot be assigned to either basis. Table 9 keeps the two distributions apart, because pooling them produces a figure that describes nothing: `hora_2022`'s 240 trials and `visual_dexterity_2022`'s 20 are experiments of comparable size recorded on different bases.

The per-cell count is the one a confidence interval attaches to, and at the sizes this field uses the interval is wider than the differences being claimed. At twenty trials per cell, two methods twenty points apart in success rate are not statistically distinguishable; Appendix E does the arithmetic, with the intervals and the two-proportion test. The modal cell in this corpus is ten trials, which is narrower still. This is the single most consequential fact about evaluation in dexterous manipulation, and it is why section 7.3 proposes a protocol rather than a league table.

Reporting has improved a little and the experiments have not. About two thirds of recent papers state a trial count, against three fifths of earlier ones, and the counts themselves have not moved. A field can get better at disclosing a number without the number itself getting any better.

The denominators also sit on different hardware. The papers that name their own hand give 79 distinct descriptions of it, unnormalised, so that figure counts strings and not designs; Figure 2 gives the distribution. A success rate on a 16-degree-of-freedom Allegro and a success rate on a 6-actuator Inspire hand are not measurements of the same thing.

Nor are the criteria the same. `dexverse_2026` counts a pick a success when the cube is "lifted at least 0.20 m above its resetting height". `bench2dex_2026` requires its terminal predicate to hold for a continuous dwell time, to reject transient contacts. `colosseum_2024` counts an episode successful "if the model completes the task fully". `dextrack_2025` reports every success rate as a pair under two threshold sets, which on one dataset gives 47 and 65 percent for the same rollouts. Among the papers that state no criterion at all, most have no success predicate to state: they report radians rotated or time-to-

fall and never define a success, which is a fact about the paper rather than a gap in this survey. Where a criterion is stated it is usually a rubric, a staged partial credit or a deferral to a benchmark's own definition rather than a threshold, and a rubric is reproducible inside a lab and not across two.

Seven quantities dissociate in the published data, which is why they have to be reported separately rather than collapsed into one number.

Task success is the first, and binary success discards the difference between a near-miss and inaction. `beyond_binary_success_2026` puts it plainly: "a policy that completes 90% of the task is clearly better than a policy that is frozen the whole time, yet their success rates would be identically 0%". In `kress_gazit_policy_eval_2024` one policy scores 17 percent overall on a pancake task while picking up the spatula and flipping the pancake in every attempt.

Robustness to perturbation has been measured, though not yet on a hand. `colosseum_2024` reports a success drop of a third to a half under single perturbation factors and more than three quarters under all of them together. In `bench2dex_2026` the leader of the matched condition is not the leader under combined shift. The ranking at the anchor is not the ranking under shift.

Generalisation to unseen objects is stated by 40 papers, and half of those test on ten objects or fewer.

Physical plausibility of the contact is the least reported of the six, at 12 percent of the papers whose notes settle the question. Section 7.2 takes it up.

Sample and wall-clock cost is reported by about a third of the corpus as an environment count and by fewer as an episode count. `robopianist_2023` is the exception that shows what disclosure looks like, giving samples per song and hours per run on named hardware.

Real-robot transfer is the one that undoes the rest, because simulated rank order is not real rank order. `autoeval_2025` scores one policy at 6 of 50 in simulation and 47 of 50 on the real arm for the same task. `suresim_2025` states the limit directly: "the simulation-to-real gap precludes rigorous statistical inferences about real-world outcomes from simulation results alone".

Reproducibility is the last, and releasing code is not the same as releasing something that settles a question. Eight papers state a training objective in their own repository that differs from the one in the paper, which section 5.6 sets out and Table 11 names file by file. That is the figure this section uses.

7.2 Physical plausibility

One quantity is specific to hands and is almost never reported: how far the hand passes into the object it is holding. Outside closed-loop control it has been a standard comparative column for years. Grasp synthesisers fail a grasp that penetrates past a stated depth `bimangrasp_2024`, retargeting methods report a maximum depth against a baseline `toporetarget_2026`, and hand-object datasets score a split on depth, intersection volume and simulation displacement `oakink_2022`. Among learned closed-loop policies, almost nobody reports it, and nobody reports it measured in the physics the policy ran in.

D-Grasp `dgrasp_2022` is the instructive case, because it does report a volume for a pose its own policy reached, and the way it does so shows where the quantity goes. Inside the simulator the paper reports zero penetration, since the meshes there are decimated for speed; the figure it prints instead comes from re-measuring a single frame against the original hand mesh and the full-resolution object mesh, with, in the paper's words, "no physical simulation involved". So the number a reader wants, how deep the hand sits in the object in the physics that trained the policy, is zero by construction, and the number reported is a geometric re-measurement on geometry the simulator never used.

The same paper supplies the reason to want the measurement. Its appendix observes that a deeply penetrating reference grasp can raise a success rate rather than lower it, because "the objects can become entangled within the hand mesh and will therefore not be able to fall down", and concludes that "the success rate metric should always be interpreted in combination with the other metrics". A success rate with no plausibility number beside it can be read two ways.

There is a reason the gap persists, and it is not difficulty. A quantity a policy optimises cannot also judge it, because the policy learns the measure rather than the property the measure stands for. `physhoi_2023` shows the failure in the clean direction: its imitation reward is maximised by not touching the object at all, because contact perturbs the reference trajectory. `dextrack_2025` shows it from the other side, defining a maximum penetration depth, applying it only to its input references, and then reporting tolerance of "severe hand-object penetrations" as a result. The remedy is to measure with something the policy never optimised.

Definitions are not shared either, which is the other half of why published figures do not compare. `grab_2020` reports an average penetration in millimetres without saying whether it is a maximum, a mean or a median; `oakink_2022` supplies the fullest published vocabulary, separating penetration depth, solid intersection volume and simulation displacement. A number without its definition is not a number a second paper can be held to.

What would close the gap is one line in a results table: the maximum and mean penetration depth over the evaluation rollouts, on a dense surface sample, computed by code that never entered the reward or the termination rule. That is the plausibility row of the protocol in Table 8. What this survey supplies there is a measurement and a count; it does not supply a threshold, and section 7.4 says where the one in the table is borrowed from and why it does not bind.

7.3 Statistical practice, and a proposed protocol

A protocol already exists for evaluating learned robot policies, and what it does not supply is the part a reader needs. `kress_gazit_policy_eval_2024` prescribes process rather than numbers: write the success criteria before the run, have someone other than their author score the runs, match initial conditions with image overlays, interleave the policies blind within one session, report counts rather than percentages, and use a posterior over the success probability instead of a point estimate. All of that is right and none of it fixes the sample size. It prescribes no minimum trial count and no interval width, and its own worked example shows why one is needed: at twenty evaluations per policy, an eighteen-point difference in success rate still leaves better than a one-in-ten chance that the worse policy is actually the better one. Multiply every count in that example by ten and the same two rates separate.

One study does put numbers on it, without saying where they come from. `lbm_careful_examination_2025` supplies them by fiat rather than derivation: 50 rollouts per task per policy per condition on hardware, 200 in simulation, blind, with randomised policy order inside per-initial-condition bundles. It replaces confidence intervals with Beta-posterior violins and corrects its pairwise tests under Bonferroni, "unless otherwise noted". Its own warning is the strongest sentence in this literature: "there is significant risk that many robotics papers are measuring statistical noise due to insufficient statistical power".

The rest fall short of their own advice. `roboarena_2025` runs hundreds of double-blind pairwise comparisons across seven institutions and thousands of rollouts, and reports neither confidence intervals nor per-policy trial counts. `colosseum_2024` evaluates hundreds of test sets with "one training seed and one evaluation seed" per baseline and gives no intervals in simulation. `autoeval_2025` calls ±10 percent "the natural variance of robot evaluations" without giving the formula behind the intervals it plots.

Two papers do give usable numbers. `suresim_2025` pairs real and simulated trials and de-biases the simulation with a rectifier, which saves up to a quarter of the hardware trials when the paired correlation is high and nothing at all when it is near zero; its decision rule is exact, that combining helps only when the rectifier variance is below the variance of the real evaluations. `beyond_binary_success_2026` gives the larger saving by scoring progress rather than success: a sequential test on graded scores cuts hardware trials by about 45 percent, and a gap that needs 80 binary trials to establish is established in 18 when the score is continuous. That is the strongest argument in the literature for grading partial progress.

Every count below is printed by `tools/make_eval_tables.py` in its derivation mode, and each axis is derived for the statistic that axis actually reports: a single rate takes a Wilson half-width, a matched comparison takes McNemar, a ratio takes the standard error of the log ratio, a correlation takes the Fisher-z interval.

Appendix E derives the counts in Table 8, and they come to this. Reporting a single rate honestly takes about 100 real rollouts per cell, which is where the worst-case interval half-width falls to ten points. Comparing two policies is a harder question: because the design is paired, the count depends on how often the two disagree rather than on the two rates alone, and separating a twenty-point difference at conventional power takes 57 matched pairs per arm, which is fewer rollouts than two independent arms would need. Simulated cells are cheap enough to run in the hundreds.

The other axes follow the same logic and Appendix E gives their counts. Perturbation axes are screened cheaply, which ranks them but cannot establish that any one of them hurt, and certified at the same cost as a single rate. Unseen-object claims resample the object rather than the trial, so the count that matters is the number of objects and not the number of rollouts. The transfer axis needs real and simulated trials paired, and separating two correlations takes several hundred pairs, which is why it belongs in a benchmark rather than in a paper. Every count there is a cap rather than a bill: a sequential test on a graded score can decide in a fraction of the trials, and a cell that stops early reports a confidence sequence rather than a fixed-sample interval, never both.

Table 8. The proposed evaluation protocol

axis	what is measured	how	minimum trial count	reported alongside it	why
Task success	fraction of episodes meeting a criterion, and a graded rubric score in [0,1] over equally weighted milestones	criterion written before the run by the designer and scored by someone else; terminal predicate held for a 0.5 s dwell; initial conditions matched by image overlay; policies interleaved blind in one session	100 real and 200 sim per policy per task per condition for an absolute rate at a 10-point interval; a comparison alone needs 57 matched pairs per arm (McNemar, 50 against 70 percent, discordance 0.3)	raw counts, the criterion verbatim, the rubric, the initial-condition protocol, and an interval whose kind is named: Wilson for a cell run to a fixed n, an anytime-valid confidence sequence (WSR, as in `beyond_binary_success_2026`) for a cell stopped early by a sequential test, because a Wilson interval computed at a data-dependent stopping time is not a 95 percent interval. Intervals are marginal, not simultaneous: a paper comparing k policies corrects its k(k−1)/2 pairwise tests to a global 95 percent level, as `lbm_careful_examination_2025` does, or says it has not	a rate without a denominator and an interval cannot be compared; the rubric separates a near-miss from inaction; the design is paired, so the count that governs it is paired
Robustness	success under each perturbation axis as an absolute rate; the ratio to the unperturbed anchor is reported descriptively and is not tested at the screening count	axes that must not change the action and axes that must are scored separately; ranges stated; anchor replayed exactly	40 per axis per policy in sim to rank the axes; 101 per axis and 101 at the anchor before any claim that a named axis hurt	both absolute rates with their Wilson intervals, the ratio without an interval unless the certifying count was run, the axis ranges, and which split each axis is in	the ranking at the anchor is not the ranking under shift; at 40 per arm a fall from 0.50 to 0.30 is a ratio of 0.60 with a 95 percent interval of 0.34 to 1.06, which contains 1
Unseen objects	mean over held-out objects of per-object success	objects drawn from a stated distribution disjoint from training; bootstrap over objects, not trials	20 objects at 5 trials for a screening claim; 93 objects for a 10-point claim	the object list, per-object rates, the bootstrap interval, the object-count caveat, and the within-object trial count, since the 93 is a Wilson width that treats one object as one Bernoulli draw	the resampling unit is the object, so trial counts overstate the precision
Physical plausibility	maximum and mean hand-object penetration depth per rollout, and the fraction of frames above 2 mm	dense surface sample against the object mesh or SDF, computed on the policy's own rollouts by code that never entered the reward or the termination rule	100 rollouts, simulation only	interval over rollouts, sample density, threshold, solver depenetration settings, and at least one rendered rollout	a measure the policy optimised is not evidence about the policy
Sample and wall-clock cost	environment steps and wall-clock to the checkpoint that produced the headline number	counted to that checkpoint, not to the end of training; GPU model and count stated	3 training seeds	the range over seeds, and the statement that 3 seeds is a range and not an interval	training variance is not rollout variance and the two are routinely conflated
Real-robot transfer	paired real and simulated outcomes on matched initial conditions, their correlation, and the rectifier variance against the variance of the real evaluation	the same 100 initial conditions run in both; correlation on the pairs; variance ratio bootstrapped over the same pairs	the 100 real trials paired to 100 sim, which fixes the correlation to about ±0.10; about 457 pairs for a claim that separates two correlations 0.1 apart	the correlation with its Fisher-z interval, the variance ratio with its bootstrap interval, and which of the two decisions the count supports	a simulator earns a real-world claim only through its measured paired correlation; at 100 pairs $\hat{\rho} = 0.70$ carries 0.58 to 0.79, enough to say a simulator tracks reality at all and not enough to separate `suresim_2025`'s 0.70 regime from its 0.59 one
Reproducibility	code, checkpoints, and the exact config that produced the headline run	diff the paper's stated objective against the released config and name the file and line of every disagreement	not a trial count	the named file and line for each disagreement, or an explicit statement that none was found	16 of the 62 code-releasing method rows carry a true contradiction between the paper's stated objective and the released code, and 38 carry a recorded disagreement of any kind (§5.8)

7.4 Feasibility and limitations

The proposed protocol has not been run. In the twelve-method comparison assembled for this survey, no method supplied all the measurements needed for a complete protocol row. DeXtreme `dextreme_2022`, for example, reports a success criterion and a confidence interval, but uses ten trials. This limits retrospective comparison; it does not imply that the reported experiments failed their own stated objectives.

The bill comes first, because it decides whether any of this is adoptable. Comparing two policies on three tasks under the protocol in Table 8 comes to 942 real rollouts, which at a minute each including the reset is about 16 hours of robot time before failed resets, repairs and scoring. Appendix E itemises it. Pairing the anchor set removes about a fifth of the bill rather than the four fifths the pair counts suggest, because a pair is still two rollouts. For scale, `autoeval_2025` ran roughly 850 episodes in a day on a parallel-jaw arm with three human interventions, pausing regularly once the motors overheated; a tendon-driven multi-finger hand is more fragile, so those 16 hours are most of a week of calendar time.

That bill is large but not unprecedented. Evaluating one generalist policy against its baselines has taken more than 2500 rollouts and more than 100 hours of human labour across three institutions (`autoeval_2025`), and a recent careful examination analysed close to two thousand real rollouts across nine hardware stations. What is unprecedented is paying that cost for a single dexterous-hand paper, where the modal cell is ten trials.

Four things would have to change. Reviewers would have to reward 57 matched trials on one task over 20 unmatched trials on five, and nothing in the corpus suggests that is happening. The loop would have to be automated, and `autoeval_2025` shows it is buildable for a gripper at one to three hours of setup, while stating that it supports binary success only and no robustness axes. Somebody would have to run the penetration measure on their own rollouts, which is a choice rather than a capability: section 4.2 shows the depth is computable from the poses and the meshes in a few lines of code, and the field's own benchmark repository already ships a task that does it every step. The comparison would also have to be sequential, because the savings in `beyond_binary_success_2026` are the only reason the protocol's counts are caps rather than costs.

Four limits apply to the proposal itself. This survey re-ran no method, so every count in Table 8 is derived from an interval width, a power calculation or another paper's measurement rather than from an experiment here. The counts are worst case, so a method with a high success rate needs fewer trials for the same interval, and the paired count rests on an assumed discordance rate that no paper in this corpus reports. The perturbation axes are borrowed from benchmarks that run parallel-jaw grippers on rigid objects (`colosseum_2024`, `simpler_2024`), where the measured sensitivity to physical parameters is smaller than a multi-finger hand should be expected to show.

That sensitivity should grow with multi-finger contact, and no work in this corpus measures it. The 2 mm penetration threshold is taken from `toporetarget_2026` with no independent justification, and the captured human grasps in `grab_2020` sit above it at 3.25 mm, which makes 2 mm a simulator convention rather than a physical bound. So the three things this survey adds to the reference-versus-rollout axis are a count, a measurement method and a protocol slot for them, and a threshold is not among them. It is borrowed from one paper and reported as borrowed, and it will stay a convention until somebody measures penetration on rollouts across hands and engines and finds a value that separates behavior a physicist would accept from behavior they would not.

Seven papers in this corpus are evaluation protocols, a small denominator, and not one of them uses a dexterous hand: `suresim_2025` runs a parallel-jaw gripper and the other six state no hand. Everything proposed above is therefore assembled from work on grippers and on whole-arm tasks. What would close that, and close it cheaply, is to run the Table 8 protocol once, on one in-hand reorientation task with one 16-DoF hand, and release the rollouts and per-trial outcomes for independent analysis.

8. Conclusion: the seven claims, and what to do next

This corpus describes a field that has converged. Most of it trains with reinforcement learning on privileged state in a GPU simulator and distills the result into a vision-only student; a second branch, growing faster, learns from teleoperated demonstration instead; and almost all of it runs on four hands. What the field cannot currently do is compare its own results, and most of the seven claims below follow from that. Each is one sentence, with the section that carries its evidence and the experiment that would close it. Nothing is argued here: the denominator, the evidence and the prescription sit in the section named, so that a finding and its consequence are read together and stated once.

1. Eight hands that can be bought today or built from published designs take zero papers between them, while the corpus's own experiments concentrate on four designs (section 3.4, Figure 2).
2. Human data does not port across hands and the map is usually unstated: 33 of the 53 method papers that use human data never say how the human motion reached the robot hand (section 5.3, Table 6).
3. Twenty-one of the 28 papers that put a learned closed-loop controller on two multi-fingered hands run one policy over a concatenated two-hand observation, a choice that has been compared twice with opposite outcomes and ablated once (section 6.2, Figure 5).
4. The generalist and vision-language-action policies that do evaluate on a multi-fingered hand run it at a median of 6 actuated degrees of freedom, against 16 across the reinforcement-learning papers that state a count, and the two papers that reach the larger band on paper never say whether the hand was in the evaluation (section 5.4).
5. The evaluation-methodology literature the protocol of section 7.3 is assembled from contains no dexterous hand at all: of its seven papers in this corpus, one runs a parallel-jaw gripper and the other six state no hand (section 7.4, Table 8).
6. Eight of the 63 papers that released parseable code state, in a named file at a named commit, something other than the value their paper prints. Table 11 gives each one as a repository, a commit, a file and two values, and all of their authors were written to before this was posted: section 5.6 says that beside the finding, with the letters in `outreach/` and the route by which a disputed case is corrected (section 5.6 and Appendix C).
7. Penetration is almost never measured where it would matter. No paper here reports how far its hand sank into the object, measured in the physics its own policy ran in. The one that does report a volume for a pose its own policy reached measures it outside the simulator, on meshes the simulator never used, and reports zero inside it. On the reference-versus-rollout split that frames this survey, every paper that handles penetration at all is still measuring a reference (section 7.2, and the contact-handling bar of Figure 6).

Six of the seven are gaps in the literature. The sixth is a result about publishing practice, and this survey's own corrections to it are printed beside it in section 5.6 rather than kept in the repository.

Three of those claims carry a case worth remembering. Hardware and software have come apart on a narrower claim than the hand count first suggests: 11 of the 19 hands that take no paper are neither sold nor open and take none for that reason, which is why the first claim is eight hands and not nineteen. A reward table in a paper is a claim about a document rather than about a run, and `physhoi_2023` is the instance to keep in mind, since the term its table weights at 0.1 is set to zero in the file at the commit this survey fetched and the success criterion the paper reports is itself position-only. And the obstacle to measuring contact is not the engines: IsaacGymEnvs ships a task that computes a per-environment maximum interpenetration depth and gates the policy update on a one-millimetre threshold, so the tooling sits in the field's own benchmark repository while the number goes unreported.

What this survey cannot establish is which method is better than which. It re-runs nothing, and Section 7 argues that the published numbers do not compare. Six works are cited by metadata only, and a seventh, Ma and Dollar 2011, is on disk but unread; no claim rests on any of them. Every coverage figure here is a floor rather than a rate, for the reason Appendix A sets out.

Three things to do next week, cheapest first.

If you are publishing, generate the reward table from the config that trained the reported run, print it, and cite the commit. State the trial count, state the success predicate, and release the per-trial outcomes. None of that needs a GPU.

If you are running experiments, measure penetration depth over your evaluation rollouts on a dense surface sample, and never let that measure become a reward. `toporetarget_2026` shows what the number looks like when it is reported carefully, and what the widely used retargeters look like when it is not.

If you are choosing hardware, the corpus names four hands and no more: the Allegro, the Shadow, the Inspire RH56 family and LEAP, in that order, with the Allegro alone carrying 35 of the 113 papers. Every other hand in Tables 2 and 3 carries eight papers or fewer. An announced hand has no URDF, no datasheet that can be checked, and no paper in this corpus that used it.

Appendix A. Method

The survey is written from a corpus that lives on disk. No claim about a paper is made from memory. Each claim traces to a note, each note to a parsed markdown file, and each markdown file to a PDF or repository with a recorded hash or commit.

Selection

Six topic-specific bibliographies were assembled in parallel (simulators and physics, hands and vendors, reinforcement learning, imitation and human data, bimanual, benchmarks and evaluation), seeded with the canonical works in each area and extended by search up to 2026-09-17. Every arXiv identifier was checked by fetching the abstract page and matching the title, and entries that could not be checked that way are marked. The six lists were merged with deduplication on arXiv identifier and normalized title, giving 222 entries.

One of those 222 was added after the search. `dgrasp_2022` entered the corpus on 2026-10-06, three weeks after the cutoff, and it turned out to be the clearest instance of the finding in section 7.2. It was not hard to find: a corpus paper's own note cites it as that paper's baseline, and the search still did not surface it. A search that missed the work most relevant to one of this survey's two findings has probably missed others, and that is the plainest statement available of what this corpus is worth as a sample.

Seven further bibliography entries are not part of that corpus and are not counted anywhere in this survey. They are the prior audits and case studies section 1 positions this survey against, they carry the topic `related` in `corpus/bib_related.json`, and none of them carries a structured row. No source for them was parsed either, so each is quoted only from its abstract and its stated method.

Acquisition

PDFs were downloaded from arXiv or, for work without a preprint, from the publisher or vendor page recorded in the bibliography. Vendor pages for hands without any paper were fetched as HTML and converted to markdown, so that a specification quoted in this survey is quoted from a stored copy of the page and dated. Repositories were shallow-cloned and parsed, then deleted. What remains is one markdown per repository holding the README, a pruned file tree, the task and reward configuration files, and the bodies of reward and observation functions.

Parsing

PDFs were converted with pymupdf4llm, falling back to raw text extraction when the layout parse returned too little. The corpus manifest records, per entry, the source URL, page count, sha256 of the PDF and the converter used. Two consequences matter for reading this survey. Equations rendered as images do not survive conversion, so where a reward weight exists only inside a figure it is recorded as unreadable rather than guessed. Tables are flattened, so a number taken from a table is quoted with the table it came from and, where the flattening is ambiguous, the ambiguity is stated.

Notes Each entry was read into a structured note under a fixed template: embodiment,

learning method, objective, contact handling, evaluation, reproducibility, stated limitations, and quotable claims. Notes were written only from the parsed files. Where the source is silent the note says "not stated", and nothing was inferred from the reviewer's prior knowledge of the work. For method papers the reward or loss was quoted from the paper and, separately, from the released code, so that disagreements between the two are visible rather than smoothed over. Those disagreements turned out to be common enough to become a finding in their own right. The reading into notes was done with language-model assistance, from the parsed files only. The template's conventions are the safeguard against what that can get wrong: a quotation rather than a paraphrase, "not stated" rather than an inference, and a named source beside every value, so a note can be checked against the file it came from. Four fields carrying a headline claim were re-read against their sources afterwards, the trial counts, the success criteria, the unseen-object counts and the penetration field, and all four audits are below; the coverage statistics are reported as floors for the same reason.

Sources that could not be obtained

Six works are cited by metadata only and no claim in this survey rests on their contents. Five are behind publisher paywalls with no author-hosted copy found on 2026-09-18: Okamura et al. 2000, Bicchi 2000 (IEEE T-RO), Piazza et al. 2019, Roa and Suarez 2015, and Butterfass et al. 2001 on DLR-Hand II. The sixth is Hwangbo et al. 2018 on RaiSim, whose hosted PDF returned no body. The freely circulating PDF often taken for Bicchi 2000 is a different work, a book chapter, and is listed separately.

Ma and Dollar 2011 is a seventh case with a different cause. The fetch that failed when its note was written succeeded on 2026-09-18, so a seven-page PDF and its hash are in the manifest, but no note has been read from it. It is cited by metadata only for that reason and not because the source is unavailable.

Three of the 222 bibliography entries carry no structured row. Two are the paywalled Bicchi 2000 and DLR-Hand II entries. The third is `bicchi_grasping_chapter_2001`, which was read into a note and quoted throughout but is a book chapter rather than a work with an embodiment, a method or a result to record in a row.

The reference list of the typeset edition is shorter than the corpus, and the two numbers are different quantities. It prints 209 entries: the 202 corpus entries that some sentence, table or figure of this paper cites, plus the 7 prior-work entries from outside the corpus. The other 20 corpus entries carry a row and a note and are counted in every statistic here, but no passage in the paper names them, so they have nothing to be cited from and do not appear in the list. This edition cites by key rather than by number and prints no list, so the place to count all 222 is `corpus/bib.json`. A reader counting the typeset reference list should get 209, and a reader counting the corpus should get 222.

What this method cannot do

Mention counts over the corpus are counts of mentions, not of use: a related-work sentence counts the same as an experiment. Vendor specifications are manufacturer claims and are labeled as such throughout, and where a page has since gone offline the note says so. The corpus is large but not exhaustive, and selection by search favours work that is indexed, in English, and posted as a preprint. It also favours recent work. Of the 222 bibliography entries, 138 are dated 2024 or later and 16 predate 2018, so this is a corpus of the learned era and any claim here about a trend over time is a claim about 2022 onward.

Every coverage statistic in this survey measures what this survey's extraction captured, not what the literature reported. A structured row holds a scalar. A paper that reports a per-task count, a rubric, or a total spread across several tables produces a null, and a null is then counted as silence. The bias runs one way. Every miss converts a reporting paper into a silent one, so the field is made to look worse at reporting than it is. The size of the effect was measured on the statistic the survey leads with. Of the 34 method rows that had a real robot and no recorded trial count, 15 carried a count in plain text in their own note, dropped because the paper reports it per task and the field takes a single integer. Those 15 have since been re-extracted, which moved the stated-trial-count row from 55 to 70. The success criterion and the unseen-object count were audited the same way, rising from 79 to 98 and from 32 to 39. Both totals are one higher in the tables of this paper, because the row added after that audit states each of them. What those 19 recovered criteria say is mostly not a threshold: eleven score by

rubric or staged partial credit, five judge binary completion against a task description by eye, two defer to a benchmark's own definition, and exactly one, `pistar06_2025`, states a verbatim numeric threshold. The audit counted only cases where the note itself carried the number, so a count the note also missed is still uncounted, and 19 trial counts, 8 unseen-object evaluations whose object count the note never gives, and 4 criteria remain genuinely unsettled. Fifteen of the recovered trial counts are named in the rows: `pi0_2024` at ten trials per task, `rdt1b_2024` at 139 across seven tasks, `umi_2024` at 260, `pistar06_2025` at 750, `gemini_robotics_2025` at twenty per task, and ten more, which is 44 percent of the audited nulls and moved the "never says" figure from 34 of 89 down to 19. Read every coverage statistic in this survey as a floor rather than as a rate, and read the bars in Figure 6 the same way.

The penetration null, sampled. A null is worth what the search behind it is worth, and this survey's extraction under-counted every other field it was audited against, by twenty to forty-five percent. The 85 method rows whose contact-handling field records that the work does not address penetration were therefore sampled: 25 of the 85, drawn at random with a fixed seed, each read again in its own parsed source and its released repository rather than in the note the field was written from, since the note is the artifact under suspicion. Not one of the 25 reports a measurement of penetration depth, intersection volume or physical plausibility on its own rollouts. Zero recoveries in 25 bounds the rows that could be hiding one at 8 of the 85 at 95 percent confidence, so the twelve is a floor and twenty a ceiling; had this field under-counted at even the mildest rate the other audits found, a sample of 25 would have missed every recoverable row with probability 0.001. The nearest miss is worth naming, because it is the one a reader might count differently: `graspxl_2024` puts hand-object interpenetration to 35 human raters as one of four dimensions of a single realism score, which is a judgment of its own rollouts rather than a measurement of one, and counting it would give one recovery in 25 and a bound of 13 rows. Three near misses recurred across the sample and none of them is a measurement of a rollout: self-collision avoidance in a retargeter, a binary self-collision penalty in a released reward, and an engine's de-penetration velocity left at its default. The draw, the seed and the per-row results are in `reviews/penetration_audit.md`, and the two fields that remain unaudited are named above.

The penetration field, audited the same way. The same mechanism reaches the penetration, code-release and failure-mode fields. The penetration field has since been audited by hand, because this survey's second finding is a null over it and a null is worth what the search behind it is worth. The population is the 85 method rows whose contact-handling field records that the work does not address penetration. It excludes the 16 nulls, which are already counted as unsettled rather than as silence, so recovering one would not move the finding. Twenty-five of the 85, 29 percent, were drawn with `random.Random(20260919).sample` over the sorted population, so the draw is fixed and can be redrawn. Each sampled row was read again in `papers/md/`, in its OCR recovery where one exists, and in `code/md/` where a repository was parsed, but not in `papers/notes/`, because the note is the artifact under suspicion: a field is wrong exactly when the source says something the note did not carry. A regular expression over the whole source collected every occurrence of penetration, interpenetration, intersection, intersection volume, solid intersection, simulation displacement, contact consistency, physical plausibility, signed distance and contact depth, and of the words that are mistaken for them, and every hit was read in its context. A row counted as a recovery if its own source reported a measurement of any of those quantities on that method's own rollouts. A statement that penetration occurs, a citation whose title says physically plausible, a collision-avoidance constraint on a reference trajectory and a solver setting did not count, and the three recurring near misses of that kind are listed in section 7.2. Recoveries: none of the 25, against the 20 to 45 percent the three earlier audits recovered on the fields above. The one-sided 95 percent bound, computed hypergeometrically over the finite population, is therefore 8 of the 85, and section 7.2 states it beside the claim and names the one borderline case that a reader might count differently. Why this field held where the others did not is visible in the per-row results: the recovered trial counts and criteria were numbers present in the source and dropped by a field shaped to hold a scalar, whereas a penetration number is absent from the source altogether. The earlier audits measured a defect in this schema; this one looked for an absence in the literature. `reviews/penetration_audit.md` holds the seed, the sample and a result per row beside the sentence it rests on, and `tools/audit_penetration.py` redraws the sample and re-runs the sweep. The code-release and failure-mode fields have still not been audited this way, and their counts stay floors.

Appendix B. The full hand table with its sources

Table 2 and Table 3 split these rows by whether the hand can be bought, and truncate their cells to stay readable. This appendix prints every hand row once and at full width. Nothing here is a measurement by this survey. Every specification is a manufacturer claim, a figure in a paper, or a figure in a press article, and the `source quality` column in B.2 says which. Cells reproduce the text stored in the row, with the source's own punctuation and wording.

B.1 Specifications as the source states them

hand	maker	DoF	act. DoF	actuation	weight g	tip force N	payload	tactile	control rate	price USD	BOM USD	open HW	licence	sim model	status
agibot_omnihand_2025															
onex_neo_hand_2026	1X Technologies	25	25	quasi-direct-drive tendons via the '1X Tendon Drive' at low gear ratios (~5:1-15:1); motors in the forearm, tendons pulled through the wrist		45		high-resolution tactile sensing across fingertips and finger surfaces (normal force, contact location, shear); sensor count/technology not stated	not stated			no			announced
leap_hand_v2_adv_2025	Kenneth Shaw, Deepak Pathak, Carnegie Mellon University	21	17	tendon (PIP/DIP coupled by a single tendon per finger); motor type not stated				none mentioned	not stated	3000	3000	yes		URDF, loaded into PyBullet by the released teleoperation node	announced
paxini_dexh13_2024	PaXini Tech							1,140 ITPU multidimensional tactile processing units (press-release claim; unclear if PX-6AX-GEN3 sensor family)							announced
clone_robotics_hand_2024	Clone Robotics	27		hydraulic artificial muscle ('Myofiber'; small water-filled tubes that contract when pressurized), carbon-fiber bones and ligament-style tethers, driven by a 500 W water pump and 36 electro-hydraulic valves; each Myofiber generates up to 1 kg grip force, survived 650,000 test cycles	<910			pressure pads in the palm detect grip firmness (count not given); 70 inertial sensors track angle/speed (proprioception, not tactile)				no			internal-only
sanctuary_phoenix_hand_2024	Sanctuary AI (Sanctuary Cognitive	21		hydraulic, 'unique miniaturized				not in the press release text; sidebar-				no			internal-only

hand	maker	DoF	act. DoF	actuation	weight g	tip force N	payload	tactile	control rate	price USD	BOM USD	open HW	licence	sim model	status
	Systems Corporation)			hydraulic valves'				only headlines mention new touch/tactile sensors as a separate later announcement, not confirmed as part of this hand							
bidexhand_2025	Zhengyang Kris Weng, Center for Robotics and Biosystems, Northwestern University	21	16	cable-driven, N-configuration tendon routing (FeeTech servos, endless-loop antagonistic pull-pull); thumb CMC flexion driven via a 4-bar linkage instead of a direct tendon		2.14	4.54 kg (10 lb) lifted	none mentioned	serial to an onboard ESP32, ROS 2 position control; rate not stated			yes	MIT (code, per README); CAD/hardware licence unconfirmed (bib's claimed CC BY-NC-SA is not verified in the parsed README or file tree)	URDF, consumed by the ROS 2 and MoveIt stack; no physics simulator	open-source
ruka_v2_2026	Xinqi (Lucas) Liu, Ruoxi Hu, Alejandro Ojeda Olarte, Zhuoran Chen, Kenny Ma, Charles Cheng Ji, Lerrel Pinto, Raunaq Bhirangi, Irmak Guzey, New York University and NYU Shanghai	20	16	tendon-driven, forearm-mounted actuators; decoupled parallel 2-DoF wrist via a passive spherical ball joint; single-tendon-plus-spring abduction/adduction				optional e-flesh fingertips (magnetic touch-sensing form factor); not part of the base design	Dynamixel over USB serial; no rate stated	1500	1500	yes		MuJoCo XML in the shared v1 repo; not confirmed to carry the v2 wrist	open-source
orca_hand_2025	Clemens C. Christoph, Maximilian Eberlein, Filippos Katsimalis, Arturo Roberti, Aristotelis Sympetheros, Michel R. Vogt, Davide Liconti, Chenyu Yang, Barnabas Gavin Cangan, Ronan J. Hinchet, Robert K. Katzschmann,	17	17	tendon-driven (antagonistic fishing-line tendon pairs per joint); wrist uses a GT2 timing belt drive	1200		10.5 kg on all four fingers and 2 kg on the index finger alone, both at a fixed 600 mA motor current	yes: FSR-based binary tactile sensing on all 5 fingertips (RP-C7.6-ST), absolute detection threshold as low as 0.05 N	serial to Dynamixel or Feetech motors; rate not stated			yes		URDF-derived kinematic constants only; no URDF or MJCF file in the released tree	open-source

hand	maker	DoF	act. DoF	actuation	weight g	tip force N	payload	tactile	control rate	price USD	BOM USD	open HW	licence	sim model	status
	Soft Robotics Lab, ETH Zurich														
`leap_hand_2023`	Kenneth Shaw, Ananye Agarwal, Deepak Pathak, Carnegie Mellon University	16	16	direct-drive (Dynamixel servos, e.g. XC330-M288), joint velocity ~8 rad/s	595	19.5		none (future work only: 'we plan to develop and integrate LEAP Hand with low-cost touch sensors')	up to 500 Hz querying over USB serial; the paper's own sim-to-real policy runs at 20 Hz	2000	2000	yes	Code: MIT License; CAD: CC BY-NC-SA (non-commercial use with attribution)	URDF claimed in the paper; the released LEAP_Hand_API repo contains no URDF, xacro or MJCF	open-source
`ruka_2025`	Anya Zorin, Irmak Guzey, Billy Yan, Aadhithya Iyer, Lisa Kondrich, Nikhil X. Bhattasali, Lerrel Pinto, New York University	15	11	tendon-driven (11 Dynamixel actuators in the forearm: XM430-W210T for the thumb, XL330-M288-T for the other fingers)		2.74	6.0 kg: weight added to a curled cloth-bag grip until joint-angle error exceeds 15 degrees, best of three trials	none (stated limitation: 'lacks tactile sensing')	Dynamixel over a USB-to-serial bridge; rate not stated, data collection ran at 15 Hz	1300	1300	yes		MJCF	open-source
`dexhand_open_source_2023`	Rob Knight, The Robot Studio (design); electronics/firmware/ROS 2 by Trent Shumay, IoT Design Shop			tendon (fishing line, Sufix 832 80lb / 0.8mm kiteline); Emax ES3301/ES3302/ES3351/ES3352 micro-servos for fingers, Feetech SCS2332 or PWM servos for the wrist				none mentioned	Arduino SCServo or SBUS firmware, BLE; rate not stated	300	300	yes	CC BY-NC-SA 4.0 (README); Onshape CAD freely available	URDF, in the IoT Design Shop ROS 2 packages	open-source
`ilda_hand_2021`	Ajou University, Korea Institute of Machinery & Materials, Korea University	20	15	linkage-driven, direct linear drive (3 Maxon DCX8M motors per finger with GPX8 16:1 gearboxes driving ball screws via parallel+serial four-bar linkages)	1100	34	18 kg	6-axis F/T sensor per fingertip (5 total), force resolution 62 mN, range +/-35 N; not a distributed taxel skin	CAN to a desktop; loop rate not stated			no			prototype
`pisa_iit_softhand_2014`	Centro E. Piaggio, University of Pisa and IIT (Catalano, Grioli, Farnioli, Serio, Piazza, Bicchi)	19	1	tendon-driven adaptive synergy, single motor via differential gears (6 W Maxon RE-max21, 84:1 reduction)			holding force about 20 N along z and holding torque 2 N m, 28 N and 3.5 N m with a stronger motor	none (motor encoder only; a padded work glove supplied contact compliance during experiments)	not stated			no		URDF and Gazebo models in the ROS repo (BSD 3-Clause)	prototype

hand	maker	DoF	act. DoF	actuation	weight g	tip force N	payload	tactile	control rate	price USD	BOM USD	open HW	licence	sim model	status
faive_hand_2023	Yasunori Toshimitsu, Benedek Forrai, Barnabas Gavin Cangan, Ulrich Steger, Manuel Knecht, Stefan Weirich, Robert K. Katzschmann, Soft Robotics Lab, ETH Zurich	16	11	tendon-driven, rolling-contact joints (16 Dynamixel XC330-T288-T servos, 6 antagonistic pairs)	1100		10 kg, whole-hand downward power grasp on a dumbbell	none on the physical hand; a simulated 'fingertip force' (15-dim, critic-only/privileged) exists only in the RL training observation, not real tactile hardware	Dynamixel over serial; joint angles estimated from tendon length by EKF; rate not stated					MJCF, used directly by the released Isaac Gym training repo	prototype
boston_dynamics_atlas_hand_2026	Boston Dynamics			electric (robot is fully electric); specific hand mechanism not stated				tactile sensing in the fingers and palms, confirmed present; type and count not stated				no			prototype
daxo_muscle_v0_2025	Daxo Robotics			ultra-redundant tendon-driven; compliant structure with no rigid joints, flexible materials and tendon routing	750										prototype
figure_03_hand_2025	Figure AI							not stated on page for the hand itself; palm cameras in each hand are vision, not tactile				no			prototype
tesla_optimus_hand_2025	Tesla			V3: tendon-driven from forearm actuators, three tendons per finger through a crosstalk-managing wrist. Gen 2: in-hand actuators/sensors (unlocated in the article).				Gen 2: tactile sensing on all fingers, demonstrated matching contact location and force in an egg-pickup video; type/count not stated. V3: not stated.				no			prototype
xiaomi_cyberone_hand_2026	Xiaomi			motors located in the hand (compact motors generate heat				full-palm tactile sensing, area ~8,200 sq mm, detects				no			prototype

hand	maker	DoF	act. DoF	actuation	weight g	tip force N	payload	tactile	control rate	price USD	BOM USD	open HW	licence	sim model	status
				requiring liquid cooling); no tendon/linkage statement, though contrast with tendon-driven hands' cycle life implies a non-tendon design				pressure and contact across the whole palm not just fingertips; taxel count and type not stated							
`shadow_dexterous_hand_2005`	Shadow Robot Company	24	20	tendon-driven, motors in the forearm (20 Smart Motor nodes, Maxon motors, PWM)	4300		4 kg in a power grasp, vendor claim, protocol not stated	Shadow Tactile Fingertips (STF): 17x3 DoF Hall-effect taxels, 1000 Hz, uncalibrated; up to 5 fingertips; software also supports BioTac/MST/UBI0/PST fingertip options	EtherCAT at 1 kHz to the host, with a 5 kHz tendon-force loop inside each motor module			no		MuJoCo models for Hand E and variants, plus Gazebo, in sr_common	sold
`proception_prohand_2026`	Proception Inc (YC W25)	22		tendon-driven, motors pull cables to move the fingers				integrated skin-like sensors detecting contact, supporting grip control; same 'sensor skin' used on the companion ProGlove; type/count not stated				no			sold
`sharpa_wave_2026`	Sharpa Robotics (Sharpa Pte Ltd)	22	22		1300	20 / 12	40 kg / 24 kg (the same two unlabelled columns)	'Dynamic Tactile Array' (DTA), camera-type array at the fingertip: resolution 240x240 / 60x60, force resolution 0.02 N / 0.05 N, spatial resolution 1 mm / 2 mm, frame rate 180 fps / 30 fps, force range 0-30 N, latency 20 ms, 6-D	500 Hz over 1000BASE-T Ethernet			no		MuJoCo and Isaac Sim models	sold

hand	maker	DoF	act. DoF	actuation	weight g	tip force N	payload	tactile	control rate	price USD	BOM USD	open HW	licence	sim model	status
								F/T; number of sensors per hand not stated							
tesollo_dg5f_2024	Tesollo Inc. (Incheon HQ, Gwangmyeong R&D/factory)	20	20	one integrated actuator per joint ('high-torque actuation', absolute encoder); page does not say direct-drive	1763		pinching 2.5 kg rated and 5 kg maximum; enveloping 10 kg rated and 20 kg maximum	none standard; optional fingertip sensors (6-axis F/T, 3-axis force, or tactile) available; count/type not given	250 Hz over Modbus RTU or TCP and Ethernet TCP/IP			no			sold
unitree_dex5_2025	Unitree Robotics (Yushu Technology Co., Ltd.)	20	16	in-joint geared motor ('hollow-cup motor' + high-precision encoder + low-damping small-clearance reducer), backdrivable; page does not use the words tendon or linkage	1100	10	3.5 kg palm down and 4.5 kg palm left, on a 5 cm round hard object	Dex5-1: none. Dex5-1P: 94 pressure sensors per hand (2x5 palm + 2x3x5 finger pad + 2x3x5 fingertip + 2x3x4 finger root), range 10 g-2500 g.	1000 Hz over USB 2.0, with per-joint stiffness and damping commands			no			sold
wuji_hand_2025	Wuji Technology (founded 2019)	20	20	direct-drive rotary, backdrivable				'Multi-Axis Force/Torque Fingertip Sensing' confirmed present; no type, count or range given	1000 Hz across 20 axes over 100BASE-TX Ethernet, MIT force-position hybrid mode			no		MuJoCo and Isaac Sim models	sold
allegro_hand_v4_2016	Wonik Robotics Co. Ltd. (Seoul, South Korea); earlier versions 1.0/2.0 made by SimLab Co. Ltd.	16	16					none	333 Hz, CAN, the hand's own real-time clock (driver README)			no		URDF and xacro, left and right, in the ROS driver repo	sold
inspire_rh56dfx_2023	Beijing Inspire Robots Technology Co., Ltd.	12	6		540	10		none (this variant is the 'without tactile sensors' table; a tactile FTP variant exists per bib but is not on this page)	RS485; rate not stated			no			sold
robotera_xhand1_2024	ROBOTERA	12	12	gear-driven force-controlled joint	1100		25 kg, tracker's 'Strength'	tactile/force sensors on every fingertip	not stated	14000		no		URDF exists but is licensed; maniptrans_	sold

hand	maker	DoF	act. DoF	actuation	weight g	tip force N	payload	tactile	control rate	price USD	BOM USD	open HW	licence	sim model	status
				modules per finger segment, back-drivable			figure, over 25 kg gripping palm-up	confirmed present (senses contact, force, temperature); count and array size not stated						2025 withholds it	
shadow_dex_ee_2024	Shadow Robot Company, in collaboration with Google DeepMind	12			4100			stereo camera-based fingertip tactile sensors (hundreds of taxels each); multi-taxel 3-DoF arrays on middle and proximal phalanges; counts not stated	not stated; the page claims 'high bandwidth torque and position control loops'			no			sold
brainco_revo2_2025	BrainCo Inc.	11	6		383	15	>=20 kg	Touch variant only: multi-dimensional fingertip tactile module with 'Tactile Adaptive Control'; count/type not stated. Basic and Pro variants: none.	RS485 and CAN FD; EtherCAT on Pro and Touch; rate not stated			no			sold
linkerbot_l20_2025	LinkerBot (SDK author 'CHIUS INC')							L10/L20: per finger 4 channels (normal force, tangential force, tangential direction, proximity), 5 sites x 4 channels, 0-255 range, sensor type not stated. O6 (reseller): 16 capacitive tactile regions.	USB-to-CAN at 1 Mbit/s, optional Modbus or RS485; rate not stated			no		URDF, with PyBullet and Isaac Gym examples in the SDK	sold
psyonic_ability_hand_2021	PSYONIC (San Diego, CA, USA)		6	linkage (repo file/function names indicate a four-bar finger linkage; not stated on the vendor page itself)				30 touch sensor values streamed (FSR-based per API and URDF 'no_fsr' variant); per-finger split not stated	BLE, I2C, UART or RS485; rate not stated			no		URDF (left, right, large, small, no-FSR) plus MuJoCo and Isaac Sim paths in the vendor repo	sold

33 rows. 216 of 528 cells (40 percent) are values no source stated. A blank price is a hand with no public list price rather than a free hand.

B.2 Where each specification came from

hand	source quality	claim date	caveat recorded in the row	source URL	parsed source in the manifest	note	corpus method rows running it
agibot_omnihand_2025	unavailable	no source: both store.agibot.com URLs 404, fetched 2026-09-17			no manifest entry	papers/notes/agibot_omnihand_2025.md	1: clutterdexgrasp_2025
onex_neo_hand_2026	vendor page	vendor page, 2026-07-09			no manifest entry	papers/notes/onex_neo_hand_2026.md	0
leap_hand_v2_adv_2025	vendor page	undated project page; API repo at commit a0936196, fetched 2026-09-17	the page claims 21 DoF with 17 powered motors and the README calls the same hand 17-DOF	https://github.com/leap-hand/LEAP_Hand_V2_Adv_API	no manifest entry	papers/notes/leap_hand_v2_adv_2025.md	2: bidex_teleop_2024, dexwild_2025
paxini_dexh13_2024	press	press, CES 2026-01-07			no manifest entry	papers/notes/paxini_dexh13_2024.md	0
clone_robotics_hand_2024	press	press, 2025-11-24, on a video of 2025-11-15	'under 2 pounds' is an upper bound, and the source does not say whether the 500 W pump and 36 valves are inside it		no manifest entry	papers/notes/clone_robotics_hand_2024.md	0
sanctuary_phoenix_hand_2024	press	undated press release; trade reprint 2024-12-17			no manifest entry	papers/notes/sanctuary_phoenix_hand_2024.md	0
bidexhand_2025	paper	paper, 2025 (ICRA Dexterity workshop abstract)	the paper states 16 independently actuated DoF and 21 joints; the README states 15 servos driving 15 joints	https://github.com/wengmister/BiDexHand	5 pp, 89f39715, fetched 2026-09-17	papers/notes/bidexhand_2025.md	0
ruka_v2_2026	paper	paper, 2026	the AS5600 encoders are attachable and detachable and are read for calibration and measurement only, not in the control loop	https://github.com/ruka-hand/RUKA	20 pp, d71d6f90, fetched 2026-09-17	papers/notes/ruka_v2_2026.md	0
orca_hand_2025	paper	paper, 2025	the 19.6 N previously tabulated as a fingertip force is the newton equivalent of the 2 kg index-finger payload at a control-imposed current limit, and no source states it as a force	https://github.com/orcahand/orca_core	8 pp, f3967f2e, fetched 2026-09-17	papers/notes/orca_hand_2025.md	0
leap_hand_2023	paper	paper, 2023		https://github.com/leap-hand/LEAP_Hand_API	11 pp, ac08a57e, fetched 2026-09-17	papers/notes/leap_hand_2023.md	12: bidex_teleop_2024, bidexhd_2024, cross_embodiment_world_models_2025, dexcap_2024, dexndm_2025, dexterous_functional_grasping_2023, dextrack_2025, dexwild_2025, dreureka_2024, teledexter_2026, unidex_2026, videodex_2022
ruka_2025	paper	paper, 2025		https://github.com/ruka-hand/RUKA	13 pp, 3a19161f, fetched 2026-09-17	papers/notes/ruka_2025.md	0
dexhand_open_source_2023	project page and GitHub README	page posts 2023-08-08 to 2023-10-01; no hardware release date	no joint count is given anywhere, so no DoF figure can be quoted; the servo count is 16 finger and thumb micro-servos plus 2 wrist servos, with a third	https://github.com/TheRobotStudio/V1.0-Dexhand	no manifest entry	papers/notes/dexhand_open_source_2023.md	0

hand	source quality	claim date	caveat recorded in the row	source URL	parsed source in the manifest	note	corpus method rows running it
			optional. The $300 is 'additional total cost of components', excluding printing and the wrist servos				
ilda_hand_2021	paper	paper, accepted 2021-11-05		https://www.nature.com/articles/s41467-021-27261-0.pdf	13 pp, ed0e19c2, fetched 2026-09-17	papers/notes/ilda_hand_2021.md	0
pisa_iit_softhand_2014	paper	paper, IJRR 2014		https://www.centropiaggio.unipi.it/sites/default/files/PisaIIT_SoftHand_0.pdf	14 pp, 9c9cd17f, fetched 2026-09-17	papers/notes/pisa_iit_softhand_2014.md	0
faive_hand_2023	paper	paper, 2023	the row's open-hardware status is unconfirmed: the repo has LICENSE and LICENSE-NVIDIA files whose contents were not captured, and no open-hardware claim appears in the sources	https://github.com/srl-ethz/faive_gym_oss	7 pp, 13feb753, fetched 2026-09-17	papers/notes/faive_hand_2023.md	1: graspxl_2024
boston_dynamics_atlas_hand_2026	press	trade press, 2026-01-05 (CES); the Boston Dynamics blog of the same date has no hand content			no manifest entry	papers/notes/boston_dynamics_atlas_hand_2026.md	0
daxo_muscle_v0_2025	third-party tracker	third-party catalogue post, 2026-09-16; daxo-robotics.com returned HTTP 404	120 actuators in 750 g is about 6 g per actuator including structure, tendons, routing and skin; no DoF number is stated, and a redundant tendon architecture with no rigid joints has fewer kinematic DoF than actuators		no manifest entry	papers/notes/daxo_muscle_v0_2025.md	0
figure_03_hand_2025	third-party tracker	undated tracker page, fetched 2026-09-17; figure.ai launch page returned HTTP 404	the tracker's 'Degrees of freedom, hands / 20' sits beside 'Number of fingers / 10', so the 20 is most likely the pair and about 10 per hand; the 16-per-hand figure in circulation has no reachable source		no manifest entry	papers/notes/figure_03_hand_2025.md	0
tesla_optimus_hand_2025	press	Gen 2 video 2023-12-13; V3 patents filed 2024-10, relayed by press 2026-04	the 22 DoF in circulation is a Teslarati article's arithmetic of four DoF on each of five fingers plus two at the wrist, not a Tesla statement; Gen 2's own figure is 11 DoF. This row's actuation cell describes V3 and its tactile cell describes Gen 2	https://image-ppubs.uspto.gov/dirsearch-public/print/downloadPdf/9505134	20 pp, 93664ecb, fetched 2026-09-17	papers/notes/tesla_optimus_hand_2025.md	0
xiaomi_cyberone_hand_2026	press	press, 2026-03-30			no manifest entry	papers/notes/xiaomi_cyberone_hand_2026.md	0
shadow_dexterous_hand_2005	datasheet	specification dated 2024-12-04	the four distal joints are coupled and not independently controllable, so the '24 DoF' quoted in learning papers is the joint count	https://shadowrobot.com/wp-content/uploads/2024/12/shadow_dexterous_hand_e_technical_specification_20241204.pdf	11 pp, 66541d02, fetched 2026-09-17	papers/notes/shadow_dexterous_hand_2005.md	21: asymdex_2024, bidexgrasp_2026, bimangrasp_2024, cross_embodiment_world_models_2025, dexdeform_2023, dexman_2025, dp3_2024, eureka_2023, graspxl_2024, maniptrans_2025, mjpc_2022, objdex_2024,

hand	source quality	claim date	caveat recorded in the row	source URL	parsed source in the manifest	note	corpus method rows running it
							openai_dexterity_2018, openai_rubiks_cube_2019, pddm_2019, pgdm_2023, pianomime_2024, resdex_2024, unidex_2026, unidexgrasp_2023, unidexgrasp_pp_2023
proception_prohand_2026	press	launch post and press, 2026-06-29	the evidence is a shipping announcement, not a datasheet: no weight, force, actuated-DoF count or control rate is stated anywhere		no manifest entry	papers/notes/proception_prohand_2026.md	0
sharpa_wave_2026	vendor page	undated page, copyright 2026, fetched 2026-09-17	the page footnotes 'Specifications may vary between products', and no column of the spec table is labelled		no manifest entry	papers/notes/sharpa_wave_2026.md	6: dexteleop0_2026, egoscale_2026, metis_2025, poise_2026, simtoolreal_2026, teledexter_2026
tesollo_dg5f_2024	vendor page	undated page, copyright 2026, download links dated 2026-08		https://en.tesollo.com/wp-content/uploads/2025/02/TESOLLO-DG-5F-catalogue-En-2.pdf	no manifest entry	papers/notes/tesollo_dg5f_2024.md	0
unitree_dex5_2025	datasheet	undated page, copyright 2016-2025, fetched 2026-09-17		https://github.com/unitreerobotics/unitree_sdk2	no manifest entry	papers/notes/unitree_dex5_2025.md	0
wuji_hand_2025	vendor page	undated page, copyright 2026, fetched 2026-09-17	the page footnotes that these figures 'reflect the current Beta1 product and the SPEC will continue to iterate'		no manifest entry	papers/notes/wuji_hand_2025.md	2: toporetarget_2026, unidex_2026
allegro_hand_v4_2016	driver repo	driver repo, versions 1.0-4.0 undated, fetched 2026-09-17	allegrohand.com/v4 returned HTTP 404 and the Wonik wiki timed out; every cell here comes from the ROS driver repository, and weight, joint torque, payload and price have no reachable source	https://github.com/simlabrobotics/allegro_hand_ros_v4	no manifest entry	papers/notes/allegro_hand_v4_2016.md	35: anyrotate_2024, anyteleop_2023, asymdex_2024, castro_sap_contact_2021, cross_embodiment_world_models_2025, cyberdemo_2024, dexmachina_2025, dexpbt_2023, dexpilot_2020, dexplore_2025, dexpoint_2022, dexteritygen_2025, dexterous_handover_2025, dextrack_2025, dextreme_2022, dime_2022, dp3_2024, dynamic_handover_2023, eureka_2023, geometric_retargeting_2025, graspxl_2024, holo_dex_2022, hora_2022, hudor_2024, human2sim2robot_2025, maniptrans_2025, pang_global_planning_2022, penspin_2024, robot_synesthesia_2023, rotateit_2023, rotating_without_seeing_2023, twisting_lids_2024,

hand	source quality	claim date	caveat recorded in the row	source URL	parsed source in the manifest	note	corpus method rows running it
							unidex_2026, viserdex_2026, wm_dex_human_videos_2025
inspire_rh56dfx_2023	vendor page	undated page, fetched 2026-09-17	the vendor's own page reads 'Degrees of freedom 6, Numbers of joints 12'; the joints column here prints the joint count, so the vendor's DoF figure is the actuated-DoF column		no manifest entry	papers/notes/inspire_rh56dfx_2023.md	19: ace_teleop_2024, articulated_tools_inhand_2025, being_h05_2026, being_h0_2025, bidexgrasp_2026, dexmachina_2025, dexmimicgen_2024, dexplore_2025, dexumi_2025, dexvla_2025, humanoid_policy_human_policy_2025, humanoid_sim2real_recipe_2025, humanplus_2024, maniptrans_2025, metis_2025, okami_2024, omnih2o_2024, open_television_2024, unidex_2026
robotera_xhand1_2024	third-party tracker	undated tracker page, fetched 2026-09-17; the vendor page was not fetched	the 15 N in circulation comes from a survey table, not from the tracker or the vendor; the tracker states no fingertip force		no manifest entry	papers/notes/robotera_xhand1_2024.md	8: cross_embodiment_world_models_2025, deximit_2026, dexmachina_2025, dexora_2026, dexumi_2025, dp3_2024, force_grasp_sim2real_2026, unidex_2026
shadow_dex_ee_2024	vendor page	undated page, copyright 2026, fetched 2026-09-17			no manifest entry	papers/notes/shadow_dex_ee_2024.md	1: demostart_2024
brainco_revo2_2025	datasheet	undated page, fetched 2026-09-17			no manifest entry	papers/notes/brainco_revo2_2025.md	1: bidexgrasp_2026
linkerbot_l20_2025	vendor page	undated page, fetched 2026-09-17		https://github.com/linkerbotai/linker_hand_sdk	no manifest entry	papers/notes/linkerbot_l20_2025.md	1: being_h05_2026
psyonic_ability_hand_2021	vendor page	undated page, fetched 2026-09-17	the four-bar finger linkage rests on repository file names and the five fingers on URDF meshes; the vendor page states neither	https://github.com/psyonicinc/ability-hand-api	no manifest entry	papers/notes/psyonic_ability_hand_2021.md	7: ace_teleop_2024, asymdex_2024, bunny_visionpro_2024, cross_embodiment_world_models_2025, dexmachina_2025, hato_visuotactile_2024, unidex_2026

33 rows. 31 of 264 cells (11 percent) are values no source stated. `parsed source in the manifest` *gives page count, the first eight characters of the sha256 of the stored copy, and the date it was fetched. A hand with no manifest entry was recorded from a page that could not be stored, and its row should be read as a claim with no retrievable source.*

Appendix C. Per-paper reward term extraction

Table 5 marks nine recurring term families across the in-hand reorientation methods. This appendix is the extraction underneath it, over every method row that records a reward at all. `terms stated` counts the terms the paper itself names. `where the reward was read` gives the section of the paper or the file in the released code that the note quotes. A row with a `paper/code class` is one where the two disagree, and C.2 prints the disagreement in full. Cells and entries reproduce the text stored in the row, with the source's own punctuation and wording.

C.1 The extraction

method	yr	paradigm	terms stated	term names as the paper names them	where the reward was read	in reward matrix	code released	paper/code class	confidence
openai_dexterity_2018	2018	RL	3		Method, reward (Sec 4.2 / App C.1)	yes	no		
openai_rubiks_cube_2019	2019	RL, distillation	3		Method, reward (Sec. 6.1)	yes	no		
pddm_2019	2019	MPC					yes	contradiction	high
dexpoint_2022	2022	RL	4	reach, contact, lift, action penalty			yes	parse-limitation	low
dextreme_2022	2022	RL	6		Reward block, paper Table 2 against allegro_hand_dextreme.py and the two DR yamls	yes	yes	contradiction	high
dexvip_2022	2022	RL	4				no		
dgrasp_2022	2022	RL	4	r_x, r_q, r_c, r_reg			yes	version-skew	high
hora_2022	2022	RL, distillation	5		Block C, paper Sec 3.1/App. C against configs/task/AllegroHandHora.yaml	yes	yes	version-skew	high
mjpc_2022	2022	MPC	3				yes		
visual_dexterity_2022	2022	RL, distillation	7	c1_sparse_task, c2_dense_task, c3_fingertip_dist, c4_energy, c5_push_away_penalty, c6_table_contact, c7_finger_height	Method, Table S1 against dexenv/envs/rewards.py and dexenv/conf/task/dclaw.yaml	yes	yes	contradiction	high
aloha_act_2023	2023	BC					yes	internal-inconsistency	high
artigrasp_2023	2023	RL	4	r_p, r_c, r_reg, r_task			yes	parse-limitation	low
dexpbt_2023	2023	RL	4	r_reach, r_pick, r_targ, r_vel	Block C, paper Sec III-C/Table II against allegro_kuka_base.py and AllegroKuka.yaml	yes	yes	contradiction	high
dexterous_functional_grasping_2023	2023	RL	2	r_threshold, r_hand-obj			no		
diffusion_policy_2023	2023	BC, diffusion					yes	parse-limitation	high
dynamic_handover_2023	2023	RL	3				no		
eureka_2023	2023	RL			Method, App. G.1/G.2 LLM-authored rewards; generated code not checked into the repo	yes	yes	parse-limitation	low
pgdm_2023	2023	RL	2				no		
physhoi_2023	2023	RL	4	body motion reward, object motion reward, ig reward, cg reward			yes	contradiction	high

method	yr	paradigm	terms stated	term names as the paper names them	where the reward was read	in reward matrix	code released	paper/code class	confidence
robot_synesthesia_2023	2023	RL, distillation	6		Method, reward (Sec IV.A.3 eq. 1); no code released	yes	no		
rotateit_2023	2023	RL, distillation	6		Block C (paper Sec 3.1, App. B); no code released	yes			
rotating_without_seeing_2023	2023	RL	6	r_rot, r_vel, r_fall, r_work, r_torque, r_dist	Reward block (Sec IV.A.3, App. D Eq. 9); no code released	yes			
unidexgrasp_2023	2023	grasp-synthesis, RL, distillation	4	goal, reach, lift, move			yes	contradiction	high
unidexgrasp_pp_2023	2023	RL, distillation	4	r_reach, r_lift, r_move, r_bonus			yes	parse-limitation	low
anyrotate_2024	2024	RL, distillation	10	keypoint distance reward, rotation reward, goal bonus reward, good contact reward, bad contact penalty, angular velocity penalty, pose penalty, work penalty, torque penalty, termination penalty	Block C (App. B.1 weights); no code released	yes	no		
asymdex_2024	2024	RL	4				yes	code-absent	medium
bidexhd_2024	2024	RL, distillation	4	r_appro, r_lift, r_bonus, r_track			yes	parse-limitation	high
bimangrasp_2024	2024	grasp-synthesis, diffusion	7				no		
demostart_2024	2024	RL, distillation	1		Method, reward (App. VII-A.1); no code released	yes	no		
dexmimicgen_2024	2024	data-collection, BC	1				yes	code-absent	high
dp3_2024	2024	BC, diffusion					yes	internal-inconsistency	medium
dreureka_2024	2024	RL	4	ang_z_vel_reward, lin_vel_penalty, object_fall_penalty, deviation_penalty	Method, App. B1 Prompt 13; cube-rotation code absent from the released repo	yes	yes	code-absent	high
graspxl_2024	2024	RL	8	r_dis, r_v, r_ω, r_m, r_c, r_f, r_reg, r_anatomy			yes	parse-limitation	low
hudor_2024	2024	RL	1				no		
humanplus_2024	2024	RL, BC	8				yes	parse-limitation	medium
objdex_2024	2024	BC, RL, distillation	3	rotation, translation, joint angle			no		
omnigrasp_2024	2024	RL, distillation	3	r_approach, r_pre-grasp, r_obj			yes	parse-limitation	medium
omnih2o_2024	2024	RL, distillation	24				yes	internal-inconsistency	medium
open_television_2024	2024	teleop-system, BC					yes	code-absent	low

method	yr	paradigm	terms stated	term names as the paper names them	where the reward was read	in reward matrix	code released	paper/code class	confidence
penspin_2024	2024	RL, distillation	7		Method, paper Table 4 against penspin/tasks/allegro_hand_hora.py and configs/task/AllegroHandHora.yaml	yes	yes	contradiction	medium
pi0_2024	2024	VLA, flow					yes	version-skew	high
pianomime_2024	2024	RL+demo, distillation	2				yes	contradiction	high
resdex_2024	2024	RL, distillation	7	task, proposal, pose, reach, lift, move, bonus			yes	parse-limitation	medium
twisting_lids_2024	2024	RL	5				no		
articulated_tools_inhand_2025	2025	RL, BC, distillation	8	r_pos, r_quat, r_goal, r_timer, r_inc, r_contact, r_slip, r_act			no		
being_h0_2025	2025	VLA, BC					yes	code-absent	high
clutterdexgrasp_2025	2025	RL, diffusion, distillation	4	r_pos, r_neg, r_grasp, r_force			no		
cross_embodiment_world_models_2025	2025	world-model, MPC	2				no		
dexgraspvla_2025	2025	VLA, diffusion	1				yes	parse-limitation	high
dexmachina_2025	2025	RL	4				yes	internal-inconsistency	low
dexman_2025	2025	RL, trajopt	3				no		
dexndm_2025	2025	RL, distillation	7		Reward block (Sec. 3.1, App. A.1); no code released	yes	no		
dexplore_2025	2025	RL, distillation	7	r_j^h, r_r^h, r_p^o, r_r^o, r_d, r_c, r_energy			yes		
dexremoe_2025	2025	RL	5	csuccess, cdist, crot, cω, ca	Method, reward (Sec. III-B, Table II); no code released	yes	no		
dexteritygen_2025	2025	RL, diffusion			Method, reward/objective: the RL reward behind the pretrained controller is not quoted in the source	yes	no		
dexterous_handover_2025	2025	RL	3	r_BASE, contact_term (c·w_c), r_MANIP					
dextrack_2025	2025	RL+demo	5	r_o,p, r_o,q, r_wrist, r_finger, r_affinity			yes	parse-limitation	low
dexvla_2025	2025	VLA, diffusion	2				yes	parse-limitation	high
dydexhandover_2025	2025	RL	5	R_dist, R_obj, R_contact, P_action, P_hand			no		
geometric_retargeting_2025	2025	teleop-system	5	Ldir, Lcover, Lflat, Lpinch, Lcol			yes		
groot_n16_2025	2025	VLA					yes	version-skew	high

method	yr	paradigm	terms stated	term names as the paper names them	where the reward was read	in reward matrix	code released	paper/code class	confidence
groot_n1_2025	2025	VLA, flow					yes	version-skew	high
human2sim2robot_2025	2025	RL	1	r_obj			yes		
humanoid_policy_human_policy_2025	2025	BC	2				yes		
humanoid_sim2real_recipe_2025	2025	RL, distillation	2				no		
maniptrans_2025	2025	RL, BC, diffusion	5	r_wrist, r_finger, r_smooth, r_object, r_contact			yes	parse-limitation	low
metis_2025	2025	VLA	2	Lar, Laction			no		
pi05_2025	2025	VLA, flow					yes	code-absent	high
pistar06_2025	2025	VLA, RL, flow					yes	code-absent	high
wm_dex_human_videos_2025	2025	world-model, MPC	2				no		
being_h05_2026	2026	VLA, flow					yes	code-absent	high
dexteleop0_2026	2026	teleop-system, MPC	3				no		
force_grasp_sim2real_2026	2026	RL	7	R_torque, R_Force, R_diff, R_outer, terminal-state penalty, R_action, R_vel	Method, reward (Sec 3.5.1, Table 2); no weights survive the parse; no code released	yes	no		
poise_2026	2026	RL	4	pose reaching, goal completion, grasp maintenance, drop and actuation regularization	Method, reward (Sec. IV-D); weights lost to the parse; no code released	yes	no		
simtoolreal_2026	2026	RL	5	r_smooth, r_approach, r_lift, r_goal, b_succ			no		
teledexter_2026	2026	RL, teleop-system			Reward block (Sec. 3.1); term formulas lost to the parse; no code released	yes	no		
toporetarget_2026	2026	trajopt, RL	4	object, link-position, joint-position, action-smoothness			no		
viserdex_2026	2026	RL, distillation	10	orientation tracking, success bonus, object dropped, object distance, object velocity, joint velocity, action magnitude, action rate, joint work, joint torques	Method, reward (App. Table IX); no code released	yes	no		

78 rows. 255 of 780 cells (32 percent) are values no source stated. A blank `terms stated` with a filled `term names` column is a paper that names its terms without numbering them. `in reward matrix` marks the rows that are also in the reward-term matrix, which covers in-hand reorientation only.

C.2 Paper against released code, in full

Each entry below is the disagreement text stored in the row, unedited. The class is what Sec. 5.8 and Sec. 8.1 count. `contradiction` means the paper states one value and the shipped code demonstrably states another. `parse-limitation` means this survey's own parse could not settle it and the accusation is withdrawn. `code-absent` means the described component is not in the released repository. `version-skew` means the repository is a later generation than the paper. `internal-inconsistency` means the paper disagrees with itself and no code is implicated. Every `contradiction` entry carries an `Artefact` line: the repository, the commit `corpus/code_manifest.json` records, the file inside it and where in that file the value sits, so the entry can be checked without asking anyone. All of those authors were written to on 19 September 2026, before this survey was posted; section 5.6 says so beside the finding, gives the date the reply window closes, and states the route by which a disputed entry is corrected. The letters are in `outreach/` in the corpus, which is available from the author at the address in the byline.

contradiction, 8 rows.

- `pddm_2019` (high). Table 2 states obs-dim 46 for In-hand Reorientation while the released cube_env.py code sums to 39; the Baoding reward code includes an extra -10*wrist_too_high term absent from Table 2's printed formula. Artefact: `google-research/pddm` at `06b88cdbaf`, `pddm/envs/cube/cube_env.py` (`_get_obs`). Review: Written to on 19 September 2026 with the claim, its evidence and the sentences this survey would print. The reply window closed on 10 October 2026 with no reply. Recorded as silence rather than as agreement; the claim rests on the two public documents it names and a reader can settle it without the authors.
- `dextreme_2022` (high). Action Delta Penalty weight is -0.25 in paper Table 2 but -0.2 in the ADR yaml and -0.01 in the ManualDR yaml; the Joint Velocity Penalty in code normalises velocity by (max_velocity-vel_tolerance) unlike the paper's stated formula; code has a timeout_rew term absent from the paper's reward table; Appendix Table 12 states critic learning rate 5e-4 and KL threshold 0.16, vs body text/code values of 5e-5 and 0.016. Artefact: `isaac-sim/IsaacGymEnvs` at `aeed298638`, `isaacgymenvs/cfg/task/AllegroHandDextremeADR.yaml` (`actionDeltaPenaltyScale`). Review: Written to on 19 September 2026 with the claim, its evidence and the sentences this survey would print. The reply window closed on 10 October 2026 with no reply. Recorded as silence rather than as agreement; the claim rests on the two public documents it names and a reader can settle it without the authors. Delivery was partial: two of the three copied coauthor addresses were rejected by the recipient's mail server, so those two coauthors never received the letter. The address the letter was addressed to was not among the rejections and its copy was delivered.
- `visual_dexterity_2022` (high). Table S1 states 32000 teacher training environments, but the released config sets alg.num_envs to 8000 (parent config 16384); fallDistance differs across two shipped configs (0.24 vs 0.15, only the latter matching Table S1's threshold); the paper's Eq 8 penultimate-joint penalty (c7=-2) does not appear anywhere in the released reward code; the config carries a dead distRewardScale=-10.0 key never used in compute_reward; and the paper's table-friction lower bound (0.05) differs by a factor of 10 from the code's randomized lower bound (0.005). Artefact: `Improbable-AI/dexenv` at `ad9634e9d2`, `dexenv/conf/dclaw.yaml` (`alg.num_envs`). Review: Written to on 19 September 2026 with the claim, its evidence and the sentences this survey would print. The reply window closed on 10 October 2026 with no reply. Recorded as silence rather than as agreement; the claim rests on the two public documents it names and a reader can settle it without the authors.
- `dexpbt_2023` (high). Paper presents the reward as 4 mutually exclusive stage terms (r_reach, r_pick, r_targ, -r_vel), but code's compute_kuka_reward sums 8 named components, one of which (hand_delta_penalty) is multiplied by 0 and disabled; there is no single r_vel term in code, instead separate kuka/allegro action penalties whose exact formula is not shown. Also, the paper reports zero experiments with domain randomization, yet the shipped AllegroKuka.yaml already carries a fully specified DR schedule (disabled via randomize: False). Artefact: `NVIDIA-Omniverse/IsaacGymEnvs` at `aeed298638`, `isaacgymenvs/tasks/allegro_kuka/allegro_kuka_base.py` (`compute_kuka_reward`). Review:

R3 adversarial review: the disabled-randomisation half is withdrawn, since the note finds it consistent with the paper; the zeroed reward term stands. Written to on 19 September 2026 with the claim, its evidence and the sentences this survey would print. The reply window closed on 10 October 2026 with no reply. Recorded as silence rather than as agreement; the claim rests on the two public documents it names and a reader can settle it without the authors. One copied coauthor's mailer returned an automatic out-of-office message, which is not an answer and is not counted as one. Delivery was partial: two of the three copied coauthor addresses were rejected by the recipient's mail server, so those two coauthors never received the letter. The address the letter was addressed to was not among the rejections and its copy was delivered.

- `physhoi_2023` (high). The released compute_humanoid_reward sets the body position-velocity error and the object rotation and rotation-velocity errors to zeros_like, unconditionally, with the computation that would produce them commented out on the same lines, while Table 4 lists nonzero λ^or=0.1/λ^orv=0.01 weights for GRAB: in that file the object's orientation error is the constant zero and those weights cannot change the reward. Artefact: `wyhuai/PhysHOI` at `6095c605e2`, `physhoi/env/tasks/physhoi.py` (`compute_humanoid_reward`). Review: Written to on 19 September 2026 with the claim, its evidence and the sentences this survey would print. The reply window closed on 10 October 2026 with no reply. Recorded as silence rather than as agreement; the claim rests on the two public documents it names and a reader can settle it without the authors.
- `unidexgrasp_2023` (high). The paper describes a four-term weighted reward (r_goal + r_reach + r_lift + r_move via Table 7's omega weights) but the released compute_hand_reward implements a different threshold-gated torch.where cascade with distinct hardcoded coefficients that do not map one-to-one onto the paper's weights. Artefact: `PKU-EPIC/UniDexGrasp` at `36c9bfcf7c`, `dexgrasp_policy/dexgrasp/tasks/shadow_hand_grasp.py` (`compute_hand_reward, goal_cond branch`). Review: Written to on 19 September 2026 with the claim, its evidence and the sentences this survey would print. On 2 October 2026 a copied coauthor forwarded the letter to two further coauthors, asked them whether they recalled the cause of the difference, and copied this survey's author on that forward. No answer had arrived when the window closed on 10 October 2026. The letter was therefore read and passed on rather than ignored, and the question it asks is open: a forward is neither a confirmation of the claim nor a dispute of it, so the row is unchanged and the claim is recorded as unanswered rather than as agreed.
- `penspin_2024` (medium). The appendix states a randomised disturbance force, and the released configs/task/AllegroHandHora.yaml ships forceScale: 0.0, so no shipped configuration applies it. Artefact: `HaozhiQi/penspin` at `5035c52dc9`, `configs/task/AllegroHandHora.yaml` (`forceScale`). Review: Narrowed while the letter to its authors was being drafted, before that letter went out on 19 September 2026. The original comparison also said the released code disables the paper's tactile channel, and that half is withdrawn: the config read has numObservations 96 and enable_tactile False, which is consistent with the proprioception-only student rather than the oracle, and the paper never claims the student has tactile input. The disturbance-force half is unaffected. Written to on 19 September 2026 with the claim, its evidence and the sentences this survey would print. The reply window closed on 10 October 2026 with no reply. Recorded as silence rather than as agreement; the claim rests on the two public documents it names and a reader can settle it without the authors.
- `pianomime_2024` (high). Paper's Table 3 states 2 weighted reward terms (Key Press 2/3, Mimic 1/3), but the released code sums roughly 5 unweighted terms (key press doubled, sustain, energy and fingering hardcoded to return 0, forearm-collision) plus a separately-added mimic wrapper term. Author-confirmed: the first author states the released code is the implementation behind the paper's results and that the paper's reward description "might be imprecise" (personal communication, C. Qian, Oct 2026), so the code is the authoritative statement of what was trained. Artefact: `sNiper-Qian/pianomime` at `c4abefac8d`, `single_task/piano_with_shadow_hands_res.py` (`_set_rewards`). Review: Written to on 19 September 2026 with the claim, its evidence and the sentences this survey would print. This one is a redirect rather than a silence. A coauthor replied on 22 September 2026 on the senior author's behalf and pointed the question at the first author, Cheng Qian, who had not been written to and whom the window closed before anyone asked. So the letter reached the group and the person who could answer it was named too late for this version. The comparison stands as written, unanswered rather than agreed, and a reply from the first author would be acted on the same way any other correction is. Answered on 14 October 2026, four days after the window closed, by the first author the 22 September redirect had named. Both halves of this entry are now author-confirmed rather than inferred. On the reward, he

states that the released code is the implementation behind the paper's results and that the paper's description of the reward "might be imprecise", which is the comparison this survey drew, with the direction settled: the code is authoritative. On the F1 figures, he confirms that the conclusion's 70% is a typo produced by a bug in an older scipy and that 56% is correct, which resolves an inconsistency this survey had recorded as unreconciled. Personal communication, C. Qian, Oct 2026; nothing in it was solicited beyond the original letter and no reply was sent from this project.

internal-inconsistency, 4 rows.

- `aloha_act_2023` (high). Algorithm 1 pseudocode states L_reconst = MSE, but Sec.IV.C's prose explicitly states L1 loss is used instead; the algorithm box and the implementation text disagree; code/md captures only function signatures, not bodies, for policy.py's loss implementation
- `dp3_2024` (medium). the paper's prose states the network predicts the noise added to the data, but the shipped default config trains with prediction_type: sample (predicting the denoised action a^0 directly), not epsilon; the paper only qualifies this later in the same section (Fig. 7 ablates both).
- `omnih2o_2024` (medium). stumble weight -0.00125 (paper) vs -1250 (code); max-feet-height sign/magnitude differ (+1000 paper vs -2500 code, a penalty not a bonus); paper's exp(-c*//.//) form vs code's exp(-err^2/sigma); curriculum level-down threshold 40 (paper) vs 50 (code) Review: Withdrawn as a contradiction at the point of drafting the letter to its authors, which was never sent. Four sibling weights match the paper to the digit under a systematic x1.25 curriculum factor and only the stumble weight differs, by a factor of about a million, which is a typo signature in the paper's own table rather than evidence of a different trained objective. The hands in this work are driven open-loop from VR pose, outside the policy and outside the reward, so it is a weak fit for a dexterous-manipulation reward census in the first place.
- `dexmachina_2025` (low). paper describes a plain weighted sum lambda_task*r_task + lambda_imi*r_imi + lambda_bc*r_bc + lambda_con*r_con with unspecified weights; code implements a multiplicative task term with per-component beta decay, an unmentioned 0.1 force-penalty term, and curriculum-driven decay of auxiliary weights not described as such in the paper Review: R3 adversarial review: the multiplicative form credited to the code is printed in the paper itself

version-skew, 5 rows.

- `dgrasp_2022` (high). The README states that the released code updated RaiSim to 1.1.6 and that "training D-Grasp is more stable and faster now", so the parsed commit is not the one the paper's numbers were produced under; separately, the paper's single contact weight w_c = 1.0 (Table 5) is two coefficients in the shipped config, contact_reward 1.0 and impulse_reward 2.0, and the config's falling_reward 1.0 has no counterpart in the paper's reward description or table. Review: The version skew is the authors' own statement in the README, not an inference, which is why it is recorded at high confidence. The reward-coefficient difference is recorded beside it rather than as the charge: the two extra coefficients plausibly implement the two parts the paper describes for its contact reward, and the C++ that would settle it is outside the parse. The same pattern is recorded for artigrasp_2023, which is built on this code base.
- `hora_2022` (high). The parsed code commit's joint-noise range, disabled default disturbance force, and default cube object type differ from the paper's stated values; the code README itself says to use tag v0.0.1, not the parsed commit, to reproduce paper numbers.
- `pi0_2024` (high). The released repo (openpi, commit 215abfb2) has evolved past this paper: it also documents pi0-FAST (autoregressive) and pi0.5 checkpoints/configs not described in this paper, which describes only the flow-matching pi0 and the non-VLM pi0-small ablation.
- `groot_n16_2025` (high). code/md is the current Isaac-GR00T main branch (N1.7 generation, commit 51d4c89f), not the N1.6 checkpoint; the repo names a separate n1d6 branch for N1.6 that was not fetched. The page states N1.6's backbone is "an internal NVIDIA Cosmos-2B VLM variant," but the repo's own changelog states the Eagle backbone (nvidia/Eagle-Block2A-2B-v2) was used through N1.6 and only replaced by Cosmos-Reason2-2B in N1.7 -- whether the page's Cosmos-2B variant is a third distinct model or the page is describing N1.7 under an N1.6 headline is not resolvable from what was fetched. Review: The page attributes a Cosmos backbone to N1.6 while the repository changelog says N1.6 used Eagle and N1.7 made the switch, so the repository is a later generation than the page describes.

- `groot_n1_2025` (high). The parsed code repo (Isaac-GR00T, commit 51d4c89f) is a later N1.7 generation, not the paper's GR00T-N1-2B: VLM backbone changed from the paper's Eagle-2 to Cosmos-Reason2-2B (via Qwen3-VL); action horizon expanded from the paper's stated H=16 to 40; state/action dimensions expanded from 29 to 132; DiT layers changed from 32 to 16; embodiment scope widened to include Unitree G1, AgiBot G1, and a 'YAM' arm not described in the paper.

code-absent, 8 rows.

- `asymdex_2024` (medium). Code contains a GraspAndPlace-task reward variant (AllegroHandDualArmGraspAndPlaceAsymDex.py) with exponential hand-distance and block-alignment terms not one of the four BiDexHands tasks described in the paper body.
- `dexmimicgen_2024` (high). The environment code includes an unused 'reward_shaping' branch (if self.reward_shaping: pass) that is never described or used in the paper text; only the sparse binary completion reward is used, as the paper states.
- `dreureka_2024` (high). The released DrEureka repo (code/md) contains only forward_locomotion/ and globe_walking/ trees; the LEAP-hand cube-rotation training/deployment code and the quoted Prompt-13 reward are the LEAP-hand authors' own code, not included in the released repo, so the cube-rotation reward cannot be cross-checked against a code file.
- `open_television_2024` (low). The paper's prose states 25k training iterations, lr 5e-5, batch size 45, but the code repo's example training command instead shows 50000 epochs and an explicit kl_weight=10 not mentioned anywhere in the paper text. Review: R3 adversarial review: a behaviour-cloning paper with no reward; it compared paper iterations to a README example's epochs
- `being_h0_2025` (high). The actual downstream action-head module referenced in §4.3 (learnable query tokens, proprioceptive projector f_p, regression head f_r) is not present in the parsed code; the repo's own TODO list marks 'Training code and scripts' as unreleased, so the action-chunk-length=16 value and exact proprioception-vector fields come only from inference/eval CLI examples, not a verifiable implementation.
- `pi05_2025` (high). The paper describes a hybrid discrete-FAST-token-pretrain-then-flow-matching-post-train recipe (Eq. 1, alpha: 0 to 10), but the released repo states plainly it 'currently only support[s] the flow matching head for both pi0.5 training and inference.'
- `pistar06_2025` (high). The general openpi framework (pi0/pi0.5/FAST, flow-matching action head, ALOHA/DROID/LIBERO adapters) is released at commit 215abfb2, but the RECAP-specific pieces described in the paper -- the advantage-conditioning input, the distributional value function, the RECAP training loop (Algorithm 1), and the espresso/laundry/box-assembly checkpoints and datasets -- are not present in the parsed repo.
- `being_h05_2026` (high). Released code (commit e12ac44f) exposes only benchmark/inference/config scaffolding (policy.py, beingh_policy.py, dataset-transform configs). no MoT/MoF model definition, no rectified-flow action-expert source, and no config for any dexterous-hand embodiment; the only post-train configs present (libero_robocasa.yaml, libero_all.yaml, robocasa_human.yaml, so101_example.yaml) are all gripper-only, so the dexterous-hand claims cannot be cross-checked against released code.

parse-limitation, 14 rows.

- `dexpoint_2022` (low). The released reward reshapes the paper's reach term into an inverse-distance form, 1.0/(0.06+finger_object_dist) in place of the plain distance of Eq. 5, and adds a 2 cm lift-threshold bonus, a target-distance term and an IK controller-tracking penalty that Eq. 5 does not list. Artefact: `yzqin/dexpoint-release` at `17f1e238bb`, `dexpoint/env/rl_env/relocate_env.py` (`AllegroRelocateRLEnv.get_reward`). Review: Reclassified from contradiction to parse-limitation after the corresponding author replied, 2026-09-20. Three parts of the original claim do not stand. The lift term is not a departure: object_lift is already h_current minus h_init, set against a resting height fixed at reset, so 10*clip(object_lift,0,0.2) is the paper's formula with the stated weight plus a saturation, and this survey's parse captured the reward body but not the line defining object_lift, so the claim rested on material the snapshot did not contain. The rotation bonus is inactive at the paper's settings, since rotation_reward_weight defaults to

zero and the term is multiplied by it, so it contributed nothing to the reported numbers. The reach-term difference is real but is version skew rather than a contradiction: the inverse form is a later, lower-variance revision and the paper's reported variance is that of the plain-distance formula the paper states. The remaining extra terms have stated purposes, the 2 cm bonus blocking a height reward hack on meshes with no flat resting surface and the IK penalty correcting a SAPIEN-specific controller mismatch. The reply window closed on 10 October 2026 with no further correspondence.

- `artigrasp_2023` (low). paper's two regulariser weights (w_rh=0.5, w_ro=0.2) correspond to five separate velocity penalties in code (-0.5,-0.2,-0.5,-0.5,-0.3); the fingertip weight 12.0 and lambda 5.0 in Table 6 do not appear in the yaml; reward weights differ between curriculum phase-1 and phase-2 configs though Table 6 reports only one set Review: R3 adversarial review: the reward body is C++ and absent from the parse; the weight split is the paper's documented curriculum
- `diffusion_policy_2023` (high). the paper's core loss equations (DDPM training loss, EBM/InfoNCE, score-matching) are unrecovered images in the parsed text, and code/md is a signature-only API map, so compute_loss's body could not be verified from code; only the exact shipped noise-scheduler config (DDPMScheduler, 100 train steps) is confirmed.
- `eureka_2023` (low). The released repo's default eureka/cfg/config.yaml ships iteration:1, sample:3 (a fast-test default), not the paper's reported experimental configuration of N=5 iterations × K=16 samples × 5 independent runs, and LLM-generated reward code itself is never checked into the repo (written at runtime to a gitignored outputs/ directory). Review: R3 adversarial review: the repo README documents the paper's defaults; the cited config block is a fast-test preset
- `unidexgrasp_pp_2023` (low). The paper's four named reward weights (0.5, 0.1, 2, 10) numerically match literals in the released compute_hand_reward, but the code never labels them with the paper's variable names, and the paper's own display equations were dropped from the PDF, so the correspondence is inferred rather than confirmed by either source.
- `bidexhd_2024` (high). Table 5 weights (w_r, w_t, w1-w4, lambda_w=0.12, lambda_ft=0.48) cannot be matched to code reward lines because the code md truncates the reward function before stage-2/final sum; lambda_w and lambda_ft numerically match the code's grasp-condition thresholds (0.12, 0.12x4 fingers) rather than confirmed reward weights
- `graspxl_2024` (low). The code splits the paper's single r_reg = -w_h//u_h//^2 - w_o//u_o//^2 into four separately-weighted terms (wrist vel, wrist qvel, object vel, object qvel), and the floating-phase object-velocity coefficient (-1.5) does not match the paper's stated w_o=0.1, so the regularization weighting cannot be reconciled 1:1 with Table 9. Review: R3 adversarial review: the environment source is not in the parse and the note says the reward cannot be verified
- `humanplus_2024` (medium). code implements a superset of legged_gym reward terms (lin_vel_z, ang_vel_xy, orientation, base_height, torques, dof_vel, dof_acc, action_rate, collision, termination, dof/torque limits, tracking_lin/ang_vel, feet_air_time, stumble, stand_still, feet_contact_forces, target_jt) beyond the 8 terms documented in the paper's Table 1
- `omnigrasp_2024` (medium). The code's compute_pregrasp_reward_time hard-codes w_pos, w_rot = 0.9, 0.1, silently overriding the rwd_specs weight arguments passed into the function; separately, the paper's pre-grasp reward weights (w_hp, w_hr) are not resolvable from either the paper text or Appendix Table 7.
- `resdex_2024` (medium). The base-policy and residual-policy task files implement different variants of compute_hand_reward (whether finger/hand-distance terms are included alongside goal_hand_rew/hand_up/bonus differs), a difference the paper's Sec 4.1/4.4 text does not describe, and the released configs' env count (4096 default) and iteration counts (2500/5000) do not match the paper's stated 11,000 envs and 5,000/20,000 iterations.
- `dexgraspvla_2025` (high). Hardware-interfacing code is explicitly withheld ("Due to intellectual property constraints, we are unable to open-source the hardware-related code"); only function signatures for the controller loss (compute_loss, noise_assignment) are present in code/md, not their bodies, so the MSE/Immiscible-Diffusion loss cannot be verified against code.
- `dextrack_2025` (low). The paper gives one canonical reward-weight table (Table 3), but the released repo's task-config YAMLs contain several divergent, unreconciled reward-coefficient sets across AllegroHandTracking/… TrackingGeneralist/other variants, and it is not identifiable from the parsed source which config produced the paper's headline Table 1 numbers.

- `dexvla_2025` (high). code/md only carries class/function signatures, not bodies, for the diffusion head (modeling_scaledp.py) and no config file with the α weight was captured; the paper's diffusion loss and denoising process cannot be verified against code bodies. only the multi-head ScaleDP signature and a DDIM fp32 patch are available, so none of the paper's tables can be reproduced from what was parsed.
- `maniptrans_2025` (low). paper's PPO learning rate (5e-4) and env count (4096) differ from the shipped README/yaml defaults (2e-4, 8192) Review: R3 adversarial review: the cited values are a README override and an unused fallback; the repo's own config matches the paper

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Appendix D. The existing surveys, and what each covers

Fourteen corpus entries are themselves surveys or engine-comparison studies. Table 10 sets them on one set of columns. The columns record what each work covers, not how well. Section 1 states what this survey adds over them and does not repeat the comparison here.

`an_dexil_survey_2025` is the closest in subject, covering imitation learning for multi-fingered hands by learning family, end-effector class and demonstration source. It gives reinforcement learning no taxonomy, treats bimanual work as a single "multi-agent" subsection, compares no simulator, and defines no evaluation metric. `welte_iil_survey_2025` finds only seven dexterous works that use interactive imitation learning and carries a fifteen-hand commercial table, with no bimanual section, no benchmark table and no contact modelling. `zhao_sim2real_survey_2020` supplies the standard sim-to-real split and predates GPU-parallel simulation. `firoozi_foundation_models_2023` has zero occurrences of bimanual, tactile or in-hand.

`bai_unified_manip_survey_2025` spans all of manipulation across 212 pages. Its Sec. 4.3 on dexterous manipulation runs about 720 words, the third longest of its ten task subsections behind grasping and quadrupedal manipulation, which is a real treatment and not a passing mention. The difference is elsewhere. Its Sec. 1.2 lists dexterous manipulation among the topics that "existing surveys" cover from "narrower perspectives" and defers it to two of them.

`zhao_dexhand_survey_2026` is the most recent hand-centred survey, with a 29-hand anatomy table and a task-by-paradigm taxonomy. Its Sec. IV-C names the reference-versus-rollout split that Section 1 credits it with, and its Sec. III-F is the one bimanual subsection, 16 cited works with no coordination analysis.

`nine_physics_engines_review_2024` is the predecessor closest to the engine comparison, and it reviews Brax, Chrono, Gazebo, MuJoCo, ODE, PhysX, PyBullet, Unity and Webots for reinforcement learning research. It scores each on documentation, model and environment creation, URDF and MJCF support, and readiness for multi-agent work. It runs no benchmark of its own and says so in its Sec. V, that implementing the same scenarios across nine engines "goes beyond the scope of this paper". Its running bodies are ant-and-humanoid RL benchmarks rather than hands, and it discusses no timestep, no friction model, no contact formulation and no penetration.

Table 10. Existing surveys and what each covers

survey	yr	scope	taxonomy used	bimanual covered	hardware covered	evaluation covered	gaps it names
isaac_sim_2026	2026	one simulator's ecosystem and application domains, reviewed rather than measured	qualitative capability matrix, Table 1, over simulators	one cited GR00T task called bimanual, no hand, DoF or number	no hand named anywhere in paper or code parse	none. The paper runs no experiment of its own	computational cost, configuration complexity, learning curve
zhao_dexhand_survey_2026	2026	dexterous hands end to end: hardware anatomy, methods, datasets, directions	five task categories, each split by learning paradigm. Hardware is split by actuation, transmission and perception	one subsection, III-F, 16 cited works, no coordination analysis	Table I, 29 hands, 12 columns, secondary values	names two layers: physical plausibility including penetration before execution, success during it. No threshold, method or count	hardware feasibility, perception fusion, learning beyond benchmark-centric optimisation, industrialisation, absent evaluation standards
an_dexil_survey_2025	2025	imitation learning for multi-fingered end-effectors	IL family by end-effector class by demonstration source. RL gets none	one subsection, II.E, framed as multi-agent	hands named in prose, no hand table	no metric defined and no trial count. Calls for protocols, proposes none	contact dynamics in engines, data-collection standards, cross-hand transfer, failure datasets, end-effector morphology
bai_unified_manip_survey_2025	2025	all of robot manipulation. Dexterous manipulation is one of ten task subsections, about 720 words, and Sec. 1.2 defers it to other surveys	high-level planning, action modelling, actuation control, plus a bottleneck taxonomy of data and generalisation	bimanual means two arms. One dual-hand mention in the paper	hands named, no DoF or actuation table	success rate and checkpoint selection, six lines. Never says how success is judged for a dexterous task	no scaling law, sim-to-real for contact-rich tasks, fragmented datasets, reliability as important as success
welte_iil_survey_2025	2025	interactive imitation learning, seven dexterous works found	IIL feedback type, plus a keyword bibliometric of 326 papers	three mentions in 687 lines, no section	Table 1, 15 commercial hands, manufacturer figures	no metric defined, no benchmark table	tactile feedback, long-horizon tasks, generalisation, human-feedback interface
nine_physics_engines_review_2024	2024	nine physics engines for RL research, scored on documentation and usability	13-axis feature and usability matrix, Table II, plus citation-count popularity	none. MARL readiness is the multi-agent axis	none. Ant and humanoid RL bodies are the running examples	no benchmark of its own. Throughput claims are second-hand	no cross-engine MARL performance comparison exists in the literature
contact_models_comparison_2023	2023	LCP, CCP, RaiSim and NCP contact models re-implemented in one framework and ranked	contact model by solver, with the physical property each one violates	none	one Allegro hand as a benchmark system, a ball dropped into it	NCP criterion, self-consistency against a 1e-5 s reference, iteration cost	no fully satisfactory contact model, and gradients through simulation artifacts are unexplored
firoozi_foundation_models_2023	2023	foundation models in robot decision-making, perception and embodied AI	background, robotics, and robotics-adjacent papers, then by application	none, zero occurrences	none. Parallel-jaw end effectors throughout	none defined. Benchmarking appears as a reproducibility problem	data scarcity, variability, uncertainty, safety, real-time inference, and simulators that neglect contact physics
zhao_sim2real_survey_2020	2020	sim-to-real transfer in deep RL, eight pages, 21 works tabulated	zero-shot, system identification, domain randomisation, domain adaptation, learning with disturbances, simulator choice	none, zero occurrences	two cited hand works, no hand table	no metric defined, no trial count, no success rate	domain randomisation has no formal account, and domain adaptation assumes matched feature spaces
piazza_century_2019	2019	*source not obtained*	*source not obtained*	*source not obtained*	*source not obtained*	*source not obtained*	*source not obtained*
physics_engine_comparison_2015	2015	five engines on one shared model: speed, self-consistency, conservation, grasp stability	none. Four test systems, one comparison per test	none	one 35-DoF rig modelled on the Shadow Hand	largest timestep that holds a grasp, and a speed-accuracy Pareto curve	restricted feature subset by design, and the authors are MuJoCo's developers
roa_suarez_grasp_quality_2015	2015	*source not obtained*	*source not obtained*	*source not obtained*	*source not obtained*	*source not obtained*	*source not obtained*
ma_dollar_dexterity_2011	2011	*PDF fetched 2026-09-18 after the note was written, no note read from it*	*PDF fetched 2026-09-18 after the note was written, no note read from it*	*PDF fetched 2026-09-18 after the note was written, no note read from it*	*PDF fetched 2026-09-18 after the note was written, no note read from it*	*PDF fetched 2026-09-18 after the note was written, no note read from it*	*PDF fetched 2026-09-18 after the note was written, no note read from it*
okamura_overview_2000	2000	*source not obtained*	*source not obtained*	*source not obtained*	*source not obtained*	*source not obtained*	*source not obtained*

14 rows, one per corpus entry of class `survey`, 3 of which could not be obtained and are entered as such, and 1 of which was fetched too late to be read into a note. Every filled cell is read from `papers/notes/<key>.md` and is checked against a quotation from that note by `tools/make_survey_table.py`. A cell whose evidence has gone from the note prints empty rather than printing an unchecked claim. The columns record what each work covers, not how well, and a blank cell in `bimanual covered` or `hardware covered` is a scope decision by its authors rather than a failure.

Table 10's last rows carry the cost of the corpus. `okamura_overview_2000`, `piazza_century_2019` and `roa_suarez_grasp_quality_2015` are behind publisher paywalls with no author-hosted copy found on 2026-09-18, and `bicchi_hands_2000` is in the same position with no row at all. `ma_dollar_dexterity_2011` is a different case. The fetch that failed when its note was written succeeded afterwards, so a seven-page PDF is on disk with a recorded hash, and no note has been read from it. All five are cited by metadata only and nothing in this survey describes their contents. The open chapter `bicchi_grasping_chapter_2001` overlaps the paywalled Bicchi paper without being identical to it, so it is quoted in its own right.

Appendix E. Where the protocol's counts come from

Every count in Table 8 is derived below, and each axis is derived for the statistic that axis actually reports rather than by one convention applied to all of them: a single rate takes a Wilson half-width, a matched comparison takes McNemar, a ratio takes the standard error of the log ratio, and a correlation takes the Fisher-z interval. Section 7.3 states what these derivations conclude. This survey re-ran no method, so every count here rests on an interval width, a power calculation or another paper's measurement, and never on a measurement of our own. The counts an earlier draft quoted and this one withdrew are kept, so that a reader can see which test was the wrong one and why.

Fix the width first, then read off the count. Take a 95 percent Wilson interval on a single reported rate, at the worst case of p = 0.5. A half-width of 20 points needs 21 trials, 15 points needs 39, 10 points needs 93 and 5 points needs 381. Ten points is the coarsest width at which a single rate is worth printing, so the absolute-rate minimum is 93, rounded to 100. At 100 trials a reported 80 percent has an interval of 71 to 87 percent, and a reported 50 percent has 40 to 60. A comparison is a different question and a harder one: two rates each carrying ±10 points do not resolve a 10-point difference between them, because the difference's standard error is larger by a factor of √2, so the width argument sets a floor on what is worth reporting and not on what can be compared.

For the A/B comparison the relevant calculation is power, and the design is paired. Table 8 matches initial conditions by image overlay and interleaves the two policies in one session, so the unit is a matched pair and the count follows McNemar, which depends on the discordance rate. The share of initial conditions on which the two policies disagree, and not on the two rates alone. To separate 50 from 70 percent at α = 0.05 with 80 percent power: 37 pairs per arm at a discordance of 0.2, 57 at 0.3, 77 at 0.4 and 96 at 0.5. The protocol assumes 0.3 and asks for 57, and states the sensitivity rather than hiding it, because 0.5 is the discordance the same two rates produce when the pairing buys nothing, and at that value the paired count returns to the 93 per arm an unpaired test would need. The saving from pairing is real but smaller than the pair counts suggest, since a pair costs two rollouts: 57 pairs is 114 rollouts against 186. An earlier version of this section quoted 93, 169 and 387 per arm for gaps of 20, 15 and 10 points, which are correct for independent arms and are the wrong test for this protocol. That 93 was also the same integer as the half-width calculation in the paragraph above, which is a coincidence of the worst-case arithmetic and not a second derivation of the same number.

One hundred is a cap and not a bill, because on a graded score a sequential test reached its decision in 12 to 36 paired hardware trials in `beyond_binary_success_2026`. At 30 rollouts a continuous score already carries a half-width of ±0.36 standard deviations, which is why a graded score can stop where a binary one cannot. A cell that stops early does not report a Wilson interval. Optional stopping breaks the coverage of a fixed-n interval, which is the reason `beyond_binary_success_2026` and `suresim_2025` use anytime-valid betting intervals rather than Wilson, so Table 8 asks a cell run to a fixed 100 for a Wilson interval and a cell stopped early for a confidence sequence, and never for both. Intervals are also marginal rather than simultaneous. For example, evaluating twelve methods on seven axes produces 84 intervals. At 84 independent 95 percent intervals, four excursions are expected by construction, so a paper comparing k policies on m tasks

corrects its k(k−1)/2 pairwise tests to a global 95 percent level, as `lbm_careful_examination_2025` does, or says its intervals are not simultaneous.

Simulation is cheap, so simulated cells take 200 episodes, giving a 6.9-point half-width. Perturbation axes are screened rather than certified, and 40 per axis buys a 15-point half-width on each axis's own absolute rate, which is enough to rank the axes and pick the two worst for hardware. It is not enough for the ratio to the anchor that an earlier draft asked each cell to report. At 40 trials in each arm, a fall from a 0.50 anchor to 0.30 is a ratio of 0.60 with a 95 percent interval of 0.34 to 1.06, which contains 1: at the screening count you cannot establish that the perturbation hurt at all. Certifying that same drop takes 101 per arm at 80 percent power, or about 50 for an interval that merely excludes 1, so Table 8 now asks for absolute rates with their own intervals at 40, reports the ratio without an interval, and prescribes 101 before any claim that a named axis hurt.

For unseen objects the resampling unit is the object and not the trial, so 20 objects at 5 trials each gives 100 trials and an object-level half-width near 20 points. That 20 points is the Wilson width at n = 20 and it treats each object's outcome as a single Bernoulli draw, which the five within-object trials are not. It is the right order of magnitude and the assumption belongs in the cell. A 10-point claim about an object distribution needs about 93 objects. Seven of the 39 rows that state an unseen count reach that: 225 in `bimangrasp_2024`, 241 in `resdex_2024` and `unidexgrasp_2023`, 360 in `dexgraspvla_2025`, 500 in `dexmv_2021`, 2029 in `clutterdexgrasp_2025` and 503409 in `graspxl_2024`. For a continuous score the half-width is 1.96 standard deviations over the square root of the count, so 100 rollouts give ±0.20 standard deviations, and the unit is the rollout because frames within one are correlated.

The transfer axis is the one where 100 is least defensible. On 100 matched pairs a measured correlation of 0.70 carries a Fisher-z interval of 0.58 to 0.79, a half-width of about 0.10 that 130 pairs would be needed to guarantee. That is enough to establish that a simulator tracks reality at all, and it is not enough to separate `suresim_2025`'s useful regime from its marginal one, since those differ by about 0.11 in correlation. Both limits come within 0.05 of the estimate only at about 457 pairs. Table 8 states which of the two decisions each count supports rather than leaving a reader to assume the larger one.